Job Preferences, Labor Market Power, and Inequality

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Abstract

This paper examines how a firm's labor market power shapes, and is shaped by, its workforce, and evaluates the implications for wage inequality and welfare. Using matched worker-firm panel data from Norway (1995-2018), I develop, identify, and estimate an equilibrium model of the labor market where firms compete for workers who are heterogeneous in both their skills and preferences over wages versus non-wage job amenities. When a firm adjusts its wages, the composition of its workforce shifts, which in turn affects the slope of its labor supply curve. As a result, a firm's wage setting power varies based on which workers it employs. I use the model to draw inference about the incidence of wage markdowns and rents within and across firms, and the implications for wage inequality and sorting. Eliminating market power widens within-firm skill premia by 1.3% while compressing wage differences between firms by 16%, leading to a 4% reduction in total wage inequality and a 3.3 percentage-point (23% of the baseline) decline in the gender pay gap. Variation in wage setting power across firms also generates large allocative inefficiency, with welfare losses from labor market power estimated at 9.6% relative to the competitive benchmark.

I. Introduction

The role of firms in shaping wage inequality is a central topic for labor policy. Three growing empirical literatures analyze how the characteristics of firms influence workers' earnings. One body of research (Manning, 2021; Card, 2022; Azar & Marinescu, 2024; Kline, 2025) studies the role of imperfect competition in labor markets, where firms face upward sloping labor supply curves, allowing them to exert wage setting power. A second literature, building on work by Abowd, Kramarz & Margolis (1999), focuses on sorting, using matched worker-firm panel data to uncover the relationship between worker and firm productivity. A third literature exam-

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ines a worker's willingness to trade higher wages for better non-wage aspects of a job, finding that the trade-off varies greatly among workers (Mas & Pallais, 2017; Wiswall & Zafar, 2018).

Although often studied in isolation, the topics explored in these literatures are interconnected. In an economy with heterogeneous workers and firms, employers with different characteristics attract different types of workers: some who are highly responsive to wages and others who are less responsive, prioritizing non-wage factors. Because of this sorting, the composition of a firm's workforce can affect its wage setting power by shaping the elasticity of labor supply that it faces, raising the question: *How is a firm's wage setting power linked to its workforce?* Addressing this question would offer new insight into the determinants of worker sorting and the incidence of labor market power, with key implications for inequality and welfare.

This paper develops an empirical framework to analyze how workers choose jobs based on heterogeneous wage-amenity trade-offs, and the consequences of this sorting for firms' wage setting power. I show that the labor market power of a firm—as measured by wage markdowns and rents—depends directly on the composition of its workforce, which influences, and is influenced by, the wages and amenities the firm provides. Moreover, by failing to account for this relationship, one would overlook a key source of allocative inefficiency arising from differences in wage markdowns among heterogeneous firms. I show how to recover key economic quantities under this framework from matched worker-firm panel data, while allowing for unobserved heterogeneity in worker skills, firm productivity, technology, and amenities. I then apply my framework to administrative data from Norway, where I draw inference about the determinants of worker sorting and the distributional impacts and welfare costs of labor market power.

I begin my analysis by presenting an equilibrium model of the labor market. Building on previous models of job differentiation, such as Card et al. (2018) and Lamadon et al. (2022), I assume workers' idiosyncratic tastes are not fully priced into wages, leading firms to face imperfectly elastic labor supply curves. Yet, unlike previous work, I allow workers to differ in their marginal rates of substitution between wages and amenities. Motivated by the literature on the willingness to pay for amenities, I allow the wage-amenity trade-offs to vary systematically across skill groups, while also allowing for idiosyncratic variation within each skill group. This variation can reflect household circumstances, financial barriers, social ties, or any other unobserved factors, and I place no parametric restrictions on the distribution of workers' trade-offs.

This extension has key implications for the slope of the labor supply curve faced by a firm. It establishes a link between worker and firm characteristics, so that the elasticity of labor supply the firm faces depends on the composition of its workforce. As a result, labor supply elasticities vary across firms and are also endogenous to changes in the environment: if a firm adjusts its wages, then its workforce composition shifts, which in turn affects the elasticity of labor supply it faces. By relaxing the standard assumption of homogeneous and isoelastic labor supply curves, the model no longer implies uniform wage markdowns across firms. Moreover, to the extent that wage-amenity trade-offs correlate with workers' skills (on which firms can condition wages), markdowns can vary within, as well as across, firms. These implications do not rely on firms competing strategically for workers, as in Berger et al. (2022). Even if firms are strategically small in the market, their wage setting power still varies based on *who* they employ.

To account for multiple sources of worker sorting long emphasized in the literature (e.g., Shimer & Smith, 2000), I allow firms to be differentiated along several key margins. Specifically, I allow productivity to be specific to the firm-skill match, so a worker's skills can be valued differently at different firms. I also allow for firm-specific production technology, so firms can have different returns to scale and differ in the degree of substitutability between labor inputs. Moreover, I allow firms to have distinct non-wage amenities, which can also vary among workers within the same firm.[1] I examine how each of these firm characteristics interacts with worker preferences in equilibrium, and their implications for the incidence of labor market power.

I then turn to the question of identification, demonstrating how to learn about labor market power under the model using matched worker-firm panel data. In my analysis, identification is challenging for two reasons. First, a worker's wage markdown is not observed in the data. Observationally equivalent workers can earn different wages due to unobserved skill differences, not market power.[2] Wages may also not fully reflect a worker's compensation, as firms may have amenities that researchers do not observe. Indeed, as Rosen (1986) discusses, compensating differentials can exist even in competitive markets. A second identification challenge stems from the generality of my framework, which allows wage markdowns to vary both within and across firms, and to be endogenous to wages. This feature complicates my analysis, since instead of recovering a single markdown, I must recover entire functions of firm characteristics.

I show how to overcome these challenges to achieve identification of the model through the use of panel data by leveraging labor demand shocks to which firms are differentially exposed. These shocks can be recovered either directly from the matched worker-firm panel data, using productivity shocks inferred from firm accounting records, or from external data sources that measure exogenous shocks to firms' product demand. I leverage both methods in my analysis. One novelty of my approach is that I allow the labor demand shocks to shift wages in a way that is reflected in market aggregates, thus affecting workers' reservation wages, skill investments, and preferences. This generates spillovers on the labor supply curves faced by *all* firms, including those that do not directly experience a shock. Such spillovers prevent me from recovering the elasticity of labor supply to a firm by directly comparing firm-level outcomes before and after the shocks. Nevertheless, I show that, under weak conditions on the instrument, the elasticities are identified from a difference-in-differences comparison of wage and labor outcomes. This method involves comparing average changes in wages and employment across firms

[1] In my analysis, I do not take a stance on which amenities matter most to workers. Rather, I allow firms to have a set of amenities that are unobserved to the analyst and are assumed to be fixed over the estimation window. This approach neither imposes nor precludes the possibility that firms initially choose amenities to maximize profits.

[2] For additional discussion, see Murphy & Topel (1990), Gibbons & Katz (1992), and Gibbons et al. (2005).

with different exposures to labor demand shocks, while controlling for the firm's initial wages.

An additional methodological contribution is that, by recovering the labor supply elasticities faced by firms, I can infer the distribution of workers' wage-amenity trade-offs. This result is grounded in a revealed preference argument: holding all else fixed, firms that offer higher wages to a given skill group disproportionately attract workers who respond more strongly to wages. This compositional effect is reflected in the labor supply elasticities firms face. Leveraging this relationship, I prove there exists a one-to-one mapping between a firm's labor supply curve and the density of workers' wage-amenity trade-offs. I then derive a nonparametric estimator of this density, which can be implemented separately for different population subgroups.

Using my framework, I show how to recover the model parameters that characterize firm productivity, technology, amenities, and workers' skills and preferences. From these parameters, I can quantify the division of rents between workers and firms, and conduct counterfactual analyses that isolate the role of labor market power on earnings inequality and worker sorting.

Lastly, I estimate the model using a matched employer-employee panel dataset covering all workers and firms in Norway for the period 1995-2018. My estimates yield three key findings. First, many employers have substantial wage setting power over workers, and this power varies considerably both within and across firms. While on average, a worker's wage constitutes 86% of her marginal product, the wage-to-marginal-product ratio ranges from 70% to 95% between the 1st and 99th percentiles. Moreover, many workers and firms earn sizable rents: the average worker is willing to forgo 16% of her wage to be at her current firm, while the average firm derives 23% of its profit from wage setting power. Rents are highly dispersed in the economy, with worker rent shares ranging from below 10% to over 25% of earnings, and firm rent shares ranging from below 5% to over 50% of profit.[3] I document systematic differences in wage markdowns and rents by industry, region, worker skills and demographics, providing a detailed picture of which workers are most affected by market power. I find that, within firms, employers tend to have more wage setting power over high-skilled workers, whereas, across firms, market power is most concentrated among employers that pay lower wages to a given skill group. I also find that women are disproportionately employed at high-markdown firms. In a counterfactual that shuts down variation in workers' wage-amenity trade-offs, one fails to recover these differences, as the restricted model implies a constant wage markdown and little scope for rents to vary.

Second, given how wage markdowns vary within and across firms, I find that labor market power has sizable effects on wage inequality. To quantify these effects, I solve for the competitive (Walrasian) equilibrium allocation, eliminating labor wedges and allowing workers to real-

[3]The estimated density of worker rents broadly aligns with Caldwell et al. (2025), who find from choice experiments in Germany that 39% of workers would prefer staying at their current firm to a 20% raise elsewhere.

locate across firms in response to wage changes. Using the variance of log earnings to measure inequality, I find that the competitive allocation raises within-firm inequality by 1.3% but reduces between-firm inequality by 16%, yielding a 4% decline in total inequality. Moreover, this counterfactual reduces the gender wage gap by 3.3 percentage points, from 14.3% to 11.0%. This finding underscores the role of monopsony power in sustaining gender pay differences, building on research dating back to Robinson (1933).[4] These effects would not be captured by a framework that does not account for rich heterogeneity in workers' labor supply responses.

Third, I find that labor market power leads to substantial misallocation of workers to firms. Eliminating labor wedges would raise total welfare by 9.6% in dollar terms. In the competitive allocation, more workers would be employed at lower-productivity firms. This finding reflects that labor wedges are largest at firms that pay low wages to a given skill group, and these firms tend to be less productive. Thus, removing wedges leads to larger wage and employment gains at the bottom of the firm productivity distribution. This result runs counter to the common presumption that larger, more productive firms necessarily have more wage setting power, an idea rooted in single-buyer monopsony models dating back to Robinson (1933). However, when the labor market contains many firms, wage setting power need not increase with firm size (Manning, 2021). Indeed, my estimates reveal that firm characteristics, rather than size per se, play a key role in shaping wage setting power by determining which workers the firm attracts and how responsive they are to wages. At the sector level, I find that labor wedges are often larger in low-premium industries, as well as in rural labor markets. Eliminating wedges would therefore shift employment on net away from high-premium industries such as finance toward retail, education, and healthcare, while also raising employment in rural areas. This reallocation is highly heterogeneous across skill groups, in both size and direction. I also show that it would be overlooked in a version of the model that rules out variation in workers' wage-amenity trade-offs.

This paper contributes to three literatures. First, it contributes to a large literature on monopsony in labor markets; see Manning (2021), Card (2022), Azar & Marinescu (2024), Kline (2025), and Caldwell et al. (2026) for recent surveys.[5] Within this literature, I build on a class of static equilibrium models studied by Card et al. (2018), Azar et al. (2022), Lamadon et al. (2022), Kroft et al. (2025), Chan et al. (2025), Roussille & Scuderi (2026), and others, where imperfect competition arises because workers' idiosyncratic tastes are not fully priced into ear-

[4]For recent empirical work on monopsony and gender pay gaps, see Barth & Dale-Olsen (2009), Card et al. (2016), Sharma (2023), Caldwell & Oehlsen (2023), and Morchio & Moser (2024). This work often infers gender differences in monopsony power by estimating firm-level labor supply elasticities separately for men and women. I extend this analysis by allowing the elasticities to vary within gender and, for each gender, estimating the full distribution of workers' wage-amenity trade-offs. This distributional approach delivers a richer characterization of gender pay gaps, allowing me to assess whether gender differences in monopsony are widespread or concentrated among certain workers. I find substantial heterogeneity in the incidence of wage setting power within each gender.

[5]See also Boal & Ransom (1997), Bhaskar et al. (2002), Manning (2011), and Sokolova & Sorensen (2021).

nings. This work, which Manning (2021) terms "new classical monopsony," draws on differentiated demand models commonly used in industrial organization. A key limitation, however, is that these models often specify utility functions that strongly restrict how worker and firm characteristics interact. These restrictions constrain worker sorting in a way that sharply limits the scope for firm market power to vary.[6] I relax these restrictions to allow for fully unrestricted heterogeneity in workers' wage-amenity trade-offs. This extension is in the spirit of consumer demand models with rich preference heterogeneity (Berry et al., 1995). As I show, it greatly expands the scope for worker sorting, yielding a richer characterization of labor market power.

Second, this paper contributes to an empirical literature on sorting, drawing on the additive worker and firm effects wage model introduced by Abowd, Kramarz & Margolis (1999). This work largely focuses on skill-based sorting, seeking to uncover the relationship between worker and firm productivity (e.g. Andrews et al., 2008; Card et al., 2013; Song et al., 2018). While the model allows for sorting on workers' skills, e.g., through production complementarities, I also study a second type of sorting based on workers' wage-amenity trade-offs. I show that this feature introduces a firm-skill interaction term—the wage markdown—in the worker's wage equation, such that the same firm can apply a larger markdown to some workers and a smaller markdown to others. In doing so, the paper provides new insights into the assortativeness of the allocation of workers to firms. My estimates show that wage markdowns are match-specific: firms are not uniformly "high-markdown" or "low-markdown" for all workers, and, as a result, market power distorts the allocation of labor differently for different types of workers.

Third, this paper contributes to research on the willingness to pay for non-wage amenities, recently reviewed by Mas (2025). Experimental studies and surveys show that workers value a wide array of non-wage job attributes (Hamermesh, 1999; Pierce, 2001). Moreover, the willingness to pay for these amenities varies greatly among workers (Mas & Pallais, 2017; Wiswall & Zafar, 2018), contributing to earnings inequality (Maestas et al., 2023). My paper takes this literature in a new direction by studying how differences in workers' amenity valuations affect firm wage setting power in imperfectly competitive labor markets. I show that workers' wage-amenity trade-offs help to shape the labor supply elasticities firms face, and that this relationship delivers a new revealed preference approach to infer the distribution of the willingness to pay for amenities, building on Sorkin (2018), Hall & Mueller (2018), Lagos (2026), and others.

This paper is organized as follows. Section II presents the model along with equilibrium properties related to sorting and rent sharing. Section III describes the data and sample select-

[6] Abstracting from strategic interactions, it implies uniform wage markdowns. As discussed in Section II.E, this restriction can be relaxed by adopting a nested logit model (Lamadon et al., 2022; Azar et al., 2022) or by letting markdowns vary within a firm based on observed worker characteristics (Deb et al., 2024; Chan et al., 2025). Roussille & Scuderi (2026) also relax this assumption while restricting preference heterogeneity to a finite set of types based on observables. In my estimates, I find that allowing for fully unobserved heterogeneity in workers' preferences and productivities is essential for uncovering how wage markdowns and rents vary in the economy.

ion. Section IV develops the identification analysis. Section V outlines the estimation and reports parameter estimates. Section VI presents the main empirical results on the incidence of labor market power and the consequences for inequality and welfare. Section VII concludes.

II. Equilibrium Model of the Labor Market

In this section, I develop a theoretical framework to characterize how workers sort into jobs and the implications of these sorting patterns for a firm's wage setting power. This framework extends prior equilibrium models of the labor market, studied by Lamadon et al. (2022), Azar et al. (2022), Kroft et al. (2025) and others, by allowing workers to differ in the marginal rate of substitution between wages and non-wage aspects of jobs. I show that this extension leads firms to face heterogeneous labor supply elasticities, which reflect differences in the composition of workers at different firms. As a result, this model generates variation in the incidence of wage markdowns and rents, thus introducing a potential source of allocative inefficiency.

II.A. Environment

The economy comprises a unit measure of workers, indexed by i, and a finite number of firms, indexed by j. Every firm operates in a local labor market $m(j)$, where $\mathcal{J}_m$ is the set of firms in market m. Workers are differentially productive, and they have heterogeneous preferences over employers. Meanwhile, firms provide differentiated work environments, and they vary in their productivities and technologies. Note that, while distinguishing between labor markets will be important for the empirical analysis, it does not meaningfully impact the theoretical results. I therefore simplify notation by focusing on a single market and suppressing the label m.

Each worker has a vector of skills $X_i = (\chi_i, \varphi_i)$. These skills contain two components: a categorical *skill type* $\chi_i \in \mathcal{X}$ and a continuous *skill level* $\varphi_i \in \mathbb{R}_+$. The worker's skill type χ_i captures the non-ordinal aspects of human capital, such as training and specialization, across which firms may face imperfect substitution in production. The worker's skill level φ_i, which I treat as unobservable to the analyst throughout the empirical analysis, measures productivity within a given skill type. By modeling skills as a composite of types and levels, the framework incorporates insights from the literature on multidimensional skills (e.g., Lise & Postel-Vinay, 2020), while also allowing for human capital to vary in ways that researchers cannot observe.

I maintain two assumptions about the information structure. First, I abstract from search frictions by assuming that workers are fully informed about job opportunities and that, given posted wages, workers freely choose where to work.[7] Second, I assume that firms observe workers' skills, but only know the distribution of workers' preferences conditional on those skills. Therefore, firms may condition wages on X_i, but not on workers' idiosyncratic preferences.[8]

[7] When specifying firm production, I allow worker-firm matches to be nonproductive, which effectively narrows the relevant choice set for many workers, while maintaining the assumption that workers are fully informed.

[8] This assumption reflects information asymmetries between workers and firms. It may also account for laws

II.B. Worker Preferences

Workers have heterogeneous preferences over wages and non-wage amenities. Let $W_j(X)$ be the wage that firm j offers to workers with skills X, and let $a_j(X)$ be the component of firm j's amenity common to workers with skills X. For a worker i, the utility from working at firm j is:

$$u_{ij}\big(W_j(X_i), a_j(X_i)\big) = \beta_i \log W_j(X_i) + a_j(X_i) + \epsilon_{ij}. \tag{1}$$

In this utility function, ϵ_{ij} is a worker's idiosyncratic taste for firm j, reflecting non-pecuniary match effects such as distance from work and personal preferences over work environment. I assume $\{\epsilon_{ij}\}_{i,j}$ are i.i.d. with a continuous distribution F_ϵ taking strictly positive density on $\mathbb{R}$.

The coefficient β_i is the worker's marginal utility of log earnings. This parameter governs the marginal rate of substitution between wages and amenities, thereby capturing the trade-off between higher pay and better working conditions. Workers with lower β_i will be more willing to sacrifice earnings in exchange for better non-wage job characteristics. In Appendix B.1, I provide an explicit micro-foundation for the utility function (1), where I show that β_i can be reinterpreted as the worker's marginal rate of substitution between consumption and leisure.

In my analysis, I allow β_i to vary freely among workers. In particular, I treat β as a random variable with density f_β, and I do not impose any parametric restrictions on this density, such as assuming normality. One important aspect of this framework is that I allow for two distinct forms of heterogeneity in β. First, I allow β to correlate with a worker's skills X, so that the trade-offs workers make when choosing jobs can depend on factors that impact productivity. Second, I allow β to vary conditionally within each skill group X, so that equally skilled workers can have different wage-amenity trade-offs. This within-skill heterogeneity is particularly important, as it allows a worker's trade-off to reflect considerations beyond productivity. For example, β may depend on factors such as age, gender, household characteristics, and wealth. It may also be correlated among workers in the same household, neighborhood, or peer group.

II.C. Firm Production

Define $D_j(\chi, \varphi)$ to be the mass of workers with skills $X = (\chi, \varphi)$ that is demanded by a firm j. For each skill type $\chi \in \mathcal{X}$, the total efficiency units of labor at the firm are defined as follows:

$$L_j^{\text{eff}}(\chi) = \int \varphi D_j(\chi, \varphi) d\varphi. \tag{2}$$

Let Y_j be the firm's value added from hiring workers. Production is a constant elasticity of substitution (CES) composite of effective labor across skill types. This specification allows skill types to be imperfect substitutes, while maintaining perfect substitution within skill types. It also allows for decreasing returns to scale, so output need not rise in proportion to labor inputs.

against pay discrimination, which are common in many labor markets. See Bhaskar et al. (2002) for discussion.

$$Y_j = T_j \left(\sum_{\chi \in \mathcal{X}} \theta_{j\chi} \left(L_j^{\text{eff}}(\chi) \right)^{\rho_j} \right)^{\frac{1-\alpha_j}{\rho_j}} . \tag{3}$$

In this production function, $T_j \in \mathbb{R}_{++}$ is the firm's total factor productivity (TFP); $\theta_{j\chi} \in \mathbb{R}_{++}$ is the firm-specific efficiency of skill type χ; $(1-\rho_j)^{-1} \in [0, \infty)$ is the elasticity of substitution between skill types at the firm; and $1 - \alpha_j \in (0, 1]$ is the firm's returns to scale. For simplicity, I abstract from capital and intermediate inputs, as well as from the nature of competition in the product market. Nevertheless, as I show in Appendix B.2, this production function may be rationalized under either a perfectly competitive or an imperfectly competitive product market.

By allowing the parameters $(T_j, \{\theta_{j\chi}\}_{\chi \in \mathcal{X}}, \rho_j, \alpha_j)$ to vary freely across firms, the model accommodates a wide range of production technologies. First, firms can differ in their overall productivity through T_j. Second, firms can differ in the relative efficiency of each skill type through $\theta_{j\chi}$; for example, some firms may place more value on a college degree than others. Third, firms can differ in the substitutability of skill types through ρ_j; for example, some firms may have specific task requirements that affect their ability to interchange high- and low-skill workers. Fourth, firms can differ in their returns to scale through α_j. Throughout the analysis, I place no restrictions on the joint distribution of firm productivity, technology, and amenities.

II.D. Characterization of an Equilibrium

In equilibrium, each worker i chooses a firm to maximize utility, given wage offers $\{W_k(X_i)\}_{k=1}^J$, non-wage amenities $\{a_k(X_i)\}_{k=1}^J$, and idiosyncratic tastes $\{\epsilon_{ik}\}_{k=1}^J$. A worker with skill profile X and wage-amenity trade-off parameter β would choose to work at firm j with the probability:

$$\mathrm{P}\left(j(i) = j | \beta, X\right) = \int_{-\infty}^{\infty} \prod_{k \neq j} F_\epsilon \left(\beta \log \left(\frac{W_j(X)}{W_k(X)} \right) + a_j(X) - a_k(X) + \tilde{\epsilon} \right) f_\epsilon(\tilde{\epsilon}) d\tilde{\epsilon}. \tag{4}$$

Averaging over realizations of β, the labor that is supplied to firm j by workers with skills X is:

$$S_j(X) = \int \left(\int_{-\infty}^{\infty} \prod_{k \neq j} F_\epsilon \left(\beta \log \left(\frac{W_j(X)}{W_k(X)} \right) + a_j(X) - a_k(X) + \tilde{\epsilon} \right) f_\epsilon(\tilde{\epsilon}) d\tilde{\epsilon} \right) f_{\beta,X}(\beta, X) d\beta. \tag{5}$$

Given these labor supply curves, each firm sets wages to maximize profit. Since a firm cannot condition wages on individual preferences (β_i, ϵ_{ij}), its wage setting decisions are based on the distributions of these parameters, $\{F_{\beta|X}\}_X$ and F_ϵ. The firm's profit maximization problem is:

$$\max_{\{W_j(\chi,\varphi)\}_{\chi,\varphi}} T_j \left(\sum_{\chi \in \mathcal{X}} \theta_{j\chi} \left(\int \varphi D_j(\chi, \varphi) d\varphi \right)^{\rho_j} \right)^{\frac{1-\alpha_j}{\rho_j}} - \sum_{\chi \in \mathcal{X}} \left(\int W_j(\chi, \varphi) D_j(\chi, \varphi) d\varphi \right), \tag{6}$$

subject to the constraint that labor demand equals supply: $D_j(X) = S_j(X)$ for all $X = (\chi, \varphi)$.

I consider an equilibrium in which firms view themselves as strategically small in the labor market. That is, the firm does not internalize the effect of changing its own wages on the labor supplied to other firms. It is worth noting that this condition does not imply that workers regard firms as infinitesimal. On the contrary, there is a finite number of firms, and each worker chooses a firm by considering the entire set of offered wages and amenities. Instead, this condition applies only to firm conduct: it assumes that the firm sees itself as a marginal player, thereby ruling out strategic interactions in wage setting. This assumption also appears in Card et al. (2018), Lamadon et al. (2022), Azar et al. (2022), and Kroft et al. (2025), among others.[9] In Section V, I derive a statistical test of this assumption. Applying this test to Norwegian administrative data, I find that firm behavior is consistent with the strategically small benchmark.

Definition. Given worker skill and preference distributions $(F_{\chi,\varphi}, \{F_{\beta|\chi,\varphi}\}_{\chi,\varphi}, F_\epsilon)$, firm amenities $\{a_j(\chi,\varphi)\}_{\chi,\varphi}$ and production parameters $(T_j, \{\theta_{j\chi}\}_\chi, \rho_j, \alpha_j)$, an equilibrium is defined by the employment outcomes $j(i)$, labor supply curves $S_j(\chi,\varphi)$, and wages $W_j(\chi,\varphi)$ where:

(i) Each worker chooses the firm that maximizes utility, with utility given by equation (1).

(ii) Labor supply curves $S_j(\chi,\varphi)$ are consistent with workers' optimal decisions, as in (5).

(iii) Each firm sets wages $W_j(\chi,\varphi)$ for all $\chi \in \mathcal{X}$ and $\varphi \in \mathbb{R}_+$ to maximize profit, as defined in equation (6), ensuring that labor demand $D_j(\chi,\varphi)$ is equal to labor supply $S_j(\chi,\varphi)$.

In Lemma 1 of Appendix C.1, I establish the existence of an equilibrium with strictly positive wages and employment. Lemmas 2 and 3 further characterize properties concerning equilibrium uniqueness. Additionally, I analyze properties such as the concavity of the profit function and dynamic stability of the equilibrium. These results are useful for counterfactual analysis, as they enable me to compute equilibrium outcomes from the underlying model parameters.

II.E. Properties of the Labor Supply Curve to a Firm

A primary motivation for incorporating random coefficients in the worker's utility function is that it enables the model to capture rich substitution patterns that reflect sorting on individual heterogeneity. This subsection examines how this heterogeneity expands the scope of worker behavior in the model and what it implies about the shape of the labor supply curve to a firm.

To streamline the exposition, I focus on a version of the model where ϵ_{ij} follows a Type 1 Extreme Value (logit) distribution, $F_\epsilon(\epsilon) = \exp(-\exp(-\epsilon))$. This is the most common parameterization of ϵ_{ij} used in the literature, and it yields transparent expressions for the firm's labor supply curve and its elasticities. The labor supply curve, as specified in equation (5), becomes:

[9] Alternatively, Berger et al. (2022), Rubens (2023), Deb et al. (2024), and Chan et al. (2025) study frameworks where firms leverage their market shares when setting wages, and Jarosch et al. (2024) studies a search model with large firms. Roussille & Scuderi (2026) develop a method to adjudicate between models of firm conduct in the context of an online job search platform, a setting in which firms also compete for workers through posted job offers. The authors find that models that abstract from strategic interactions outperform those that include them.

$$S_j(X) = \int \frac{\exp\big(\beta \log W_j(X) + a_j(X)\big)}{\sum_{k=1}^{J} \exp\big(\beta \log W_k(X) + a_k(X)\big)} f_{\beta,X}(\beta, X) d\beta. \tag{7}$$

Define $I(\beta, X) = \sum_{k=1}^{J} \exp\big(\beta \log W_k(X) + a_k(X)\big)$ to be the wage index for workers with preference parameter β and skills X. From the perspective of a strategically small firm, its wage has a negligible effect on $I(\beta, X)$. Note that, although I work with the logit error structure for expositional clarity, the main insights in this subsection do not depend on this specific distribution. Appendix Properties A.9 and A.10 show the results extend to more general error structures.

II.E.1. Labor Supply under Homogeneous Effects

The standard assumption in models of job differentiation is that β is constant across workers. Under this assumption, firm wages and amenities do not interact with worker-specific preferences in the utility function. This separability delivers a relatively simple and tractable labor supply system. Yet, it imposes a strong form of homogeneity on aggregate substitution patterns that sharply restricts worker sorting. This critique has previously been raised in the context of product markets (Berry et al., 1995). As I now show, it also holds key implications for labor markets.

To see how the homogeneous effects assumption restricts worker sorting, consider two firms, j and j', with different technologies and work environments. With constant β, these firms exhibit identical substitution patterns: if wages fall at j and j' for workers with the same skills, workers induced to leave either firm are equally likely to move to any third firm k, regardless of whether k is more similar to j or to j'. This restriction contradicts the usual intuition that similar firms attract workers with similar preferences, and are thus closer substitutes for one another. It is also hard to reconcile with the sorting patterns observed in many labor markets.

One key economic quantity that is impacted by this restriction is the elasticity of labor supply faced by a firm, $\frac{\partial \log S_j(X)}{\partial \log W_j(X)}$. If β is homogeneous, then this elasticity is identical across firms:

$$\frac{\partial \log S_j(X)}{\partial \log W_j(X)} = \frac{\partial \log S_{j'}(X)}{\partial \log W_{j'}(X)}, \quad \text{for all } j, j' \in \{1, \ldots, J\}. \tag{8}$$

This property underscores the limitations of assuming homogeneous effects. It implies that all firms, regardless of their characteristics, face the same employment response to a wage change.

One way to relax these restrictions on worker sorting patterns is to allow ϵ_{ij} to be correlated across firms in a local labor market, for example, via a nested logit structure. This approach is adopted by Azar et al. (2022), Lamadon et al. (2022), and others, and it implies that workers are more likely to substitute toward firms within markets than across markets. Nevertheless, in any given labor market, the same restrictions on worker sorting patterns would still apply.[10]

[10] In principle, nests can be defined more broadly so they do not coincide with markets. However, any nesting structure would still be based on factors fully observable to the analyst, a restriction that is typically undesirable.

A second extension could be to allow β to vary based on observable worker characteristics such as education or experience.[11] Yet, a concern with this strategy is that firms are also likely to observe these characteristics and would thus factor them into a worker's skills X. Indeed, as long as β is constant conditional on skills, the implications of equation (8) remain unchanged.

II.E.2. Labor Supply under Heterogeneous Effects

If β is heterogeneous, then a firm's labor supply curve reflects differences in the composition of workers at different firms. For example, a firm offering higher wages and worse amenities is more likely to attract workers who place greater value on wages (higher β), whereas a firm offering lower wages and better amenities tends to attract workers who place greater value on amenities (lower β). Thus, firms with similar wage-amenity profiles are more likely to attract workers who make similar trade-offs, leading these firms to exhibit larger substitution effects.

This variation in workforce composition shapes the elasticity of labor supply to a firm. In particular, this elasticity can be expressed as the average value of β within the firm's workforce:

$$\frac{\partial \log S_j(X)}{\partial \log W_j(X)} = \mathrm{E}\left(\beta_i \middle| X_i = X, j(i) = j\right). \tag{9}$$

When β is heterogeneous, this elasticity varies across firms, because firms with different wage-amenity profiles attract workers who make different trade-offs. As a result, each firm employs a workforce with a distinct distribution of β. The labor supply elasticities are therefore linked to firm characteristics in a way that cannot be captured in a model with homogeneous effects.

Another key consequence of heterogeneous effects is that the firm's labor supply elasticity is endogenous to wages. If a firm posts a higher wage, then it attracts workers who are more responsive to wages (higher β), which in turn leads the firm to face a more elastic labor supply. To formalize this property, I analyze the second derivative of the labor supply curve to a firm:

$$\frac{\partial^2 \log S_j(X)}{\partial [\log W_j(X)]^2} = \mathrm{Var}\left(\beta_i \middle| X_i = X, j(i) = j\right). \tag{10}$$

This second derivative is strictly positive if and only if $\mathrm{Var}(\beta_i | X_i = X) > 0$. Additionally, it is increasing in $\mathrm{Var}(\beta_i | X_i = X)$, since greater heterogeneity in β expands the scope for sorting.[12]

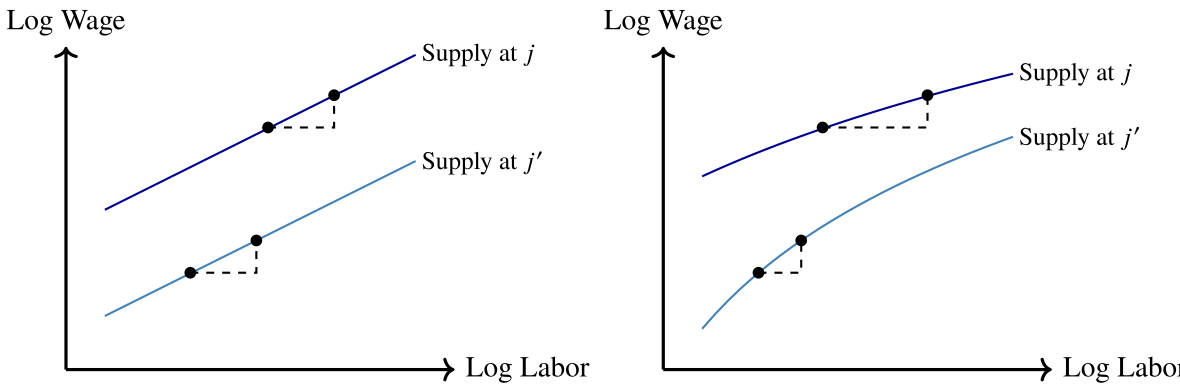
Figure 1 illustrates these implications by contrasting firm-level labor supply curves under homogeneous and heterogeneous effects. Consider two firms, j and j', where firm j pays high

[11] Examples of this approach include Deb et al. (2024) and Chan et al. (2025). The former uses a multinomial logit discrete choice model, and the latter uses a representative agent CES model. As shown in Verboven (1996), Proposition 2, both modeling approaches generate labor supply systems that are isomorphic at the market level.

[12] In Property A.1 of Appendix A, I derive the third, fourth, and fifth derivatives of a firm's labor supply curve under the logit model. I show that these quantities depend on the higher-order moments of the distribution of β.

wages but has worse amenities, and firm j' pays low wages but has favorable amenities. Under homogeneous effects (Figure 1.a), both firms would face the same employment response to an internal wage change, meaning that an $X\%$ change in wages yields the same percent change in labor at each firm. Under heterogeneous effects (Figure 1.b), these firms would face different employment responses. For the same $X\%$ wage change, the high-wage, low-amenity firm sees a larger employment response, while the low-wage, high-amenity firm sees a smaller response.

Figure 1: Illustration of Firm-Specific Labor Supply Responses



(a) Homogeneous Effects (b) Heterogeneous Effects

Notes. This figure depicts the labor supply curves $\log S_j(X)$ and $\log S_{j'}(X)$ for two firms, j and j', respectively. Panel (a) presents the case where $\text{Var}(\beta_i \mid X_i = X) = 0$. Panel (b) presents the case where $\text{Var}(\beta_i \mid X_i = X) > 0$.

II.F. Wage Determination and Sorting in Equilibrium

In equilibrium, firms face upward-sloping labor supply curves, allowing them to set wages below workers' marginal products. Let $L_j(X)$ be the equilibrium employment of skill group X at firm j, and let $\boldsymbol{\varepsilon}_j(X) = \frac{\partial \log L_j(X)}{\partial \log W_j(X)}$ be the labor supply elasticity firm j faces. A worker's wage is:

$$W_j(\chi, \varphi) = \underbrace{\frac{\boldsymbol{\varepsilon}_j(\chi, \varphi)}{1 + \boldsymbol{\varepsilon}_j(\chi, \varphi)}}_{\textit{Markdown}} \times \underbrace{\varphi T_j (1 - \alpha_j) \theta_{j\chi} \left(L_j^{\text{eff}}(\chi)\right)^{\rho_j - 1} \left(\sum_{\chi' \in X} \theta_{j\chi'} \left(L_j^{\text{eff}}(\chi')\right)^{\rho_j}\right)^{\frac{1-\alpha_j}{\rho_j} - 1}}_{\textit{Marginal Product of Labor}}. \quad (11)$$

There are three main factors that drive systematic worker sorting in the model. One factor is that workers' marginal products differ across firms. These differences arise due to variation in firm productivity $(T_j, \{\theta_{j\chi}\}_\chi)$ and technology (ρ_j, α_j), which have the potential to generate large dispersion in worker productivity within and across firms. Note that, with production complementarities, $\rho_j < 1$, and match-specific efficiency, $\theta_{j\chi}$, wages need not be log-additively separable into worker and firm components, as assumed in much of the sorting literature following Abowd et al. (1999). Relaxing this restriction allows for richer sorting, since the same firm can have a high marginal product for some workers and a low marginal product for others.

A second factor driving sorting is the non-wage amenity $a_j(X)$. These amenities generate vertical firm differentiation, as some firms can have strictly better amenities than others. They also create horizontal differentiation, as workers in the same firm can encounter different amenities. Since the amenities can depend arbitrarily on firm production parameters T_j, $\{\theta_{j\chi}\}_\chi$, ρ_j, and α_j, the model places no restriction on how they relate to a worker's marginal product.[13]

A third factor driving sorting is the wage markdown $\frac{\varepsilon_j(X)}{1+\varepsilon_j(X)}$. These markdowns vary across firms due to differences in workforce composition, reflecting an interaction between worker preferences and firm characteristics. A low-productivity, high-amenity firm is more likely to attract workers who are less responsive to wages, which allows the firm to set a larger markdown. Note that, in the model, wage markdowns are skill-specific, since firms can condition wages on skills. Therefore, workers with different skills can face different markdowns even at the same firm. This further relaxes the log-additivity restriction of two-way fixed effects models, since a given firm can apply a larger markdown to some workers and a smaller markdown to others.

II.G. Rents, Pass-throughs, and Allocative Inefficiency

I conclude this section by showing how the structural parameters in the model map into key economic quantities, such as rents earned by workers and firms, the pass-through of productivity shocks to earnings, and measures of welfare loss due to allocative inefficiency. To simplify these derivations, I assume a logit error structure for ϵ_{ij}, setting $F_\epsilon(\epsilon) = \exp(-\exp(-\epsilon))$. In each of the derivations in the Appendix, I carefully highlight which parts of the analysis rely on this assumption, and I also show how the results extend to alternative parameterizations of F_ϵ.

Worker Rents

In the economy, firms do not observe workers' individual preferences, β_i and ε_{ij}, which means they cannot price-discriminate based on each worker's reservation wage. Thus, an equilibrium involves surpluses—or rents—for inframarginal workers. These rents represent the earnings that workers are willing to forgo to be indifferent between their current firm and their next-best alternative. Let R_i^w denote a worker i's rents at her current firm. I define these rents as follows:

$$u_{ij}\big(W_j(X_i) - R_i^w, a_j(X_i)\big) = \max_{j' \neq j(i)} u_{ij'}\big(W_{j'}(X_i), a_{j'}(X_i)\big). \tag{12}$$

In a model where β is constant, the scope for worker rents to vary is sharply limited. Property A.4 in Appendix A shows that under homogeneous effects, a worker's rents as a share of wages, $R_i^w / W_{j(i)}(X_i)$, are statistically independent of both her skills X and firm j. Thus, a worker's employer and skills have no impact on the fraction of wages she earns as rents. This places a strong restriction on the incidence of rents in the economy, and it runs counter to the idea that workers at different firms or with different skills tend to benefit differently from their wage contracts.

[13] Hence, the process by which firms choose amenities is unrestricted. Identification requires only that unobserved amenities be fixed over the estimation window; initial amenity choices do not affect the estimates. If amenities are costly to the firm, the markdown formula is unchanged when value added is measured net of amenity costs.

When β is heterogeneous, the share of wages that a worker earns as rents varies systematically across firms and skill groups. To characterize this variation, I derive an expression for the average rent share, $\mathrm{E}(R_i^w | j(i) = j, X_i = X)/W_j(X)$, earned by workers with skills X at firm j.

$$\frac{\mathrm{E}(R_i^w | j(i) = j, X_i = X)}{W_j(X)} = \mathrm{E}\left(\frac{1}{1+\beta_i} \,\middle|\, j(i) = j, X_i = X \right). \tag{13}$$

This average rent share depends on the distribution of wage-amenity trade-offs, β, among workers who sort into the firm. Because the distribution can differ by skill, workers at the same firm can earn systematically different rent shares. Because it can also differ across firms conditional on skill, equally skilled workers could tend to earn different rent shares at different employers.

Property A.5 of the Appendix analyzes how average worker rent shares depend on firm characteristics. It shows that, conditional on skills, higher-paying firms attract a larger share of wage-responsive workers, who are more likely to switch firms in response to small pay differences. So, although workers at these firms may earn higher rents in levels, their rents may also constitute a smaller share of wages. In the empirical analysis, I find that neglecting this compositional effect leads to an inaccurate assessment of where rents are concentrated in the economy.

Firm Rents

Firms also capture rents in equilibrium. These rents come in the form of excess profit that firms obtain by exerting wage setting power. To measure firm rents, I study a counterfactual where the firm is a price-taker, facing a perfectly elastic labor supply curve. The firm then maximizes:

$$\max_{\{D_j^{\mathrm{pt}}(\chi,\varphi)\}_{\chi,\varphi}} T_j \left(\sum_{\chi \in \mathcal{X}} \theta_{j\chi} \left(\int \varphi D_j^{\mathrm{pt}}(\chi,\varphi) d\varphi \right)^{\rho_j} \right)^{\frac{1-\alpha_j}{\rho_j}} - \sum_{\chi \in \mathcal{X}} \left(\int W_j^{\mathrm{pt}}(\chi,\varphi) D_j^{\mathrm{pt}}(\chi,\varphi) d\varphi \right), \tag{14}$$

subject to the labor supply constraint: $D_j^{\mathrm{pt}}(X) = S_j^{\mathrm{pt}}(X)$ for all $X = (\chi, \varphi)$. In this counterfactual, the only difference in the firm's behavior is that it does not exercise its wage setting power. Therefore, this counterfactual is distinct from an equilibrium where all firms are price takers.

I define a firm j's rents as the difference between its realized profits and its counterfactual profits, given by $R_j^e = \Pi_j - \Pi_j^{\mathrm{pt}}$. In Property A.6 of the Appendix, I show that these rents equal:

$$R_j^e = Y_j \times \left[\sum_{\chi \in \mathcal{X}} \int \left(\frac{1 + \alpha_j \boldsymbol{\varepsilon}_j(\chi,\varphi)}{1 + \boldsymbol{\varepsilon}_j(\chi,\varphi)} \right) \left(\frac{\theta_{j\chi}(L_j^{\mathrm{eff}}(\chi))^{\rho_j}}{\sum_{\chi'} \theta_{j\chi'}(L_j^{\mathrm{eff}}(\chi'))^{\rho_j}} \right) \left(\frac{\varphi L_j(\chi,\varphi)}{\int \varphi' L_j(\chi,\varphi') d\varphi'} \right) d\varphi - \alpha_j \left(\frac{Y_j^{\mathrm{pt}}}{Y_j} \right) \right], \tag{15}$$

where Y_j^{pt} is the firm's value added as a price-taker. In this model, Y_j^{pt} does not appear to have a closed-form expression. Nevertheless, I show how to solve for this counterfactual quantity from the structural parameters in the model using a gradient descent algorithm. This procedure relies on the concavity of the firm's profit function, which is established in Appendix A.4.

In a model with homogeneous β, there is little scope for firm rents to vary. As shown in Property A.6, homogeneous effects implies that all firms will capture the same rents as a share of profit, R^e_j/Π_j, provided that they share the same returns to scale and that labor is perfectly substitutable.[14] Thus, for a simplified production technology (common α_j, $\rho_j = 1$), all firms receive the same fraction of profits as rents, regardless of how their characteristics might differ.

When β is heterogeneous, a firm's rents depend on its workforce composition and therefore on the full set of firm characteristics that influence sorting. Moreover, because the firm's market power is endogenous to wages, the relationship between firm productivity T_j and rents R^e_j becomes ambiguous. On the one hand, a more productive firm can capture higher profits. On the other hand, it can raise wages and attract workers with more wage-elastic labor supply, which erodes firm market power and makes labor more costly. Therefore, while productivity can raise a firm's profit, it may also reduce the share of that profit that constitutes rents. In the limit, as $\boldsymbol{\varepsilon}_j(X) \to \infty$ for all X, a firm's profit converges to that of a price-taker, with $R^e_j \to 0$.

In my analysis, it is difficult to empirically distinguish between firm rents measured before and after non-wage amenities are determined. This task would require detailed information (or assumptions) on how firms choose and pay for amenities, which is not available in the administrative data. Thus, for identification, I assume that firms cannot adjust amenities over the estimation window. This assumption does not restrict how amenities relate to firm productivity or technology, nor does it restrict how firms may initially choose amenities to maximize profit.

Pass-through of TFP Shocks to Wages

Next, I examine the impact of a hypothetical shock to firm productivity on workers' wages, as measured by the elasticity $\frac{\partial \log W_j(X)}{\partial \log T_j}$. This object is commonly used to learn about rent sharing, as it reflects how an increase in a firm's value added would be passed on to workers in the form of higher pay.[15] In Property A.7 of Appendix A, I show that $\left[\frac{\partial \log W_j(X)}{\partial \log T_j}\right]_X = (I - \mathbf{K}_j)^{-1}\mathbf{1}$, where:

$$\mathbf{K}_j[X, X'] = \boldsymbol{\varepsilon}_j(\chi', \varphi') \Bigg[(1 - \alpha_j - \rho_j) \left(\frac{\theta_{j\chi}(L^{\text{eff}}_j(\chi))^{\rho_j}}{\sum_{\chi''} \theta_{j\chi''}(L^{\text{eff}}_j(\chi''))^{\rho_j}} \right) \left(\frac{\varphi L_j(\chi, \varphi)}{\int \varphi'' L_j(\chi, \varphi'') d\varphi''} \right) - \mathbb{1}\{\chi = \chi'\}(1 - \rho_j) \frac{\varphi L_j(\chi, \varphi)}{\int \varphi'' L_j(\chi, \varphi'') d\varphi''} \Bigg] + \mathbb{1}\{X = X'\} \frac{\frac{\partial^2 \log L_j(X)}{\partial [\log W_j(X)]^2}}{\boldsymbol{\varepsilon}_j(X)[1 + \boldsymbol{\varepsilon}_j(X)]}. \quad (16)$$

In a model where β is homogeneous, the pass-through of productivity shocks to wages can only vary due to differences in firm technology. Indeed, as shown in Property A.7, homogeneous effects implies that $\frac{\partial \log W_j(X)}{\partial \log T_j}$ is constant whenever firms exhibit the same returns to scale and labor is perfectly substitutable. In other words, all workers, regardless of their firm or skills, would experience the same proportional wage change in response to a firm productivity shock.

[14] As the firm approaches constant returns to scale, $\alpha_j \to 0$, the profit that it could earn as a price-taker tends to zero. Therefore, any profit that the firm manages to capture would come in the form of rents: $\lim_{\alpha_j \to 0} \Pi_j = R^e_j$.

[15] For example, Card et al. (2018) explicitly interprets the pass-through $\frac{\partial \log W_j(X)}{\partial \log T_j}$ as a "rent-sharing elasticity".

When β is heterogeneous, the pass-through $\frac{\partial \log W_j(X)}{\partial \log T_j}$ varies systematically across firms and skill groups because employers face different labor supply elasticities $\boldsymbol{\varepsilon}_j(X)$. Moreover, given how workers sort on individual heterogeneity, the wage markdown can amplify the impact of firm productivity on earnings. To see why, consider that when a productivity shock drives up wages at a firm, it shifts the firm's workforce composition toward workers with more elastic labor supply. This reduces the firm's market power and drives wages up further. As a result, wage changes at a firm can be self-reinforcing.[16] Note that, in principle, the pass-through may be greater than one, although my empirical results show that, in practice, it is well below one.

Welfare and Allocative Inefficiency

I consider a measure of social welfare that incorporates firm profits. Specifically, I assume that profits are redistributed to workers in direct proportion to their wages. This approach ensures that the redistribution does not distort workers' labor supply decisions. Welfare is defined as:

$$\mathcal{W} = \mathrm{E}\left(\max_j \left\{u_{ij}\left(W_j(X_i), a_j(X_i)\right)\right\}\right) + \log\left(\sum_{j=1}^{J} \Pi_j\right). \tag{17}$$

To measure the welfare cost of market power, I compare welfare in the monopsonistic economy to what would be achieved by a social planner who optimally assigns workers to firms, solving: $\mathcal{W}^* = \max_{\{j(i)\}_i} \mathcal{W}$. I assume that the planner has full knowledge of worker and firm fundamentals, but firms continue to set wages based on their limited information of (β_i, ϵ_{ij}). In Property A.8 of Appendix A, I prove that the planner's allocation coincides with the competitive (Walrasian) allocation, where firms act as price-takers facing perfectly elastic labor supply curves. I also show how to solve for welfare under the optimal allocation from the model's structural parameters, allowing me to evaluate the welfare cost of imperfect competition in the economy.

In an economy with constant wage markdowns, labor market power distorts only the level of wages, but not the relative wages across firms. As a result, there is no misallocation of labor across firms because all workers sort into the same firms they would choose absent monopsony power. Even with a nonemployment margin, a constant markdown may generate underemployment, but no cross-firm misallocation. If wage markdowns vary across firms, however, there is a potential for misallocation of labor. This allocative inefficiency occurs, for example, when

[16] Another way to visualize this effect is to analyze the wage elasticity of the markdown $\frac{\boldsymbol{\varepsilon}_j(X)}{1+\boldsymbol{\varepsilon}_j(X)}$, which equals:

$$\frac{\partial \log \frac{\boldsymbol{\varepsilon}_j(X)}{1+\boldsymbol{\varepsilon}_j(X)}}{\partial \log W_j(X)} = \frac{\frac{\partial^2 \log L_j(X)}{\partial [\log W_j(X)]^2}}{\boldsymbol{\varepsilon}_j(X)[1+\boldsymbol{\varepsilon}_j(X)]}, \quad \text{where} \quad \frac{\partial^2 \log L_j(X)}{\partial [\log W_j(X)]^2} \geq 0.$$

This elasticity is positive, implying that wage changes at a firm are amplified through the markdown. In principle, this can generate multiple solutions to (11), potentially yielding two local optima—one with lower wages and employment but larger markdowns, and another with higher wages and employment but smaller markdowns. This phenomenon is discussed by Manning (2010) in the context of spatial agglomeration. In my setting, I show that multiplicity does not arise as long as $\mathrm{Var}(\beta \mid X)$ is bounded relative to $1-\alpha_j$. I also show that, even if multiple solutions to (11) exist, there is almost surely a unique global profit maximum (Appendix C.1, Lemma 2).

a worker who is most productive at one firm j earns a higher wage at another firm k because firm j has a larger markdown than firm k. These discrepancies can alter workers' rankings over firms and distort their labor supply decisions. Consequently, heterogeneous markdowns introduce a welfare cost of labor market power that is not present in the case of constant markdowns.

III. Data and Empirical Context

I now describe the data sources and institutional context for the empirical analysis, as well as the main variables and sample selection criteria. Appendix D.1 provides additional discussion.

III.A. Data Sources

I use a matched employer-employee panel dataset from Norway, containing information about workers and firms over the period 1995-2018. The data is constructed by linking annual business tax records to a national employment register from the Norwegian Labour and Welfare Administration. It encompasses all workers who are not self-employed, and it covers all firms in the country, both public and private. Additional details on worker demographics and financial assets come from individual tax filings and social security registers. All administrative data are maintained by Statistics Norway and are subject to comprehensive quality control measures.[17]

At the worker level, the data report annual earnings, scheduled hours, and job spell duration. Earnings include a worker's salary, bonuses, overtime, vacation, and severance pay, as well as other employer payments.[18] Monetary variables are adjusted for inflation and expressed in 2014 U.S. dollars. The data also reports worker education, experience, and specialization, such as college major and vocational training. I use these variables to define a worker's skill type χ. In the baseline specification, χ is the interaction of education and experience, with education divided into three categories (no high school, high school but no college, and college), and experience into two categories ($<$10 years in the workforce and $\geq$10 years in the workforce). This classification yields six skill types. Appendix D also considers alternate definitions of χ. Note that the empirical analysis will allow both the observed skill types χ and unobserved skill levels φ to evolve over time during the estimation window. Lastly, the data contains rich demographic information on workers and their families, which I use for cross-sectional comparisons.

Workers in the data are linked to establishments, which are subunits (or branches) of firms. I measure employment at the establishment-level, and I allow a firm to treat labor at different establishments as imperfectly substitutable inputs in production.[19] This approach allows for

[17] See Fagereng et al. (2021) for discussion. Bruun et al. (2021) also give an independent quality evaluation.

[18] In-kind benefits are also included in earnings; meanwhile, parental leave and sickness benefits are excluded.

[19] Specifically, if the firm j operates multiple establishments $k \in \mathcal{K}_j$, then I extend the production function so that $Y_j = T_j[\sum_{k\in\mathcal{K}_j}\sum_{\chi\in\mathcal{X}}\theta_{jk\chi}(L^{\text{eff}}_{jk}(\chi))^{\rho_j}]^{(1-\alpha_j)/\rho_j}$, where $L^{\text{eff}}_{jk}(\chi) = \int \varphi D_{jk}(\chi,\varphi)d\varphi$ is the efficiency units of labor of skill type χ demanded by firm j for establishment k. This extension does not meaningfully affect the theoretical results or identification analysis. Thus, to ease notation, the ongoing discussion will continue to

richer patterns of sorting and wage determination. For example, if an oil and gas firm operates a processing plant in rural Norway and a corporate office in Oslo, then these two branches are treated as distinct entities, both by workers and by the managing firm. I assign establishments to local labor markets based on industry and location. The Central Register of Establishments and Enterprises classifies them into 10 broad industries and 46 commuting zones, as defined in Bhuller (2009). I define a market as the intersection of an industry and a commuting zone.

All accounting data is measured at the firm level and is collected from corporate income statements and balance sheets. A firm's value added is defined as the difference between revenue and non-labor expenses, which include operating costs, spending on intermediate inputs, and capital depreciation. The firm's profit is defined as revenue minus total costs, where total costs account for wages paid to workers. All variables are measured prior to interest and taxes.

III.B. Institutional Context

Norway is a country with low unemployment and high labor force participation. Over the period I study, unemployment averaged below 4%, and employment among adults aged 25 to 55 was about 84%. Female employment is high by international standards: women make up 47% of the workforce, although part-time work remains more common among women, with 28% of employed women working part-time compared with 12% of men. Wage inequality is also low relative to other countries. In the private sector, the 90/10 wage ratio rose from 2.4 in 2000 to 3.1 in 2018 (Nilsen, 2020), still well below the U.S. ratio of roughly 5. At the same time, the raw gender wage gap remained stable, with women earning about 15% less than men on average.

Wage setting in Norway combines centralized coordination with substantial local discretion. Union membership varies sharply by sector, from about 20% in trade and services to 80% in the public sector. For workers covered by collective bargaining agreements, wages are set through a two-tier system: sectoral bargaining establishes wage floors, while local adjustments are made at the firm level. In the sample, less than 4% of workers are constrained to their wage floor. Additionally, Bhuller et al. (2022) show that wage changes above negotiated floors vary greatly at the worker-firm level and respond to firm productivity. These features suggest there is meaningful scope for firms to set wages. In Appendix Figure A.6, I show that the estimates remain largely unchanged when restricting the sample to sectors with low union membership.

III.C. Sample Construction

I restrict the estimation sample to full-time workers, defined as those working at least 35 hours per week at a given establishment. Employment is defined as the number of full-time workers. To calculate workers' earnings, I restrict the sample to individuals aged 25 to 55 who work at

focus on a case where each firm operates one establishment ($|\mathcal{K}_j| = 1$). As a robustness check in Appendix Figure A.5, I assess the sensitivity of the parameter estimates to restricting the sample to single-establishment firms.

the same establishment for a full year and who do not hold a higher-paying job elsewhere. Following Song et al. (2018), I also impose a minimum annual earnings threshold, set to $15,000.

My instrumental variables design relies on utilizing a balanced panel of firms. Therefore, for this part of the analysis, I restrict the estimation sample to firms that remain operational for nine consecutive years. During this period, the firm must report positive revenue and maintain at least five full-time employees. I also require the firm to continue operating the same set of establishments, with each establishment remaining in the same industry and commuting zone.

Table 1 reports descriptive statistics for the main estimation sample used in the firm panel. The sample covers over 1.3 million unique full-time workers and more than 31,000 unique establishments. All six skill types are well represented, with nearly 30% of workers holding a college degree. Panel C documents sizable dispersion in firm revenues, value added, and profits. The distributions, plotted in Appendix Figure A.1, closely match those in the full, unrestricted sample, suggesting that the firm panel remains representative of the broader firm population.

Table 1: Summary Statistics—Main Estimation Sample

Panel A. Population Counts	Unique Identifiers	Unique Observations
Number of Full-time Workers	1,325,863	8,303,075
Number of Establishments	31,252	475,614
Number of Firms	23,066	274,687
Number of Local Markets	457	10,563
Panel B. Worker Skill Types	Share of Workforce	Mean Log Earnings
Lower Secondary, <10 Years Exp.	6.82%	10.92
Upper Secondary, <10 Years Exp.	19.03%	11.12
College Graduate, <10 Years Exp.	14.76%	11.33
Lower Secondary, ≥10 Years Exp.	21.37%	11.10
Upper Secondary, ≥10 Years Exp.	23.74%	11.30
College Graduate, ≥10 Years Exp.	14.28%	11.57
Panel C. Firm Accounting Variables	Mean	Std. Dev.
Log Revenues	15.82	1.29
Log Total Costs	15.78	1.27
Log Value Added	14.64	1.17
Log Profit	12.82	1.60

Notes. This table presents descriptive statistics for the baseline estimation sample. All monetary quantities are expressed in log 2014 US dollars and are adjusted for inflation.

IV. Identification Analysis

In this section, I demonstrate how to bring the model to data. In particular, I show how to recover key economic quantities of interest, such as the distribution of wage markdowns and rents, and measures of allocative inefficiency, from the matched employer-employee dataset. Although many of these quantities can be recovered without full knowledge of the model's structural parameters, others—particularly those based on counterfactual outcomes—do rely on the full set of parameters. For this reason, my goal is to achieve full point identification of the model.

My identification analysis follows two steps. First, I show how to recover the firm-specific labor supply elasticities $\boldsymbol{\varepsilon}_j(X)$ using panel data, along with instrumental variables that shift firms' labor demand. Second, with estimates of these elasticities in hand, I prove identification of firm production parameters $(T_j, \{\theta_{jX}\}_X, \rho_j, \alpha_j)$, amenities $a_j(X)$, and the joint density $f_{\beta,X}$ of worker skills and wage-amenity trade-offs. This approach avoids imposing a log-additive wage structure, and thus does not require estimating an AKM model, restricting attention to connected worker-firm sets, or assuming exogenous mobility. Instead, identification leverages instruments to recover labor supply elasticities within and across firms, and then draws on equilibrium relationships between wages, employment, and value added to infer structural parameters.

IV.A. Identification of Firm-Specific Labor Supply Elasticities

IV.A.1. Ideal Experiment

I begin by laying out the comparative statics that identify the labor supply elasticities. Consider a hypothetical, firm-specific labor demand shock that shifts a firm's TFP in logs from $\log T_j$ to $\log T_j + \delta_j$. Assume all other aspects of the economy remain unchanged. For any skill group X, let $\big(W_j(X), L_j(X)\big)$ and $\big(W_j'(X), L_j'(X)\big)$ denote the firm's wage and employment before and after the shock, respectively. As δ_j tends to zero, these outcomes can be used to recover $\boldsymbol{\varepsilon}_j(X)$.

$$\lim_{\delta_j \to 0} \Big[\big(\log L_j'(X) - \log L_j(X)\big) - \boldsymbol{\varepsilon}_j(X) \times \big(\log W_j'(X) - \log W_j(X)\big) \Big] = 0. \tag{18}$$

This condition follows from two key properties of the model. First, under the equilibrium characterization, firms view each other to be strategically small. Therefore, they do not internalize wage changes that are confined to a single rival firm in the economy. This delivers an exclusion restriction, whereby an isolated shock to T_j impacts firm j's labor supply $L_j(X)$ only through changes to its own wage $W_j(X)$. The second property of the model, established in Lemma 2 of Appendix C.1, is that the solution to the firm's first order condition is locally stable, which means that iterating on the equilibrium wage condition (11) converges to a fixed point for nearby initial values. This property matters for the comparative statics because it ensures that a small, isolated labor demand shock does not create large jumps in a firm's wages or employment.

Based on equation (18), the ideal experiment to recover the elasticities $\boldsymbol{\varepsilon}_j(X)$ would be to introduce an isolated, infinitesimal TFP shock to a single firm, measure the resulting change in that firm's wages and employment, and repeat this exercise separately for every firm in the economy. In practice, however, this experiment is not feasible, since I cannot independently manipulate each firm's labor demand while holding the rest of the economy fixed. Empirical labor demand shifters, such as export-demand shocks (Garin & Silvério, 2024), value-added shocks (Lamadon et al., 2022), and procurement demand shocks (Kroft et al., 2025) typically affect many firms at once, even if exposure is idiosyncratic across firms. Therefore, these shifters could generate non-marginal changes in the wage distribution, altering how workers sort across firms and creating spillovers onto firms that are not directly exposed, which violate the exclusion restriction underlying equation (18). Moreover, even if it were possible to replicate the comparative statics in (18), doing so would require measuring each firm's wages and employment separately for each skill group $X = (\chi, \varphi)$. This task poses an additional challenge, since worker skill levels φ are not directly observed by the researcher. In the remainder of this subsection, I show how to overcome these challenges using panel data and instrumental variables.

IV.A.2. Econometric Model

I now describe the empirical framework and assumptions underlying the identification results. Throughout this discussion, I adopt the logit error structure, $F_\epsilon(\epsilon) = \exp(-\exp(-\epsilon))$, where firm-specific labor supply curves are given by equation (7). To ease notation, I use lowercase letters to denote variables that are in logs: $w_j(X) = \log W_j(X)$, $\ell_j(X) = \log L_j(X)$, and $y_j = \log Y_j$.

Description of Empirical Framework for a Generic Instrument

Consider the evolution of equilibrium outcomes over two time periods: τ_0 and τ_1. After period τ_0, an event occurs in which firms are differentially exposed to labor demand shocks, which enter the model as shifts in the distribution of T_j. Wages and labor then adjust to the new equilibrium by period τ_1. Let $w_{j\tau_0}(X)$ and $\ell_{j\tau_0}(X)$ denote log wages and employment for skill group X at firm j in period τ_0, and let $w_{j\tau_1}(X)$ and $\ell_{j\tau_1}(X)$ be the corresponding outcomes in period τ_1. For expositional clarity, I focus on a case with binary instruments, defining Z_j as an indicator of whether firm j experiences a TFP shock. Firms are *treated* if $Z_j = 1$ and *untreated* if $Z_j = 0$.[20]

Figure 2 illustrates the evolution of wages and labor for a given firm j, delineating which aspects of the model remain fixed and which evolve over time. Point A shows the firm's initial outcome, while B and B' depict the firm's post-period potential outcomes when untreated and treated, respectively. Importantly, between τ_0 and τ_1, wages and labor adjust at all firms, including untreated firms. These adjustments occur because, unlike in the ideal experiment, TFP shocks are not confined to only one firm. Rather, they take place at many firms in the economy. As a result, wage changes among treated firms affect market aggregates by shifting workers' wage indices, $I(\beta, X)$, which in turn reshape the labor supply curves faced by every firm, even those that are untreated. Figure 2 depicts this spillover as a rightward shift in firm j's labor supply curve, though depending on the nature of the shocks, this curve can shift in any direction.

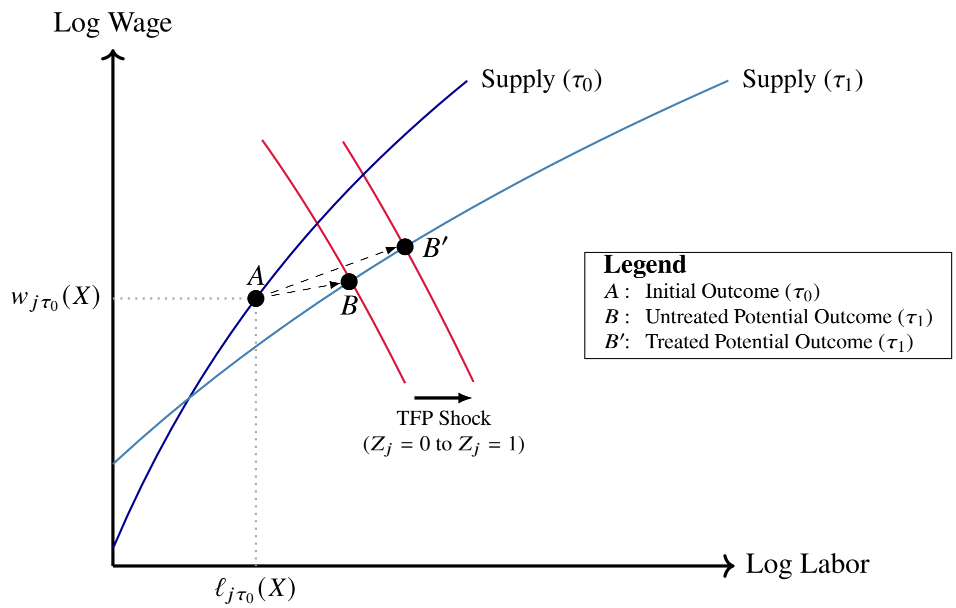
Contributing to these dynamics, I allow worker skills and preferences to evolve over time. These shifts can arise exogenously or in direct response to firms' labor demand shocks; I do not restrict whether or how they occur. In particular, I permit the skill distribution F_X and the

[20]While I focus on a binary instrument for exposition, the subsequent identification results extend to multivalued or continuous Z_j, where identification comes from comparing firms exposed to shocks at different intensities.

conditional distributions of wage-amenity trade-offs $F_{\beta|X}$ to change from τ_0 and τ_1. I do not restrict the evolution of these distributions, placing no constraint on how $(F_{X|\tau_0}, \{F_{\beta|X,\tau_0}\}_X)$ relates to $(F_{X|\tau_1}, \{F_{\beta|X,\tau_1}\}_X)$. While I also allow workers' taste shocks to evolve, I assume that the cross-sectional distribution retains a logit structure in each period, $F_\epsilon(\epsilon_{ij\tau}) = \exp(-\exp(-\epsilon_{ij\tau}))$. To achieve this, I assume that $\epsilon_{ij\tau}$ is a Markov process with independent innovations across i.

Given the evolution of workers' wage indices, skills, and preferences over time, firms face different labor supply curves in periods τ_0 and τ_1. To measure the elasticity of labor supply to firm j, it is therefore necessary to compare the firm's potential outcomes within the same time period. In period τ_0, the firm's treated and untreated potential outcomes are the same, because no TFP shocks have yet occurred. However, in period τ_1, the firm's potential outcomes differ, as illustrated by points B and B' in Figure 2. This difference reflects the comparative statics from the ideal experiment, capturing how a firm j's wages and employment respond to a TFP shock that is confined only to that firm. If it were possible to compare both potential outcomes for a firm in period τ_1, then one could compute the firm's labor supply elasticity in that period.

Figure 2: Illustration of Potential Outcomes for a Firm



Notes. This figure illustrates how the potential outcomes of a firm j respond to an event that shifts TFP at many firms. In each period $\tau \in \{\tau_0, \tau_1\}$, workers' wage indices, skills, and preferences are given by $\{I_\tau(\beta, X)\}_{\beta,X}$ and $\{F_{\beta,X|\tau}\}_X$. All firm characteristics $(\{\theta_{j\mathcal{X}}\}_\mathcal{X}, \rho_j, \alpha_j, \{a_j(X)\}_X)$ remain fixed; only T_j changes for treated firms.

Assumptions on Treatment Assignment

Throughout my identification analysis, I impose an assumption about how the treatment status of a firm, denoted by Z_j, depends on that firm's characteristics. Specifically, I assume that Z_j is statistically independent of the firm's technology (ρ_j, α_j), the relative efficiencies $\{\theta_{j\mathcal{X}}\}_\mathcal{X}$ of each skill type, and the differences in amenities $\{a_j(X) - a_j(X')\}_{X,X'}$ between skill groups.

Assumption I (*Instrument Independence*). $Z_j \perp \left(\rho_j, \alpha_j, \{\theta_{j\mathcal{X}}\}_\mathcal{X}, \{a_j(X) - a_j(X')\}_{X,X'}\right)$.

Note that this assumption does not require Z_j to be fully idiosyncratic. In particular, I allow Z_j to depend on a firm's productivity through the baseline TFP. Additionally, while Z_j must be independent of the amenity differences within a firm, I do not restrict how it relates to firms' average amenities. Therefore, exposure to the shock Z_j need not be random: because a firm's treatment status can depend on its own productivity and amenities, treated and untreated firms may exhibit systematically different wages, employment, and output. Moreover, Assumption I can be relaxed to allow for independence conditional on observables. In estimation, I condition on a firm's local labor market, so the assumption need only hold within industry-by-region cell.

IV.A.3. Identification Results

I now establish the identification of firm-specific labor supply elasticities. As described above, these elasticities can vary over time due to changes in workers' wage indices, skills, and preferences. In any period τ, I can write $\boldsymbol{\varepsilon}_{j\tau}(X)$ as an (X,τ)-specific function of the log wage $w_{j\tau}(X)$:

$$\boldsymbol{\varepsilon}_{j\tau}(X) = \int \beta \times \frac{\left[\exp\left(\beta w_{j\tau}(X)\right)/I_\tau(\beta, X)\right] f_{\beta|X,\tau}(\beta|X,\tau)}{\int \left[\exp\left(\beta' w_{j\tau}(X)\right)/I_\tau(\beta', X)\right] f_{\beta|X,\tau}(\beta'|X,\tau)d\beta'} d\beta. \tag{19}$$

My analysis proceeds in two parts. First, I prove identification in the case where workers' skills X are observed. This allows me to focus on the main identification challenge: recovering elasticities $\boldsymbol{\varepsilon}_{j\tau}(X)$ from labor demand shocks to firms, while accounting for how these shocks affect market aggregates. Second, I show how my results extend to the case where workers' skills are not fully observed, in particular where the skill level φ is unknown to the researcher. Here, I establish conditions such that φ is identified from matched worker-firm panel data, and I derive an estimand for firms' labor supply elasticities that may be applied even with unobserved skills.

Identification with Observed Skills

When workers' skills X are observed in the data, the elasticities $\boldsymbol{\varepsilon}_{j\tau_1}(X)$ are identified through a difference-in-differences (DiD) comparison of firms' wage and employment outcomes. To motivate this design, it is important to recognize that one could not recover these elasticities from either a purely cross-sectional comparison of treated and untreated firms or from a comparison of outcomes within the same firm over time. A cross-sectional comparison fails because treated and untreated firms are likely to face systematically different labor supply curves, for example if Z_j correlates with $T_{j\tau_0}$ or $\mathrm{E}(a_j(X)|j)$. A within-firm panel regression fails because Z_j creates spillovers that can affect workers' wage indices, skills, and preferences. These spillovers imply that the same firm faces different labor supply curves in the pre- and post-periods.

I propose to recover the elasticity $\boldsymbol{\varepsilon}_{j\tau_1}(X)$ from a ratio of DiD estimands that condition on a firm's pre-period wage. Let $\left(\ell_{j\tau_0}(X), w_{j\tau_0}(X)\right)$ and $\left(\ell_{j\tau_1}(X), w_{j\tau_1}(X)\right)$ be the log employment and log wage for skill group X at firm j in periods τ_0 and τ_1, respectively. I define the estimand:

$$\mathrm{DiD}_{\tau_0,\tau_1}(w|X) = \frac{\mathrm{E}\left[\ell_{j\tau_1}(X)-\ell_{j\tau_0}(X)\middle|Z_j{=}1, w_{j\tau_0}(X){=}w\right] - \mathrm{E}\left[\ell_{j\tau_1}(X)-\ell_{j\tau_0}(X)\middle|Z_j{=}0, w_{j\tau_0}(X){=}w\right]}{\mathrm{E}\left[w_{j\tau_1}(X)-w_{j\tau_0}(X)\middle|Z_j{=}1, w_{j\tau_0}(X){=}w\right] - \mathrm{E}\left[w_{j\tau_1}(X)-w_{j\tau_0}(X)\middle|Z_j{=}0, w_{j\tau_0}(X){=}w\right]}. \tag{20}$$

The numerator is a DiD estimand of changes in log labor, while the denominator is a DiD estimand of changes in log wages. Figure 3 illustrates how the observed movements in wages and labor among treated and untreated firms are used to form $\mathrm{DiD}_{\tau_0,\tau_1}(w|X)$. The estimand is read as the horizontal red segment, the labor DiD, divided by the vertical red segment, the wage DiD.

To show that this estimand recovers the labor supply elasticity, I first prove that a common trends assumption holds. Specifically, I show that, conditional on a firm's initial wage $w_{j\tau_0}(X)$, the change in untreated potential outcomes, $w_{j\tau_1,0}(X)-w_{j\tau_0,0}(X)$ and $\ell_{j\tau_1,0}(X)-\ell_{j\tau_0,0}(X)$, is a deterministic function of the parameters Γ_j, where $\Gamma_j = \left(\rho_j, \alpha_j, \{\theta_{jX}\}_X, \{a_j(X) - a_j(X')\}_{X,X'}\right)'$. Assumption I guarantees that a firm's treatment status Z_j does not depend on Γ_j. Therefore, treated and untreated firms with the same initial wages must, on average, see the same change in their untreated potential outcomes. By this reasoning, I establish the following proposition.

Proposition 1. Suppose Assumption I holds. Then, given a firm j's pre-period wage $w_{j\tau_0}(X)$, the change in untreated potential outcomes is mean independent of the firm's treatment status:

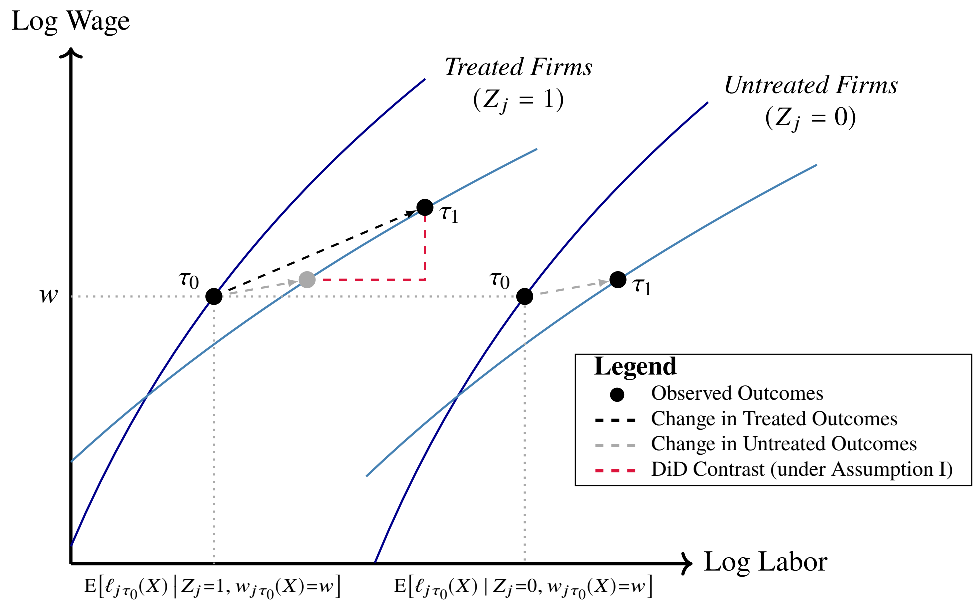
$$\mathrm{E}\left[\ell_{j\tau_1,0}(X) - \ell_{j\tau_0,0}(X)\middle|Z_j=1, w_{j\tau_0}(X)=w\right] = \mathrm{E}\left[\ell_{j\tau_1,0}(X) - \ell_{j\tau_0,0}(X)\middle|Z_j=0, w_{j\tau_0}(X)=w\right] .$$

$$\mathrm{E}\left[w_{j\tau_1,0}(X) - w_{j\tau_0,0}(X)\middle|Z_j=1, w_{j\tau_0}(X)=w\right] = \mathrm{E}\left[w_{j\tau_1,0}(X) - w_{j\tau_0,0}(X)\middle|Z_j=0, w_{j\tau_0}(X)=w\right] .$$

Proposition 1 is central to my analysis. It implies that, by using a DiD research design, I can reproduce the comparative statics from the ideal experiment. In particular, I can recover the mean difference between treated and untreated potential outcomes in the post-period for any treated firm offering initial wage w. Using this result, together with the fact that the elasticity $\boldsymbol{\varepsilon}_{j\tau_1}(X)$ is itself an (X,τ_1)-specific function of the firm's wage (see equation (19)), I establish that $\boldsymbol{\varepsilon}_{j\tau_1}(X)$ is identified from the estimand $\mathrm{DiD}_{\tau_0,\tau_1}(w_{j\tau_0}(X)|X)$ for any firm j and skill group X. Thus, by evaluating $\mathrm{DiD}_{\tau_0,\tau_1}(w|X)$ over the entire wage distribution, I can recover the labor supply elasticity faced by every firm in the economy. In practice, as I discuss in Section V, this task relies on an overlap condition, which requires the presence of both treated and untreated firms throughout the wage distribution: $\mathrm{P}\left(Z_j=1|w_{j\tau_0}(X)=w\right) \in (0,1)$ for all $w \in \mathrm{supp}(w_{j\tau_0}(X))$.

Before proceeding, it is useful to review how this identification result generalizes previous pass-through approaches to measuring firm-specific labor supply elasticities (e.g., Lamadon et al., 2022; Kroft et al., 2025). First, whereas previous work typically assumes that elasticities are constant, or constant conditional on observables, I allow them to vary within and across firms and to depend endogenously on the firm's own wages. The object of identification is therefore not a single elasticity $\boldsymbol{\varepsilon}$, but the distribution of firm- and skill-specific elasticities $\boldsymbol{\varepsilon}_{j\tau}(X)$, which can itself evolve as the wage distribution changes. Second, whereas previous studies consider isolated firm-level shocks that leave the rest of the economy unaffected, I allow shocks to affect many firms simultaneously, generating spillovers to workers' wage indices, skills, and preferences. Such spillovers violate the stable unit treatment value assumption and thus complicate identification. I overcome this challenge by showing that under the equilibrium framework, the spillovers are absorbed into a time trend within a well-specified DiD estimand. Third, unlike previous work, I allow exposure to the shock to depend on unobserved firm characteristics through baseline TFP and average amenities. Hence, Z_j need not be randomly assigned.

Figure 3: Illustration of DiD Comparison of Firm-Level Outcomes



Notes. This figure shows how average changes in log wages and employment among treated and untreated firms define $\mathrm{DiD}_{\tau_0,\tau_1}(w \mid X)$ for a given log wage w and skill group X. The dark blue curves depict pre-period labor supply; the light blue curves depict post-period labor supply. Under Assumption I, untreated potential outcome changes, $w_{j\tau_1,0}(X) - w_{j\tau_0,0}(X)$ and $\ell_{j\tau_1,0}(X) - \ell_{j\tau_0,0}(X)$, are mean independent of Z conditional on $w_{j\tau_0}(X) = w$.

Identification with Unobserved Skills

I now extend my results to allow workers' skills to have unobserved components. In particular, I assume that the researcher cannot directly measure a worker's skill level φ and thus cannot directly distinguish between a firm's labor inputs. To recover the labor supply elasticities in this case, I derive conditions under which φ can be inferred from matched worker-firm panel data. These conditions imply that φ is identified, allowing me to apply the same IV design as above.

Going forward, I make an assumption about the relationship between a worker's skill level φ and the other parameters in the model. This assumption consists of three main restrictions.

Assumption II (*Restrictions on Unobserved Skill Levels*)**.**

II.1. $\varphi \perp \beta | \chi, \tau$.
II.2. $a_j(\chi, \varphi) = a_{j\chi} + a_{\chi\varphi}$.
II.3. $\mathrm{E}(\log \varphi | \chi, \tau) = \mathrm{E}(\log \varphi | \chi)$.

Assumption II.1 states that, conditional on a worker's skill type χ and time period τ, the trade-off between wages and amenities, as summarized by β, does not depend on the skill level φ. This assumption does not impose that β is fully independent of workers' skills, as I still allow β to depend on the skill type χ. Additionally, it does not rule out the possibility that β varies along unobserved skill dimensions; instead, it only precludes systematic variation based on φ.

Assumption II.2 asserts that a firm's non-wage amenities can be decomposed into a firm-by-skill-type component, $a_{j\chi}$, and a skill-level component, $a_{\chi\varphi}$. This separability implies that amenities do not vary across firms based on unobserved dimensions of worker skills. To better interpret the two amenity components, I normalize $\mathrm{E}(a_{\chi\varphi}|\chi) = 0$ without loss of generality. This normalization allows me to interpret $a_{j\chi}$ as the average amenity at firm j for skill type χ.

Assumption II.3 requires that the average log skill level $\mathrm{E}(\log\varphi|\chi,\tau)$ within a given skill type χ is invariant over time τ. This assumption does not restrict how the variance or higher-order moments of $\log\varphi$ evolve, nor does it restrict the evolution of observable skill types, $F_{\chi|\tau}$. Going forward, I normalize $\mathrm{E}(\log\varphi \mid \chi) = 0$ for each χ. This normalization is without loss of generality, since the relative efficiencies $\theta_{j\chi}$ can always be redefined to absorb $\mathrm{E}(\log\varphi \mid \chi)$.[21]

To establish identification, I first examine the implications of Assumption II for workers' labor supply decisions and wage determination. The key implication of this assumption is that the labor supply elasticities faced by a firm do not vary based on unobserved skill dimensions: $\boldsymbol{\varepsilon}_{j\tau}(\chi,\varphi) = \boldsymbol{\varepsilon}_{j\tau}(\chi)$. Consequently, wage markdowns at a firm would differ across skill types χ, but not across skill levels φ within a given χ. This property implies the following result.

Proposition 2. Suppose that Assumption II holds. Then, for any worker at firm j with skills (χ,φ) in period τ, the wage can be represented as $W_{j\tau}(\chi,\varphi) = \varphi\,\psi_{j\chi\tau}$, where $\psi_{j\chi\tau}$ is a firm-skill-time specific term, and φ is a worker-specific component satisfying $\mathrm{E}(\log\varphi|j,\chi,\tau) = 0$.

Proposition 2 provides the necessary structure for computing worker skill levels φ from linked worker-firm panel data. In particular, I can recover φ from cross-sectional wage differences:

$$\log\varphi = \log W_{j\tau}(\chi,\varphi) - \mathrm{E}\left[\log W_{j\tau}(\chi,\varphi) \;\middle|\; j,\chi,\tau\right]. \tag{21}$$

To simplify notation, it will be useful to measure wages in efficiency units. Define the effective wage for workers with skill type χ at firm j in period τ as $W^{\mathrm{eff}}_{j\tau}(\chi) = W_{j\tau}(\chi,\varphi)/\varphi$. Using this notation, I can reinterpret the elasticity of labor supply to a firm in terms of the effective wage:

$$\boldsymbol{\varepsilon}_{j\tau}(\chi) = \int \beta \times \frac{\left[\exp\left(\beta w^{\mathrm{eff}}_{j\tau}(\chi)\right)/I^{\mathrm{eff}}_{\tau}(\beta,\chi)\right] f_{\beta|\chi,\tau}(\beta|\chi,\tau)}{\int\left[\exp\left(\beta' w^{\mathrm{eff}}_{j\tau}(\chi)\right)/I^{\mathrm{eff}}_{\tau}(\beta',\chi)\right] f_{\beta|\chi,\tau}(\beta'|\chi,\tau)d\beta'} d\beta, \tag{22}$$

where $I^{\mathrm{eff}}_{\tau}(\beta,\chi) = \sum_{k=1}^{J}\exp(\beta w^{\mathrm{eff}}_{k\tau}(\chi) + a_{k\chi})$ is the effective wage index for workers with preference parameter β and skill type χ at time τ. The elasticity $\boldsymbol{\varepsilon}_{j\tau_1}(\chi)$ is identified from a DiD

[21] Specifically, if $\mathrm{E}(\log\varphi|\chi) \neq 0$ for some skill type χ, then I can redefine $\log\theta_{j\chi}$ as $\log\theta_{j\chi} + \rho_j\,\mathrm{E}(\log\varphi|\chi)$.

estimand $\text{DiD}_{\tau_0,\tau_1}(w|\chi)$ evaluated at the effective log wage $w^{\text{eff}}_{j\tau_0}(\chi)$. I define this estimand as:

$$\text{DiD}_{\tau_0,\tau_1}(w|\chi) = \frac{\mathrm{E}\left[\ell_{j\tau_1}(\chi)-\ell_{j\tau_0}(\chi)\middle|Z_j=1, w^{\text{eff}}_{j\tau_0}(\chi)=w\right]-\mathrm{E}\left[\ell_{j\tau_1}(\chi)-\ell_{j\tau_0}(\chi)\middle|Z_j=0, w^{\text{eff}}_{j\tau_0}(\chi)=w\right]}{\mathrm{E}\left[w^{\text{eff}}_{j\tau_1}(\chi)-w^{\text{eff}}_{j\tau_0}(\chi)\middle|Z_j=1, w^{\text{eff}}_{j\tau_0}(\chi)=w\right]-\mathrm{E}\left[w^{\text{eff}}_{j\tau_1}(\chi)-w^{\text{eff}}_{j\tau_0}(\chi)\middle|Z_j=0, w^{\text{eff}}_{j\tau_0}(\chi)=w\right]}. \quad (23)$$

IV.A.4. Sources of Instruments

Up to this point, I have not yet specified how to construct the instruments Z_j. In the analysis, I consider two approaches: internal instruments, constructed directly from the matched worker-firm panel using model-implied restrictions, and external instruments, which draw on outside data sources. Both can be used within the framework, but they rely on different assumptions.

Internal Instruments

My main specification uses internal instruments, constructed by specifying a process for firm productivity shocks. Following Guiso et al. (2005) and Lamadon et al. (2022), I assume that firm TFP follows an autoregressive process with a unit root. In each period τ, log TFP equals:

$$t_{j\tau} = \bar{t}_j + \tilde{t}_{j\tau}, \quad \text{where} \quad \tilde{t}_{j\tau} = \tilde{t}_{j\tau-1} + u_{j\tau}. \quad (24)$$

To ensure that the productivity shocks do not lead to large jumps in wages and labor, I assume that $u_{j\tau}$ is small relative to the log wage levels at firm j. I also make the following assumption.

Assumption III (*Identification Assumptions for Internal Instruments*)**.**
III.1 *(Instrument Relevance)***.** $\text{Var}(u_{j\tau}) > 0$.
III.2 *(Instrument Independence)***.** $u_{j\tau} \perp \left(\rho_j, \alpha_j, \{\theta_{j\chi}\}_\chi, \{a_j(X) - a_j(X')\}_{X,X'}\right)$.

Assumption III.1 requires that the productivity shocks exhibit variation across firms, implying that the internal instrument is relevant. Assumption III.2 requires that the internal instrument satisfies Assumption I, which is sufficient to ensure that the common trends assumption holds.

Under this setup, I construct the internal instrument from short-run changes in a firm's log value added, $\Delta y_{j\tau}$, defined within a given calendar year τ. For integers $p, p' \geq 2$, I define the pre-period as $\tau_0 = \tau - p$ and the post-period as $\tau_1 = \tau + p'$, corresponding to p years before and p' years after τ, respectively. Under Assumption III, I derive the following moment condition:

$$\mathrm{E}\left[\Delta y_{j\tau}\left[\left(\ell_{j\tau_1}(\chi)-\ell_{j\tau_0}(\chi)\right)-\text{DiD}_{\tau_0,\tau_1}(w|\chi)\left(w^{\text{eff}}_{j\tau_1}(\chi)-w^{\text{eff}}_{j\tau_0}(\chi)\right)\right] \;\middle|\; w^{\text{eff}}_{j\tau_0}(\chi) = w\right] = 0. \quad (25)$$

This moment condition corresponds to an IV regression of long run differences in log labor on long run differences in log effective wages, instrumented by short run differences in log value added, while controlling for effective wages in the pre-period. In implementing this procedure, I define the treatment status Z_j by splitting firms at the median of the distribution of $\Delta y_{j\tau}$.[22]

[22] This setup can be extended to allow for transitory measurement error in firm value added and employment. This extension requires that the errors are both mutually independent and independent of firm-level parameters.

External Instruments

To complement the analysis using internal instruments, I also use an external instrument based on plausibly exogenous product demand shocks from public procurement auction outcomes. This supplementary analysis draws on data covering the universe of product, construction, and service contracts procured by the Norwegian government from 2003 to 2018. This data comprises about 30,000 contract awards across all industries and regions. The estimation sample is defined using the same dataset and following similar protocols as those used to analyze the internal instrument. Further details about the procurement data are provided in Appendix D.2.

To construct the external instrument, I adopt the research design of Kroft et al. (2025) and de Frahan et al. (2024), where a firm is classified as treated ($Z_j = 1$) in the year τ it wins its first procurement contract. A firm is untreated ($Z_j = 0$) if it wins its first contract in a later year $\tau' > \tau_1$ or if it never wins. Since the instrument is constructed from outside data sources, identification does not rely on specifying the evolution of firm TFP. However, the instrument's main limitation is that the auction outcomes only pertain to a subset of firms and are observed over a more limited time period, from 2003 to 2018. For common trends to hold, I maintain Assumption I, which requires that a firm's auction outcome is independent of the parameters $\big(\rho_j, \alpha_j, \{\theta_{j\chi}\}_\chi, \{a_j(X)-a_j(X')\}_{X,X'}\big)$, although it can depend freely on $T_{j\tau_0}$ and $\mathrm{E}(a_j(X)|j)$.

IV.B. Identification of Technology, Amenities, and Preferences

I now show how to recover the structural parameters characterizing firm productivity, technology, and non-wage amenities, as well as the distribution of worker preferences. These quantities are needed to compute worker and firm rents, and they also allow me to run counterfactuals.

Firm Productivity and Technology

I begin by proving the identification of the production parameters $(T_{j\tau}, \{\theta_{j\chi}\}_\chi, \rho_j, \alpha_j)$ for any arbitrary firm j and any time period τ. Consistent with my previous analysis, I assume that the firm's TFP parameter $T_{j\tau}$ can evolve over time, while the parameters $(\{\theta_{j\chi}\}_\chi, \rho_j, \alpha_j)$ remain fixed. In addition, I normalize the firm-skill-specific efficiency $\theta_{j\chi}$ by setting $\theta_{j\chi^*} = 1$ for some skill type $\chi^* \in \mathcal{X}$, since I can only recover these terms up to scale, relative to the firm's TFP.

Identification of firm technology relies on the ability to recover the labor supply elasticities faced by the firm at two different time periods. This task is feasible in my setting because I have access to a 24-year panel, enabling me to replicate the analysis from Section IV.A over multiple years. For periods τ and τ', I define $\boldsymbol{\varepsilon}_{j\tau}(\chi)$ and $\boldsymbol{\varepsilon}_{j\tau'}(\chi)$ as the labor supply elasticities for skill type χ at firm j. If these elasticities are identified, then I can compute a worker's marginal product from the structural wage equation (11). Specifically, for $\tilde{\tau} \in \{\tau, \tau'\}$, I define:

$$\mathrm{MPL}^{\mathrm{eff}}_{j\tilde{\tau}}(\chi) = \frac{1 + \boldsymbol{\varepsilon}_{j\tilde{\tau}}(\chi)}{\boldsymbol{\varepsilon}_{j\tilde{\tau}}(\chi)} \times W^{\mathrm{eff}}_{j\tilde{\tau}}(\chi) \tag{26}$$

as the effective marginal product for skill type χ at firm j, adjusting for workers' skill levels φ. I can then recover the substitution parameter ρ_j from changes over time in the relative effective marginal products and labor shares of different skill types. This result is formalized below.

Proposition 3. For each firm j, the elasticity of substitution between different skill types is:

$$(1-\rho_j)^{-1} = \frac{\log\left(\frac{L^{\text{eff}}_{j\tau}(\chi)}{L^{\text{eff}}_{j\tau}(\chi')}\right) - \log\left(\frac{L^{\text{eff}}_{j\tau'}(\chi)}{L^{\text{eff}}_{j\tau'}(\chi')}\right)}{\log\left(\frac{\text{MPL}^{\text{eff}}_{j\tau}(\chi)}{\text{MPL}^{\text{eff}}_{j\tau}(\chi')}\right) - \log\left(\frac{\text{MPL}^{\text{eff}}_{j\tau'}(\chi)}{\text{MPL}^{\text{eff}}_{j\tau'}(\chi')}\right)}, \quad \text{for any } \chi, \chi' \in \mathcal{X}.$$

Proposition 3 expresses the elasticity of substitution $(1-\rho_j)^{-1}$ in terms of identified model quantities. This expression reflects how the relative demand for two labor inputs depends on the relative productivity of those inputs, as measured by the elasticity of $L^{\text{eff}}_{j\tau}(\chi)/L^{\text{eff}}_{j\tau}(\chi')$ with respect to $\text{MPL}^{\text{eff}}_{j\tau}(\chi)/\text{MPL}^{\text{eff}}_{j\tau}(\chi')$. In my framework, this relationship may be firm-specific.

Once ρ_j is identified, I show that $\{\theta_{j\chi}\}_\chi$, α_j, and T_j can also be recovered from the firm's effective marginal products, labor shares, and value added. I establish the following result.

Proposition 4. For any firm j, the parameters $\{\theta_{j\chi}\}_\chi$, α_j, and T_j are defined by the equations:

$$\theta_{j\chi} = \exp\left[\log\left(\frac{\text{MPL}^{\text{eff}}_{j\tau}(\chi)}{\text{MPL}^{\text{eff}}_{j\tau}(\chi^*)}\right) + (1-\rho_j)\log\left(\frac{L^{\text{eff}}_{j\tau}(\chi)}{L^{\text{eff}}_{j\tau}(\chi^*)}\right)\right].$$

$$1-\alpha_j = \exp\left[\log \text{MPL}^{\text{eff}}_{j\tau}(\chi) - y_{j\tau} - \log(\theta_{j\chi}) + (1-\rho_j)\log L^{\text{eff}}_{j\tau}(\chi) + \log \sum_{\chi' \in \mathcal{X}} \theta_{j\chi'}\left(L^{\text{eff}}_{j\tau}(\chi')\right)^{\rho_j}\right].$$

$$T_{j\tau} = \exp\left[y_{j\tau} - \frac{1-\alpha_j}{\rho_j}\log \sum_{\chi \in \mathcal{X}} \theta_{j\chi}\left(L^{\text{eff}}_{j\tau}(\chi)\right)^{\rho_j}\right].$$

Note that recovering the substitution parameter ρ_j is the only step that requires knowledge of labor supply elasticities for two periods. If ρ_j is already known, or if one assumes perfect substitutability ($\rho_j = 1$), then $\{\theta_{j\chi}\}_\chi$, α_j, and T_j are identified from a single period of estimates.

Firm Amenities

Next, I prove identification of the parameters $a_{j\chi}$, which correspond to the average amenities at each firm j for any skill type χ. In my analysis, these terms are only identified up to a location normalization, because common shifts do not affect workers' labor supply decisions. Therefore, I normalize the amenities by defining a reference firm j^* at which $a_{j^*\chi} = 0$ for all $\chi \in \mathcal{X}$.

To recover firm amenities, I use a revealed preference approach. This method rests on the idea that—holding wages fixed—firms that offer better amenities to a given skill type attract more workers of that type. Therefore, by comparing employment across firms in a counterfactual where all firms pay the same wages, I can infer the cross-firm differences in amenity values.

To implement this strategy, I use the labor supply elasticity curves $\boldsymbol{\varepsilon}_{j\tau}(\chi, w)$, which are identified for each skill type χ by the estimand $\mathrm{DiD}_{\tau_0,\tau_1}(w|\chi)$. Integrating along these curves allows me to recover the labor supplied to any firm j under any counterfactual wage w, holding all other wages fixed. In doing so, I can then compare the employment at firm j to that of the reference firm j^* under a counterfactual in which both pay the same wages. This comparison allows me to infer the amenity value $a_{j\chi}$. The following proposition formalizes this argument.

Proposition 5. Under Assumptions I and II, the average amenity for skill type χ at firm j is:

$$a_{j\chi} = \ell_{j\tau}(\chi) + \int_{w^{\text{eff}}_{j\tau}(\chi)}^{w^{\text{eff}}_{j^*\tau}(\chi)} \boldsymbol{\varepsilon}_{j\tau}(\chi, w)dw - \ell_{j^*\tau}(\chi).$$

Worker Preferences

I conclude this section by demonstrating the identification of the densities $f_{\beta|\chi,\tau}$, which characterize the distribution of workers' marginal rates of substitution between wages and amenities. This identification result is achieved through revealed preference, by establishing a direct link between the distribution of workers' preferences and the shape of a firm's labor supply curve.

To understand the source of identification, first consider how the distribution of β impacts a firm's labor supply curve. If β is homogeneous, then the curve is isoelastic, and the elasticity of labor supply to a firm reduces to β. By contrast, if β is heterogeneous, then the curve is nonlinear, as its elasticity depends on a firm's wage. As equations (9) and (10) reveal, the nature of this wage-dependence carries information about the moments of the distribution of β. For example, greater curvature in the firm's labor supply curve indicates greater dispersion in β.

In the following proposition, I demonstrate that this relationship extends even further. In particular, I prove that the elasticity curve $\boldsymbol{\varepsilon}_{j\tau}(\chi, w)$ uniquely determines the density $f_{\beta|\chi,\tau}$.

Proposition 6. Given wages and labor $\{W^{\text{eff}}_{j\tau}(\chi), L_{j\tau}(\chi)\}_{j=1}^{J}$, there is a one-to-one mapping between the labor supply elasticity curve $\boldsymbol{\varepsilon}_{j\tau}(\chi, w)$ and the density function $f_{\beta|\chi,\tau}(\beta|\chi,\tau)$.

Proposition 6 implies that the density $f_{\beta|\chi,\tau}$ is identified under the same assumptions needed to recover the labor supply elasticity curve $\boldsymbol{\varepsilon}_{j\tau}(\chi, w)$. The proof leverages the fact that a firm's labor supply curve can be represented as a Laplace transform whose inverse is proportional to $f_{\beta|\chi,\tau}$.[23] This property ensures that the labor supply curve can be inverted to recover the density of β. Note that this identification result is fully nonparametric with respect to this density.

[23]Any two functions share the same Laplace transform only if they differ on a set of Lebesgue measure zero.

V. Estimation Strategy and Results

In this section, I describe the estimation procedure and present estimates of firm-specific labor supply elasticities and structural parameters. The results show that there is substantial variation in labor supply elasticities both within and across firms, implying that wage setting power varies considerably in the economy. I also document sizable dispersion in firm productivity, technology, and non-wage amenities, providing insight into how these factors shape worker sorting.

My estimates reveal two key patterns. First, labor supply is less elastic among more educated and experienced workers. This pattern suggests that, within firms, there is greater scope to exert wage setting power over workers with observably higher skills. Second, across firms, labor supply is less elastic at firms that pay lower wages to a given skill group. This pattern supports the selection mechanism captured by the model: fixing skills, lower-paying firms attract workers who are less sensitive to wages. The estimation nowhere imposes this relationship, so the result offers substantive support for the mechanism rather than reflecting any built-in structure.

V.A. Estimation of Firm-Specific Labor Supply Elasticities

Estimation Procedure and Specifications

My estimation procedure is based on the identification strategy in Section IV.A, and it consists of two steps. First, I estimate worker skill levels by implementing a two-way fixed effects estimator, where I regress log wages on a worker fixed effect γ_i and a fixed effect $\psi_{\chi(i,\tau),j(i,\tau),\tau}$ for the skill type, firm, and year. I then define the permanent component of the skill level as $\exp(\gamma_i)$ and the effective wage $W^{\text{eff}}_{j\tau}(\chi)$ as $\exp(\psi_{\chi(i,\tau),j(i,\tau),\tau})$.[24] Table 2, Panel C, reports the variance of γ_i by education and experience, with the full distributions plotted in Appendix Figure A.14.

Second, I estimate the elasticities of labor supply to each firm by computing a sample analogue of the estimand $\text{DiD}_{\tau_0,\tau_1}(w|\chi)$. This estimand corresponds to a ratio of DiD estimands for wages and labor, controlling for a firm's pre-period effective wage. Since there are finitely many firms in the data, I rarely observe multiple firms that pay exactly the same effective wages to workers. Thus, for the purposes of implementation, I use a nonparametric kernel estimator:

$$\widehat{\text{DiD}}_{\tau_0,\tau_1}(w|\chi) = \frac{\sum_j K_{1,j}(w|\chi)\big(\ell_{j\tau_1}(\chi) - \ell_{j\tau_0}(\chi)\big) - \sum_j K_{0,j}(w|\chi)\big(\ell_{j\tau_1}(\chi) - \ell_{j\tau_0}(\chi)\big)}{\sum_j K_{1,j}(w|\chi)\big(w^{\text{eff}}_{j\tau_1}(\chi) - w^{\text{eff}}_{j\tau_0}(\chi)\big) - \sum_j K_{0,j}(w|\chi)\big(w^{\text{eff}}_{j\tau_1}(\chi) - w^{\text{eff}}_{j\tau_0}(\chi)\big)}, \quad (27)$$

where $K_{1,j}(w|\chi)$ and $K_{0,j}(w|\chi)$ are Kernel weights for treated and control firms, respectively. For $z \in \{0,1\}$, the weights are defined as $K_{z,j}(w|\chi) = \frac{\mathbf{1}\{Z_j = z\}\kappa_h(w^{\text{eff}}_{j\tau_0}(\chi) - w)}{\sum_k \mathbf{1}\{Z_k = z\}\kappa_h(w^{\text{eff}}_{k\tau_0}(\chi) - w)}$, where κ_h is a kernel function with bandwidth h. This estimator allows me to compare firms with similar pre-period wages, thus ensuring tractability in finite samples. In Appendix D.3, I prove the estimator is consistent, and I derive its asymptotic properties, establishing that it is capable of inference.

[24]The estimates are robust to defining effective wages $W^{\text{eff}}_{j\tau}(\chi)$ using the average wages in each (χ, j, τ)-cell.

In the baseline specification, I use a Gaussian kernel function to define the weights $K_{z,j}$.[25] Appendix Figures A.8 and A.9 show that the estimates are robust to alternative kernels, including the Uniform kernel, which corresponds to a standard nonparametric binning estimator. In addition, Appendix Figures A.10 and A.11 show that the results are stable across bandwidth choices. These robustness checks validate the estimator's ability to approximate $\text{DiD}_{\tau_0,\tau_1}(w|\chi)$.

My primary estimates are based on internal instruments, following the identification analysis in Section IV.A.4. For each year τ that TFP shocks occur, I define Z_j by splitting firms at the median of the distribution of $\Delta y_{j\tau}$. I then define the pre- and post-periods to be $\tau_0 = \tau - 2$ and $\tau_1 = \tau + 3$, respectively. I estimate the model repeatedly using all years of data, with the post-period τ_1 ranging from 2002 to 2017. As a robustness check, I also report estimates from the external instruments, defined from public procurement auction wins in Norway. These supplementary results draw on the research design of de Frahan et al. (2024), who also study Norwegian administrative data. I report the external IV estimates in Appendix Figures A.12 and A.13, together with reduced-form estimates and placebo checks showing no evidence of pre-trends. Although these estimates are noisier, reflecting the smaller set of firms used to define treated and control groups, they display the same broad patterns as the baseline IV estimates. In particular, the estimated elasticities span a similar range and preserve the same cross-firm relationship: conditional on skills, labor supply is generally more elastic at higher-wage firms.

In implementing my approach, I include fixed effects for local labor markets m, where m is the interaction between industry and commuting zone, as defined in Section III. For comparison, I also estimate the model without fixed effects, treating Norway as a single labor market; those results are reported in Appendix Figure A.4. I average the estimates across all available years of data to improve precision. Since I allow the labor supply elasticities to evolve over time, the estimates can be interpreted as an average of these quantities over multiple years.

Description of Empirical Findings

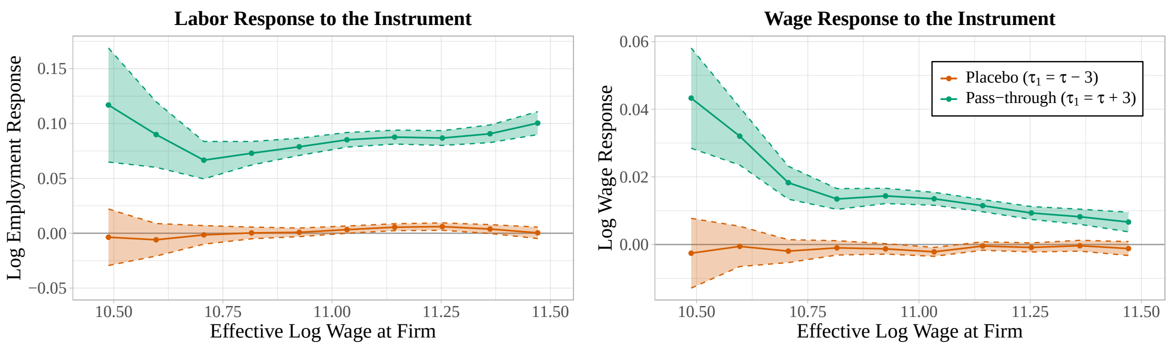
Figure 4 presents reduced form estimates, averaged across worker skill types and plotted over the distribution of effective wages. These estimates correspond to the numerator and denominator of the estimator in equation (27). The green line reports the pass-through for $\tau_1 = \tau + 3$, while the orange line reports placebo estimates based on pre-trends, with $\tau_1 = \tau - 3$. The placebo effect is tightly centered around zero, lending credibility to the IV design. Appendix Figures A.2 and A.3 report reduced form estimates by skill type, as well as event study plots of the pass-through before and after the shock; these plots likewise show no evidence of pre-trends.

The reduced form estimates reveal two distinct patterns. First, the employment response to the instrument lies consistently between 7% and 11% across the effective wage distribution. Second, the earnings response ranges from 1% to 5%, and declines with firms' effective wages.

[25] Specifically, I define the kernel to be $\kappa_h(u) = \frac{1}{\sqrt{2\pi}} \exp\left[-\frac{1}{2}(u/h)^2\right]$, where the parameter h is the bandwidth.

Based on equation (16), this pattern could suggest that higher-paying firms tend to have lower returns to scale. In addition, it could mean that these firms face higher labor supply elasticities.

Figure 4: Average Pass-Through by Workers' Effective Wages



Notes. This figure plots reduced-form estimates, averaged across skill types and years for $\tau_1 \in \{2002, \ldots, 2017\}$ using sample size weights. Market fixed effects are included. Standard errors are based on 500 bootstrap samples.

Figure 5 presents IV estimates of the firm-specific labor supply elasticities, disaggregated by worker college attainment and experience. The estimated elasticities vary greatly across firms, ranging from about 1.5 to 10 for some skill types. This dispersion strongly rejects a model with constant β. It also points to substantial heterogeneity in wage setting power across firms.

One key pattern that emerges from the estimates is that firms paying lower wages to a given skill type tend to face smaller labor supply elasticities. This relationship holds across all skill types, and it aligns with the model's selection mechanism that, holding skills fixed, low-paying firms are more likely to attract workers who are less wage-responsive. This pattern implies that wage setting power is more concentrated at the bottom end of the effective wage distribution.[26]

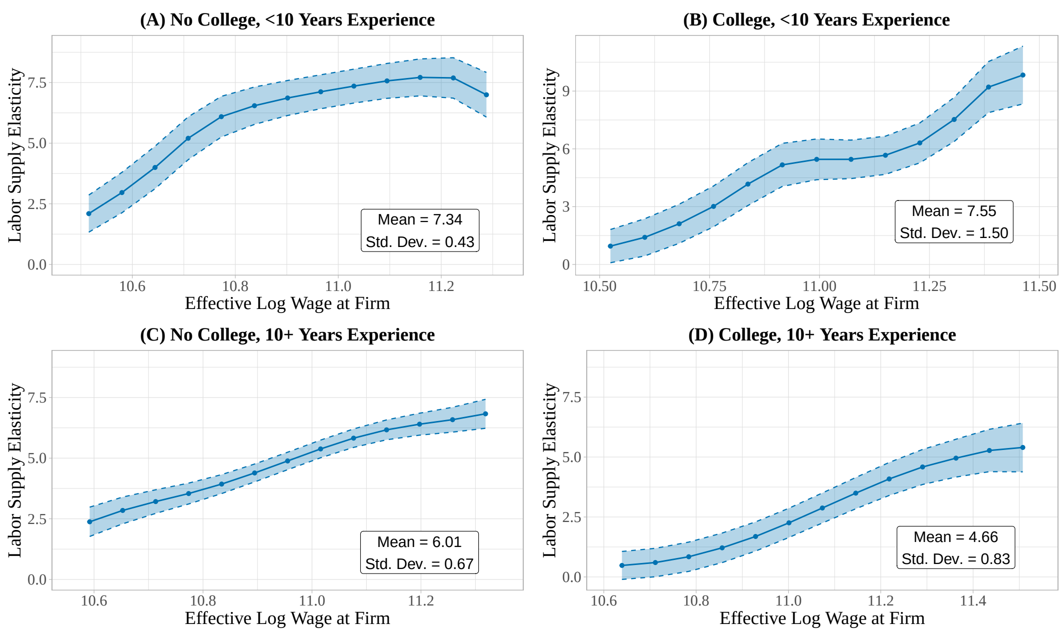
The elasticity estimates also differ meaningfully by skill type. Labor supply to the firm is, on average, less elastic for more experienced workers, especially those with a college degree. These elasticities are also more dispersed among less experienced, college-educated workers. These patterns persist even after controlling for worker tenure at the firm. Under the equilibrium framework, the estimates imply that, within firms, there is greater wage setting power over higher skilled workers, but that there is also sizable cross-firm variation within every skill type.

To further validate my approach, I conduct two robustness checks. First, in Appendix Table A.1, Panel A, I test whether Z_j affects labor on the intensive margin by changing hours. This test addresses the concern that firms may respond to labor demand shocks by adjusting hours rather than wages, a violation of the assumption that the pass-through operates through wages. The estimated effect of Z_j on hours is tightly centered around zero, supporting the IV design.

[26] This finding may seem surprising given that low-wage firms and sectors often have high turnover. However, the relevant object is not the level of turnover but its sensitivity to wages. Consistent with this distinction, Bassier et al. (2022) find lower separation elasticities for low-wage sectors, even while these sectors exhibit high turnover.

Next, I test the assumption that firms are strategically small in the labor market. A testable implication of this assumption is that firms paying the same wages to workers with the same skills would face the same labor supply elasticities. By contrast, if firms were strategically large, two firms paying the same wage to the same types of workers can face different elasticities if they differ in market share. Guided by this property, Table A.1, Panel B, tests whether a firm's labor supply elasticity varies with its local market share after controlling for effective wages. I find no evidence of market share effects: the estimated relationship between labor supply elasticities and firm market share is statistically insignificant and turns positive after adding market fixed effects, which suggests that the data is consistent with the strategically small benchmark.

Figure 5: IV Estimates of Firm-Specific Labor Supply Elasticities



Notes. This figure plots IV estimates of firm-specific labor supply elasticities, averaged over years 2002-2017 using sample size weights. No College Degree combines the Lower Secondary and Upper Secondary skill types. All moments are population-weighted. Market fixed effects included. Standard errors are based on 500 bootstrap samples.

V.B. Estimation of Structural Parameters in the Model

Next, I present estimates of the structural parameters in the model, including firm productivity and technology $(T_{j\tau}, \{\theta_{j\chi}\}_{\chi}, \rho_j, \alpha_j)$, non-wage amenities $a_{j\chi}$, and the densities $f_{\beta|\chi,\tau}$ governing workers' preferences. These estimates are summarized in Table 2 and in Figures 6 and 7.

Firm Productivity, Technology, and Amenities

To estimate firm production parameters, I apply the formulas in Propositions 3 and 4 separately for each firm j and year τ in the data, giving me a unique set of estimates for every combination of (j, τ).[27] In Figure 6, I plot the marginal distributions of estimated TFP $T_{j\tau}$, returns to scale

[27]Recall that recovering the substitution parameter ρ_j for a firm j requires me to compare estimates over two

$1-\alpha_j$, and the substitution parameter ρ_j, averaged over all years. The means and variances are reported in Table 2. I find substantial dispersion in firms' returns to scale and substitution parameters, indicating that differences in firm technology play a key role in wage determination.

To estimate the value of non-wage amenities, I follow Proposition 5 by integrating firms' estimated labor supply elasticities with respect to wages. This exercise allows me to compare how much labor each firm would attract in a counterfactual where firms pay the same wages.[28] Figure 6 plots the distribution of firms' average amenities, and Table 2 reports the variances by worker type; Appendix Figure A.16 plots the full type-specific distributions. I find that the variance of non-wage amenities across firms is broadly similar among worker types, with the exception of less experienced, college educated workers, who face more variation in amenities.

Table 2: Structural Parameter Estimates—Means and Variances

	Notation	Mean	Variance
Panel A. Firm Production			
Log Total Factor Productivity	$\log T_{j\tau}$	12.940 (0.026)	1.187 (0.011)
Returns to Scale	$1-\alpha_j$	0.700 (0.006)	0.035 (0.001)
Substitution Parameter	ρ_j	0.746 (0.018)	0.463 (0.055)
Panel B. Firm Amenities	$a_{j\chi}$		
No College, < 10 Years Exp.		—	1.598 (0.124)
College, < 10 Years Exp.		—	2.120 (0.244)
No College, ≥ 10 Years Exp.		—	1.552 (0.074)
College, ≥ 10 Years Exp.		—	1.390 (0.079)
Panel C. Worker Skill Levels	$\log \varphi_i$		
No College, < 10 Years Exp.		—	0.062 (0.011)
College, < 10 Years Exp.		—	0.073 (0.021)
No College, ≥ 10 Years Exp.		—	0.064 (0.014)
College, ≥ 10 Years Exp.		—	0.076 (0.020)
Panel D. Worker Preferences	β_i		
No College, < 10 Years Exp.		7.270 (0.789)	5.204 (0.939)
College, < 10 Years Exp.		7.647 (1.120)	16.767 (4.911)
No College, ≥ 10 Years Exp.		6.075 (0.334)	7.340 (1.206)
College, ≥ 10 Years Exp.		4.898 (0.601)	9.269 (2.500)

Notes. This table reports cross-sectional means and variances of estimated model parameters, averaged across years. Means are omitted for parameters identified only up-to-location.

Worker Preferences

To estimate the density $f_{\beta|\chi,\tau}$, I draw on Proposition 6, which shows that $f_{\beta|\chi,\tau}$ is uniquely determined by the labor supply curve to a firm. Therefore, it is identified as the unique solution to:

$$L_{j\tau}(\chi, W) = \int \frac{\exp\left(\beta \log W + a_{j\chi}\right)}{I_\tau^{\text{eff}}(\beta, \chi)} f_{\beta|\chi,\tau}(\beta|\chi,\tau) f_{\chi|\tau}(\chi|\tau) d\beta, \tag{28}$$

different time periods, τ and τ'. In my implementation, I set τ' to be $\tau - 1$, corresponding to the previous year.

[28]For each skill type χ, amenities are normalized relative to a "reference firm" j^*, where $a_{j^*\chi} = 0$. In my analysis, j^* is defined as the firm offering an average wage equal to the median of the distribution in the sample.

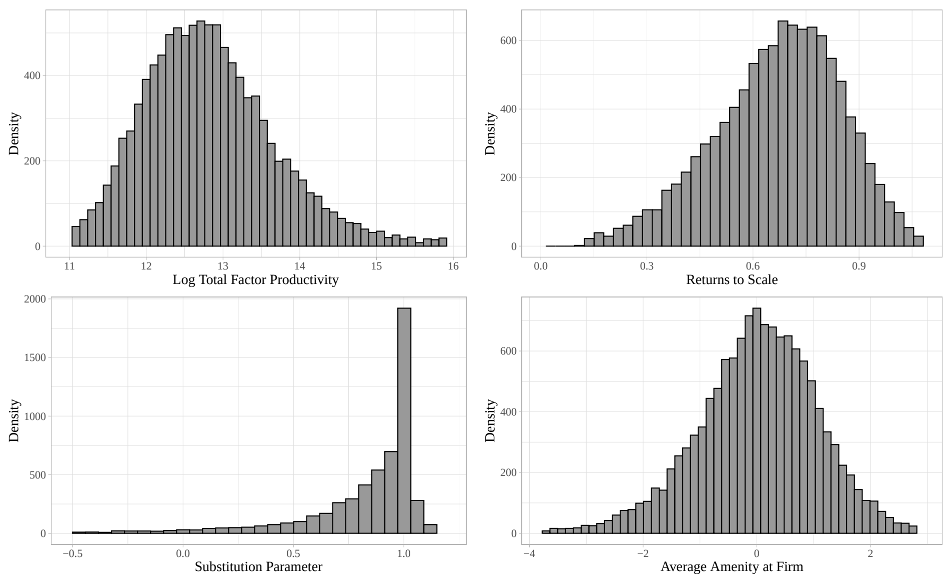
for all $W > 0$, where the labor supply curve $L_{j\tau}(\chi, W)$, the effective wage index $I_\tau^{\text{eff}}(\beta, \chi)$, the density of skill types $f_{\chi|\tau}$, and the non-wage amenities $a_{j\chi}$ can all be recovered from the data.

To make the estimation problem tractable, I introduce two important modifications. First, since the density function $f_{\beta|\chi,\tau}$ is an infinite dimensional parameter, I approximate it using a finite dimensional basis expansion. Specifically, I replace $f_{\beta|\chi,\tau}$ with a function $\hat{f}_{\beta|\chi,\tau}$, where:

$$\hat{f}_{\beta|\chi,\tau}(b) = \sum_{k=1}^{d_\phi} \phi_k \hat{f}_{\beta|\chi,\tau}^{(k)}(b), \quad \text{for } \phi = (\phi_1, \ldots, \phi_{d_\phi}) \in \Phi. \tag{29}$$

In this expression, ϕ is a d_ϕ-dimensional vector of unknown coefficients with support Φ, and $\{\hat{f}_{\beta|\chi,\tau}^{(k)}\}_k$ are basis functions known to the researcher. By defining the basis in this way, I can approximate $f_{\beta|\chi,\tau}$ with a function $\hat{f}_{\beta|\chi,\tau}$ that is parametrized by a finite dimensional vector ϕ.

Figure 6: Histograms of Estimated Firm Productivity, Technology, and Amenities



Notes. This figure plots histograms of the estimated (log) total factor productivity $\log T_{j\tau}$, the returns to scale $1-\alpha_j$, the substitution parameter ρ_j, and the average amenities $\mathrm{E}(a_{j\chi}|j)$ across firms in the estimation sample.

Second, rather than imposing an infinite number of constraints, I evaluate equation (28) on a finite grid of effective wages $\mathbf{W} \subset \mathbb{R}_+$, noting that identification is achieved as $|\mathbf{W}| \to \infty$. This modification allows me to recast (28) as a convex optimization problem, minimizing the ℓ^2-distance between the estimated labor supply curves and the predicted supply curves implied by the density $\hat{f}_{\beta|\chi,\tau}$. Specifically, the estimation procedure involves choosing ϕ^* to solve:

$$\phi^* = \operatorname{argmin}_\phi \sum_{W \in \mathbf{W}} \left[\phi' \hat{h}_{j\tau}(\chi, W) - \hat{L}_{j\tau}(\chi, W)\right]^2. \tag{30}$$

where $\hat{L}_{j\tau}(\chi, W)$ is the estimated labor supply at $W_{j\tau}^{\text{eff}}(\chi) = W$ and $\hat{h}_{j\tau}(\chi, W) \in \mathbb{R}^{d_\phi}$ denotes a vector with entries $\hat{h}_{j\tau,k}(\chi, W) = \int \left[\exp\left(\beta \log W + \hat{a}_{j\chi}\right) / \hat{I}_\tau^{\text{eff}}(\beta, \chi)\right] \hat{f}_{\beta|\chi,\tau}^{(k)}(\beta|\chi, \tau) \hat{f}_{\chi|\tau}(\chi|\tau) d\beta$.[29]

[29] Note that equation (30) can be written in matrix form as $\phi^* = \operatorname{argmin}_\phi \left\{\phi^\top \hat{h}^\top \hat{h} \phi - 2\phi^\top \hat{h}^\top \hat{L} + \hat{L}^\top \hat{L}\right\}$, where

In the baseline specification, I construct a basis using mixtures of $d_\phi = 18$ Gamma density functions. This type of density is shown to be particularly well suited for nonparametric estimation of functions that take support on the positive real line (e.g., Wiper et al., 2001). As a robustness check, I investigate the sensitivity of the estimates to the choice of basis function, the basis size d_ϕ, and the number of points $|\mathbf{W}|$ in the wage grid.[30] I find that the estimates are robust across alternative basis functions and that they stabilize once the basis size and the wage grid become sufficiently rich. Moreover, I find that the wage and employment distributions implied by the estimated densities $\hat{f}_{\beta|\chi,\tau}$ closely match those in the sample, indicating good fit.

Figure 7: Density of Workers' Individual Labor Supply Elasticities

Notes. This figure plots the estimated density f_β, corresponding to the distribution of workers' marginal rates of substitution between wages and amenities. The confidence region is bootstrapped using 500 bootstrap samples.

Figure 7 plots the estimated density of β across all workers, while Table 2, Panel D, reports the means and variances by worker skill type. On average, more educated and experienced workers possess lower values of β, implying a greater willingness to pay for non-wage amenities. Nevertheless, there is substantial heterogeneity within each worker type. In Appendix Figure A.15, I plot the full distributions by skill type, and I find considerable overlap across groups.

VI. Evaluating the Incidence and Consequences of Labor Market Power

I now present the main findings from the empirical framework, quantifying imperfect competition in the labor market and evaluating its impacts on wage inequality, sorting, and welfare.

VI.A. Wage Markdowns and Rent Sharing

I begin by quantifying the incidence of labor market power, as measured by wage markdowns and the division of rents between workers and firms. Table 3 reports key moments of the es-

$\hat{L} \in \mathbb{R}^{|\mathbf{W}|}$ and $\hat{h} \in \mathbb{R}^{|\mathbf{W}| \times d_\phi}$. This framing of the problem is more convenient for computational implementation.

[30] For example, I also approximate $f_{\beta|\chi,\tau}$ using a linear sieve constructed from Laguerre polynomials. These functions are capable of approximating any square-integrable function with positive support (see Chen, 2007).

timated distributions—means, standard deviations, and a variance decomposition into within- and between-firm components—and compares them to the corresponding moments implied by a restricted model with constant β.[31] This comparison allows me to isolate the role of heterogeneous labor supply responses in shaping both the level and dispersion of market power. In Appendix Table A.4, I also report estimates from two intermediate specifications: one that sets β constant conditional on worker skill types and one that imposes a common labor supply elasticity within each local labor market. These supplementary estimates allow me to isolate the role of within-skill and within-market heterogeneity in workers' labor supply responses.

On average, I find that workers are paid 86.5% of their marginal product. However, wage markdowns vary greatly both within and across firms, ranging from about 70% to 95% at the 1st and 99th percentiles. About 60% of the variation in markdowns is within firms, while the remaining 40% is between firms. In the model, within-firm variation reflects systematic differences in labor supply elasticities across worker skill types, while between-firm variation reflects selection: conditional on skills, firms attract workers with different labor supply responses.

Table 3: Estimates of Wage Markdowns and Rents

	Full Model: Heterogeneous β			Restricted Model: Homogeneous β		
	Mean	Std. Dev.	Variance Decomp. (Within Firm / Between Firm)	Mean	Std. Dev.	Variance Decomp. (Within Firm / Between Firm)
Wage to MPL Ratio	0.865	0.036	62.46% / 37.54%	0.874	0.000	——
	(0.007)	(0.010)		(0.007)		
Worker Rents						
Dollars per Worker	\$12,976	\$3,946		\$10,305	\$1,818	
	(800)	(1,292)		(580)	(118)	
Share of Earnings	16.1%	6.2%	68.29% / 31.71%	12.6%	1.2%	100.00% / 0.00%
	(1.0)	(1.4)		(0.7)	(0.1)	
Firm Rents						
Dollars per Worker	\$8,375	\$3,031		\$7,990	\$2,853	
	(796)	(1,096)		(622)	(169)	
Share of Profits	23.0%	20.1%	0.00% / 100.00%	21.6%	19.1%	0.00% / 100.00%
	(1.2)	(0.4)		(1.1)	(0.5)	

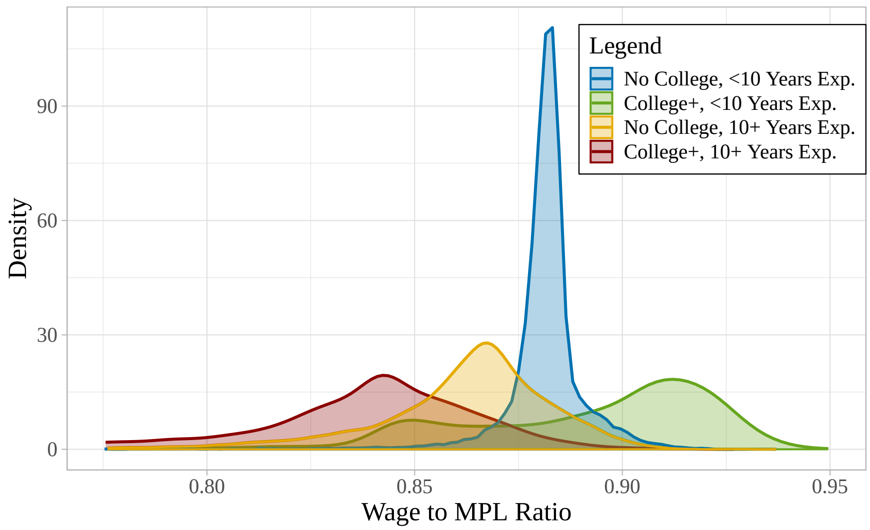
Notes. This table shows estimated wage markdowns and rents under two model specifications. The main specification allows for unrestricted variation in workers' wage-amenity trade-offs (β), and the restricted specification sets β constant across workers, equal to the mean $E(\beta)$ under the full model. Variance decompositions show the share of variation in each quantity attributable to within-firm and between-firm components, calculated via the Law of Total Variance: $\text{Var}(X) = E(\text{Var}(X|j)) + \text{Var}(E(X|j))$. Standard errors are bootstrapped using 500 replications.

To further characterize the incidence of wage markdowns, Figure 8 plots the distributions by worker education and experience. Markdowns are larger, on average, for more experienced workers, especially those with a college degree, and more dispersed among college-educated workers. Appendix Table A.2 reports means and variances by industry and region. Industry di-

[31] In the restricted model, I set β to the economy-wide mean from the full model, estimated as $E(\beta) \approx 6.9$. I also consider an alternative specification that sets β equal to the IV estimand from a constant-effects model (e.g., Lamadon et al., 2022). I compute this estimand as an unconditional version of (20), yielding $\beta^{\text{IV}} \approx 6.4$. The inequality $\beta^{\text{IV}} < E(\beta)$ is consistent with the fact that, under heterogeneous effects, linear IV recovers a weighted average of elasticities that is attenuated relative to the unweighted mean (Angrist et al., 2000; Gaarder et al., 2026).

fferences are modest, with markdowns ranging from 85% to 88% on average. Markdowns are somewhat smaller in high-premium industries such as finance and larger in healthcare, education, and retail. Meanwhile, regional differences are more pronounced: the wage-to-marginal-product ratio averages 84.5% in rural areas versus 87.5% in urban areas. This gap is consistent with greater market power in rural labor markets. Over the estimation period, rural wages were on average 11% lower than urban wages.[32] Differences in markdowns explain approximately 8% of this raw wage gap, with the remainder reflecting productivity differences and sorting.[33]

Figure 8: Density of Wage Markdowns by Education and Experience



Notes. This figure plots the estimated density of wage markdowns by worker college attainment and experience. Estimates are averaged across years $\tau_1 \in \{2002, \dots, 2017\}$ using sample size weights.

I find that many workers and firms capture substantial rents. The average worker is willing to pay \$12,976 (about 16% of earnings) to be at her current firm, while the average firm captures \$8,375 per worker (about 23% of profits) through wage setting power. Appendix Figure A.19 plots the estimated distributions of worker and firm rents. This figure shows that rents are highly dispersed: worker rent shares range from less than 10% to above 25% of earnings, while firm rent shares range from below 5% to above 50% of profits. Also, Figures A.18 and A.20 show that worker rent shares tend to be larger for more educated and experienced workers, and that firm rent shares tend to be larger for firms with lower estimated productivity T_j.

Finally, I benchmark these estimates against those from the restricted model with constant β. This restriction rules out all heterogeneity in workers' wage markdowns, within and across

[32]Bennett et al. (2022) document a large decline in Norway's urban wage premium over the past sixty years.

[33]See Manning (2010), Hirsch et al. (2022), and Azkarate-Askasua & Zerecero (2025) for more evidence on the role of monopsony in generating regional wage differences. My framework builds on previous work by estimating an entire distribution of wage markdowns in each region, rather than a single region-specific markdown.

firms. It also forces worker rent shares to be independent of one's skills X and firm j. Moreover, it implies that firm rent shares cannot depend on workforce composition, so any variation is driven by differences in technology (α_j, ρ_j). Figure A.19 compares the densities of rents in the restricted model to those in the full model. The marked differences between these densities show that allowing for heterogeneous labor supply responses is key for quantifying rents. Even under less restrictive benchmarks where β is assumed constant conditional on observables, Table A.4 shows that the model still falls short in capturing the incidence of labor market power.

VI.B. Consequences for Wage Inequality and the Gender Wage Gap

To evaluate the implications of labor market power for wage inequality, I compare the equilibrium outcomes in the monopsonistic economy to counterfactual outcomes that would arise in a competitive (Walrasian) economy, in which firms set wages equal to workers' marginal products. I solve for the counterfactual equilibrium, taking into account how workers reallocate across firms in response to wage changes. Table 4 reports these counterfactual estimates and compares them to the corresponding estimates implied by a restricted model with constant β. In Table A.5, I also show results for the two intermediate specifications: one that fixes β within skill types and one that imposes a common labor supply elasticity in each local labor market.

I find that labor market power has a large impact on the structure of wage inequality, as measured by the variance of log earnings. Eliminating labor wedges would raise within-firm inequality by 1.3% but reduce between-firm inequality by 16%. These offsetting effects produce a modest 3.9% decline in total inequality. This finding reflects that, within firms, markdowns are larger for more educated and experienced workers, while, across firms, they are larger at firms that pay low wages to a given skill group. Removing labor wedges therefore widens within-firm skill premia while compressing wage differences between firms. Models that restrict β to be constant, either unconditionally or conditional on observables, would largely overlook these effects, since they strongly limit the extent to which wage markdowns can vary across workers.

Next, I apply my framework to study the gender wage gap. Several recent papers, including Card et al. (2016), Sharma (2023), and Caldwell & Oehlsen (2023), infer gender differences in monopsony power by estimating labor supply elasticities separately for men and women. I extend this analysis by allowing the elasticities to vary within gender and, for each gender, estimating the full distribution of workers' wage-amenity trade-offs. This distributional perspective delivers a richer characterization of gender wage gaps, allowing me to assess whether gender differences in monopsony are widespread or concentrated in certain parts of the labor market.

In my framework, the gender wage gap can arise through three channels. The first is human capital: if skills differ by gender, men and women may have different marginal products at the same firms. The second is sorting: if men and women differ in their preferences over wages versus non-wage amenities, they may sort differently between high- and low-wage firms even absent market power, as emphasized in Sorkin (2017). The third is monopsony: women may face more wage setting power than men. The model assumes that firms cannot condition pay on gender. So, conditional on worker skills, gender differences in market power arise due to firm composition, with men and women differentially employed at firms with different markdowns.[34]

Table 4: Counterfactual Estimates—Measuring Impacts on Wage Inequality and Welfare

	Full Model: Heterogeneous β			Restricted Model: Homogeneous β		
	Monopsonistic Labor Market	Competitive Labor Market	% Difference in Dollars	Monopsonistic Labor Market	Competitive Labor Market	% Difference in Dollars
Panel A. Market Aggregates						
Mean Earnings (Log Dollars)	11.291	11.393	10.74%	11.291	11.400	11.52%
		(0.010)			(0.009)	
Mean Profit (Log Dollars)	14.687	14.334	-29.73%	14.687	14.439	-21.91%
		(0.014)			(0.008)	
Panel B. Earnings Inequality						
Variance in Log Earnings						
Total	0.102	0.098	-3.92%	0.102	0.102	0.00%
		(0.004)				
Between Firms	0.025	0.021	-16.00%	0.025	0.025	0.00%
		(0.003)				
Within Firms	0.076	0.077	1.32%	0.076	0.076	0.00%
		(0.001)				
Gender Wage Gap	14.3%	11.0%	-23.1%	14.3%	14.3%	0.00%
		(1.2)				
Panel C. Allocative Inefficiency						
Total Welfare (Log Dollars)	13.111	13.204	9.65%	12.864	12.864	0.00%
	(0.211)	(0.223)		(0.064)	(0.064)	
Labor-TFP Correlation	0.182	0.156	-14.26%	0.180	0.180	0.00%
	(0.006)	(0.014)		(0.005)	(0.005)	

Notes. This table considers how key economic quantities, such as measures of wage inequality and welfare, differ between the monopsonistic labor market and a counterfactual in which all firms are price-takers, so that the labor wedge in the first-order conditions is removed. Results are reported for two model specifications: the full model, which allows for unrestricted variation in workers' wage-amenity trade-offs (β), and a restricted model where β is constant, equal to the mean $E(\beta)$ under the full model. Standard errors are bootstrapped using 500 replications.

To analyze the gender wage gap, I take three steps. First, I examine whether women systematically face more wage setting power than men. To do so, I compare the estimated distribution of wage markdowns across firms, summarized in Table 3, with the gender composition of each firm, evaluating how the incidence of markdowns varies by gender. The estimates, shown in Figure 9(a), indicate that, although there is substantial variation within each gender, women

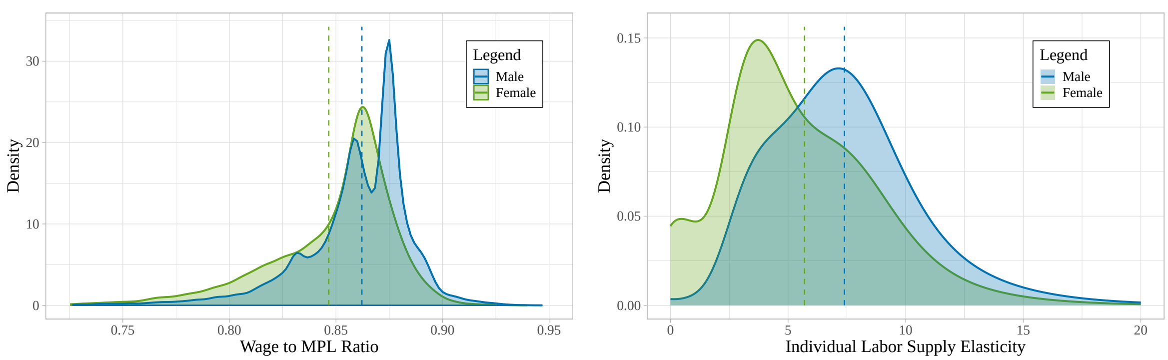
[34]This approach is distinct from Barth & Dale-Olsen (2009), who also study the gender wage gap in Norway, but assume (in their baseline model) that employers can set gender-specific markdowns within the same firm. This assumption implies larger gender differences in markdowns, and hence a larger role for monopsony in the gender pay gap, than a model in which firms condition wages only on skills. My paper adopts the latter as a more natural benchmark in the Norwegian setting, given strong legal restrictions on gender-based pay discrimination.

tend to face larger wage markdowns than men: on average, the wage-to-marginal-product ratio is 84% for women and 87% for men. This gap is primarily driven by between-firm rather than within-firm differences, with women disproportionately employed at high-markdown firms.

Second, I estimate the gender-specific densities of β, applying the estimator from equation (29) separately for men and women. The resulting densities, plotted in Figure 9(b), reveal that there is substantial heterogeneity in wage-amenity trade-offs within each gender; however, on average, women place greater value on amenities than men. This finding is consistent with evidence from choice experiments (Mas & Pallais, 2017; Wiswall & Zafar, 2018) that many women exhibit a higher willingness to pay for non-wage job attributes like workplace flexibility.

Third, I evaluate how the gender wage gap would change if labor wedges were eliminated, taking into account how men and women reallocate differently in response to wage changes. As reported in Table 4, Panel B, removing labor wedges reduces the gender wage gap by 3.3 percentage points, from 14.3% to 11%. Decomposing this effect into within- and between-firm components shows that the reduction is driven primarily by changes in wages between firms.

Figure 9: Individual Labor Supply Elasticities and Wage Markdowns by Gender



(a) Density of Wage Markdowns (b) Density of Individual Labor Supply Elasticities

Notes. Figure 9(a) plots densities of estimated wage markdowns by gender. Figure 9(b) plots densities of estimated wage-amenity trade-offs, f_β, by gender. I compute f_β by estimating firm-specific labor supply curves separately for men and women and applying the estimator in (29). Dashed lines indicate the subgroup-specific means.

VI.C. Consequences for Worker Sorting

I now examine how labor market power affects worker sorting. These effects depend critically on where wage markdowns are concentrated in the economy. If markdowns are identical across firms, then market power would not distort the allocation of labor. By contrast, if some firms set larger markdowns than others, this can distort workers' rankings over employers and lead them to choose firms they would not select in a competitive labor market. Quantifying misallocation therefore necessitates a rich characterization of how wage markdowns vary in the economy, which requires a framework that allows for heterogeneous labor supply responses.

To see how workers would reallocate in the absence of market power, recall that wage markdowns are largest at firms that pay lower wages to a given skill group. These firms tend to be those where labor is less-productive (low T_j). Indeed, Appendix Figure A.18 plots wage markdowns against firm TFP and finds that markdowns are larger, on average, at the bottom of the TFP distribution. Given this relationship, removing labor wedges would yield larger wage and employment gains for low-productivity firms. In line with this prediction, I find that the correlation between log employment ℓ_j and log TFP t_j is higher in the present allocation (0.182) than in the competitive allocation (0.156). Figure 10 illustrates the effect by plotting, over the firm productivity distribution, the predicted change in firms' labor shares from removing labor wedges. In the competitive allocation, more workers are employed at low-productivity firms.

To further decompose the allocative effects of market power, it is important to understand how the ranking of wage markdowns across firms differs by skill type. If the same firms are uniformly "high markdown" or "low markdown" for all workers, then removing labor wedges would shift each firm's employment in the same direction for all skill types. However, if the ranking of markdowns is skill-specific, then removing wedges could generate more complex reallocation patterns, with the same firm employing more of some workers and fewer of others.

I assess this question empirically by comparing how the ranking of wage markdowns across firms differs by skill type. Appendix Figure A.21 reports rank-rank plots of firm-level markdowns for all pairs of skill types and reports Spearman rank correlations as a summary measure of the similarity of these rankings. Rankings are positively correlated across skill types, but far from identical. Especially for college versus non-college workers, many firms rank as high markdown for one group but low markdown for the other. Appendix Table A.3 reports formal tests of equality in the rankings. For all skill type pairs, the null of a common firm-level markdown ranking is rejected, with especially strong rejections for college versus non-college workers.

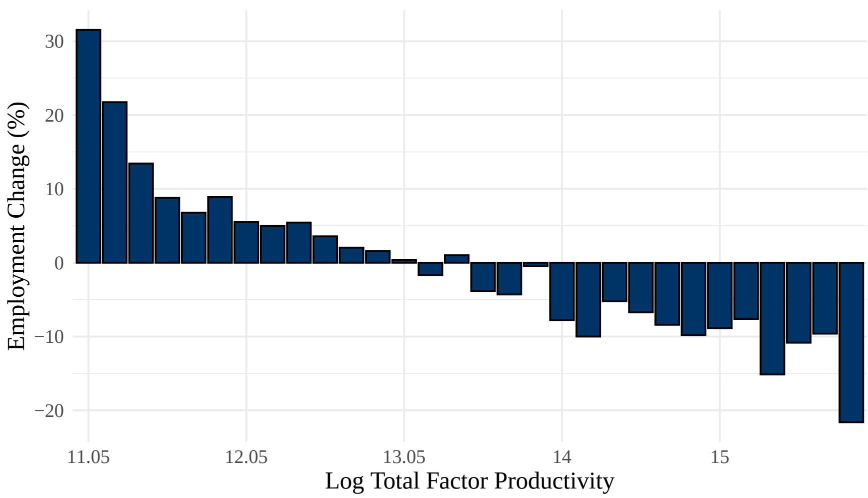
I then examine, by skill type, the implied reallocation of labor when wedges are removed. Appendix Figure A.22 plots these reallocations by industry and region. For some sectors, the direction of reallocation is uniform across skill types: finance and business services contract, while retail, education, and healthcare expand; employment also rises in rural areas. In other sectors, such as trade, manufacturing, and construction, the direction of reallocation is skill-specific, with the same sector expanding for some groups while contracting for others. These results highlight the importance of allowing wage markdowns to vary within and across firms. Imposing a common markdown distribution would misrepresent the nature of misallocation.

VI.D. Welfare Costs and the Role of Policy

Lastly, I assess the welfare cost of labor market power, taking into account firm profits and the value of non-wage amenities to workers. In the model, welfare losses arise from misallocation of workers to firms, driven by differences in firms' wage markdowns. In a version of the model with constant β, markdowns do not vary, leaving no scope for market power to affect welfare.

I estimate large efficiency gains from eliminating labor wedges, with total welfare increasing by 9.65% in dollar terms (Table 4, Panel C).[35] In principle, a planner could implement the first-best allocation through a redistributive tax system that offers wage-specific subsidies to workers, with the shape of the subsidy schedule determined by the distribution of preferences. In practice, however, such a finely targeted policy is hard to envision. Nevertheless, because labor wedges are not purely idiosyncratic but instead differ systematically based on worker, firm, and regional characteristics, second-best policies that target earnings along observable dimensions may compress the dispersion in labor wedges and help to reduce misallocation.

Figure 10: Estimated Reallocation of Labor by Eliminating Labor Wedges



Notes. This figure depicts the implied reallocation of labor resulting from eliminating labor wedges. It plots the estimated percent change in employment across different groups of firms, grouped by their estimated log TFPs.

One policy tool that is often debated as a response to labor market power is the minimum wage. The welfare effect of this policy depends not just on the overall level of market power but also on where it is located; namely, how pervasive it is among workers who would be affected by a minimum wage. If monopsony only affects high-wage workers, then a minimum wage is unlikely to target distortions and may reduce welfare. If instead it is concentrated among low-wage workers, then a minimum wage may have more scope for welfare gain. In Norway, sectoral minimum wages bind for very few workers: less than 4% of those in the sample. Changes to minimum wages would therefore impact workers at the very bottom of the wage distribution.

Since the model abstracts from nonemployment and firm entry and exit, I do not run a full counterfactual to study the equilibrium effects of a minimum wage, as in Berger et al. (2025) and Haanwinckel (2025). Instead, I examine the estimated incidence of labor wedges across the wage distribution to provide suggestive evidence on the policy's potential welfare effects. The

[35] In interpreting these results, it is important to recall that, while firms may choose amenities $a_j(X)$ initially, $a_j(X)$ is held fixed in the counterfactuals. With richer data on amenities and a credible instrument, an interesting extension would be to allow firms to adjust $a_j(X)$ in response to the counterfactual changes. This task requires further information—or assumptions—about how firms choose and pay for the amenities provided to workers.

results are a priori ambiguous: more educated and experienced workers face larger markdowns on average, but conditional on skill, lower paying firms tend to set larger markdowns. Appendix Figure A.23 reconciles these two patterns by plotting the estimated means and the 1st and 99th percentiles of workers' wage markdowns across the realized wage distribution. The figure shows that markdowns are highly dispersed at every wage level. Lower paid workers do not uniformly face larger or smaller markdowns than higher paid workers. On average, however, markdowns are larger at the bottom of the wage distribution, suggesting that wage setting power is more concentrated among the workers most likely to be directly affected by a minimum wage.

VII. Conclusions

This paper has developed an empirical framework to analyze how a firm's labor market power is linked to the composition of its workforce, and the consequences of this relationship for worker sorting, wage inequality, and welfare. I showed that, in a labor market with heterogeneous workers and firms, the slope of a firm's labor supply curve depends on the types of workers it employs. As a result, employer wage setting power can vary both within and across firms, and is endogenous to firm characteristics. I proved identification of the model from matched worker-firm panel data using instruments based on firm-level labor demand shocks, while allowing for unobserved heterogeneity in worker skills, firm productivity, technology, and amenities. I then estimated the model using Norwegian administrative data, drawing inference about the incidence of wage markdowns and the division of rents among workers and firms. The estimates revealed substantial heterogeneity in wage setting power, with key implications for the structure of wage inequality, the gender wage gap, and allocative efficiency. These effects would be overlooked by frameworks that do not allow for rich heterogeneity in workers' labor supply responses.

This paper has provided a first step toward understanding the link between firm workforce composition and wage setting power; yet, there is much room for future work. In particular, I do not take a stand on the economic forces underlying workers' wage-amenity trade-offs or on the extent to which they can be affected by policy. Studying these forces would connect policies that shape household constraints, e.g., through access to child care or transportation access, to the structure of wages and allocative efficiency. In addition, the paper abstracts from strategic competition where firms leverage their market shares in setting wages. In the Norwegian setting, I provide support for this benchmark by testing whether a firm's market share affects its labor supply elasticity, finding no significant effect. This test, however, does not rule out strategic interactions in all settings. An extension of the framework might examine how a firm's wage setting power is influenced not only by its workforce, but also by its local market share.

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Appendix

This appendix accompanies the paper "Job Preferences, Labor Market Power, and Inequality". It contains derivations of economic quantities, model extensions, proofs of equilibrium properties and identification results, details about the data preparation and estimation, and robustness analyses. All notation is consistent with the rest of the paper.

A. Derivation of Equilibrium Quantities

A.1. Firm Labor Supply

Given any set of wage offers $\mathbf{W}(X) = \{W_k(X)\}_k$, a worker with skills X and marginal utility of (log) earnings β would choose to work at firm j with probability $\mathrm{P}(j(i) = j|\beta, X)$, where:

$$
\begin{aligned}
\mathrm{P}(j(i) = j|\beta, X) &= \mathrm{P}\Big(u_{ij}\big(W_j(X_i), a_j(X_i)\big) > \max_{k\neq j}\big\{u_{ik}\big(W_k(X_i), a_k(X_i)\big)\big\} \,\Big|\, \beta_i = \beta, X_i = X\Big) \\
&= \mathrm{P}\Big(\beta \log W_j(X) + a_j(X) + \epsilon_{ij} > \beta \log W_k(X) + a_k(X) + \epsilon_{ik}, \forall k \neq j\Big) \\
&= \mathrm{P}\left(\epsilon_{ik} < \beta \log\left(\frac{W_j(X)}{W_k(X)}\right) + a_j(X) - a_k(X) + \epsilon_{ij}, \forall k \neq j, \forall k \neq j\right) \\
&= \int_{-\infty}^{\infty} \mathrm{P}\left(\epsilon_{ik} < \beta \log\left(\frac{W_j(X)}{W_k(X)}\right) + a_j(X) - a_k(X) + \epsilon_{ij}, \forall k \neq j \,\middle|\, \epsilon_{ij} = \tilde{\epsilon}\right) f_{\epsilon_{ij}}(\tilde{\epsilon}) d\tilde{\epsilon} \\
&= \int_{-\infty}^{\infty} \prod_{k\neq j} \mathrm{P}\left(\epsilon_{ik} < \beta \log\left(\frac{W_j(X)}{W_k(X)}\right) + a_j(X) - a_k(X) + \epsilon_{ij} \,\middle|\, \epsilon_{ij} = \tilde{\epsilon}\right) f_{\epsilon_{ij}}(\tilde{\epsilon}) d\tilde{\epsilon} \\
&= \int_{-\infty}^{\infty} \prod_{k\neq j} F_\epsilon\left(\beta \log\left(\frac{W_j(X)}{W_k(X)}\right) + a_j(X) - a_k(X) + \tilde{\epsilon}\right) f_\epsilon(\tilde{\epsilon}) d\tilde{\epsilon}.
\end{aligned}
$$

Here, the second-to-last equality holds because $\{\epsilon_{ij}\}_{i,j}$ are independent, and the final equality holds because $\{\epsilon_{ij}\}_{i,j}$ are identically distributed. The mass of workers with skills X at firm j is:

$$
S_j(X) = \int \left(\int_{-\infty}^{\infty} \prod_{k\neq j} F_\epsilon\left(\beta \log\left(\frac{W_j(X)}{W_k(X)}\right) + a_j(X) - a_k(X) + \tilde{\epsilon}\right) f_\epsilon(\tilde{\epsilon}) d\tilde{\epsilon}\right) f_{\beta,X}(\beta, X) d\beta.
$$

Under a logit error structure, where $F_\epsilon(\epsilon) = \exp(-\exp(-\epsilon))$, the choice probability becomes:

$$
\mathrm{P}(j(i) = j|\beta, X) = \frac{\exp\big(\beta \log W_j(X) + a_j(X)\big)}{\sum_{k=1}^{J} \exp\big(\beta \log W_k(X) + a_k(X)\big)}.
$$

Defining $I(\beta, X) = \sum_{k=1}^{J} \exp\big(\beta \log W_k(X) + a_k(X)\big)$ to be the worker's wage index, I obtain:

$$S_j(X) = \int \frac{1}{I(\beta, X)} \exp\big(\beta \log W_j(X) + a_j(X)\big) f_{\beta,X}(\beta, X) d\beta.$$

Using this parameterization for F_ε, I will now derive the firm-specific labor supply elasticity $\frac{\partial \log S_j(X)}{\partial \log W_j(X)}$, as well as the higher-order derivatives of $\log S_j(X)$. Consider the following property.

Property A.1. Assume $F_\epsilon(\epsilon) = \exp(-\exp(-\epsilon))$. From the perspective of a strategically small firm, the first five derivatives of the log labor supply curve with respect to the log wage equal:

$$\begin{aligned}
\frac{\partial \log S_j(X)}{\partial \log W_j(X)} &= \mathrm{E}\big(\beta | X, j(i) = j\big) \\
\frac{\partial^2 \log S_j(X)}{\partial \left[\log W_j(X)\right]^2} &= \mathrm{Var}\big(\beta | X, j(i) = j\big) \\
\frac{\partial^3 \log S_j(X)}{\partial \left[\log W_j(X)\right]^3} &= \mathrm{E}\big([\beta - \mathrm{E}(\beta | X, j(i) = j)]^3 \big| X, j(i) = j\big) \\
\frac{\partial^4 \log S_j(X)}{\partial \left[\log W_j(X)\right]^4} &= \mathrm{E}\big([\beta - \mathrm{E}(\beta | X, j(i) = j)]^4 \big| X, j(i) = j\big) - 3\,\mathrm{Var}\big(\beta | X, j(i) = j\big)^2 \\
\frac{\partial^5 \log S_j(X)}{\partial \left[\log W_j(X)\right]^5} &= \mathrm{E}\big([\beta - \mathrm{E}(\beta | X, j(i) = j)]^5 \big| X, j(i) = j\big) \\
&\quad - 10\,\mathrm{Var}\big(\beta | X, j(i) = j\big)\,\mathrm{E}\big([\beta - \mathrm{E}(\beta | X, j(i) = j)]^3 \big| X, j(i) = j\big).
\end{aligned}$$

Proof. If the firm views itself as strategically small, then it does not internalize the impact of its own wage on each worker's wage index. Specifically, it sets $\partial I(\beta, X)/\partial W_j(X) = 0$ for all β and X. Under this assumption, the following relationships hold for any function $g : \mathbb{R} \to \mathbb{R}$:

$$\begin{aligned}
\frac{\partial\, \mathrm{E}(g(\beta)\, \mathrm{P}(j(i) = j | \beta, X) \mid X)}{\partial \log W_j(X)} &= \int \frac{\partial}{\partial \log W_j(X)} \left[\frac{g(\beta) \exp\big(\beta \log W_j(X) + a_j(X)\big)}{I(\beta, X)}\right] f_{\beta|X}(\beta | X) d\beta \\
&= \int \frac{\beta g(\beta) \exp\big(\beta \log W_j(X) + a_j(X)\big)}{I(\beta, X)} f_{\beta|X}(\beta | X) d\beta \\
&= \mathrm{E}\left(\beta g(\beta)\, \mathrm{P}(j(i) = j | \beta, X) \mid X\right).
\end{aligned} \tag{A.1}$$

$$\begin{aligned}
\frac{\mathrm{E}(g(\beta)\, \mathrm{P}(j(i) = j | \beta, X) \mid X)}{\mathrm{P}(j(i) = j | X)} &= \frac{\sum_{k=1}^{J} \mathrm{E}(g(\beta)\, \mathrm{P}(j(i) = j | \beta, X) \mid X, j(i) = k) \times \mathrm{P}(j(i) = k | X)}{\mathrm{P}(j(i) = j | X)} \\
&= \frac{\mathrm{E}(g(\beta) \mid X, j(i) = j) \times \mathrm{P}(j(i) = j | X) + \sum_{k \neq j} 0 \times \mathrm{P}(j(i) = k | X)}{\mathrm{P}(j(i) = j | X)} \\
&= \mathrm{E}(g(\beta) \mid X, j(i) = j)\,.
\end{aligned} \tag{A.2}$$

To ease notation, let $s_j(X) = \log S_j(X)$ and $w_j(X) = \log W_j(X)$. Also, define $\tau_{jX,s}$ such that:

$$\tau_{jX,s}(w) = \mathrm{E}\big(\beta^s \exp\big(\beta w + a_j(X)\big)/I(\beta, X)|X\big).$$

Equation (A.1) implies $\partial\tau_{jX,s}(w_j(X))/\partial w_j(X) = \tau_{jX,s+1}(w_j(X))$, while equation (A.2) implies $\tau_{jX,s}(w_j(X))/\tau_{jX,0}(w_j(X)) = \mathrm{E}(\beta^s|X, j(i) = j)$. Using these properties, I derive the following:

First Derivative. The first derivative of a firm's labor supply curve $\partial s_j(X)/\partial w_j(X)$ is:

$$\frac{\partial s_j(X)}{\partial w_j(X)} = \frac{\partial \log\big(\tau_{jX,0}(w_j(X))\big)}{\partial w_j(X)} = \frac{\tau_{jX,1}(w_j(X))}{\tau_{jX,0}(w_j(X))} = \mathrm{E}\big(\beta|X, j(i) = j\big).$$

Second Derivative. The second derivative of a firm's labor supply curve $\partial^2 s_j(X)/\partial w_j^2(X)$ is:

$$\begin{aligned}
\frac{\partial^2 s_j(X)}{\partial w_j^2(X)} &= \frac{\partial\big(\tau_{jX,1}(w_j(X))/\tau_{jX,0}(w_j(X))\big)}{\partial w_j(X)}\\
&= \frac{\tau_{jX,2}(w_j(X))\tau_{jX,0}(w_j(X)) - \tau^2_{jX,1}(w_j(X))}{\tau^2_{jX,0}(w_j(X))} = \frac{\tau_{jX,2}(w_j(X))}{\tau_{jX,0}(w_j(X))} - \left(\frac{\tau_{jX,1}(w_j(X))}{\tau_{jX,0}(w_j(X))}\right)^2\\
&= \mathrm{E}\big(\beta^2|X, j(i) = j\big) - \mathrm{E}\big(\beta|X, j(i) = j\big)^2 = \mathrm{Var}\big(\beta|X, j(i) = j\big).
\end{aligned}$$

Third Derivative. The third derivative of a firm's labor supply curve $\partial^3 s_j(X)/\partial w_j^3(X)$ is:

$$\begin{aligned}
\frac{\partial^3 s_j(X)}{\partial w_j^3(X)} &= \frac{\partial}{\partial w_j(X)}\left(\frac{\tau_{jX,2}(w_j(X))\tau_{jX,0}(w_j(X)) - \tau^2_{jX,1}(w_j(X))}{\tau^2_{jX,0}(w_j(X))}\right)\\
&= \frac{\tau_{jX,3}(w_j(X))\tau^3_{jX,0}(w_j(X)) - 3\tau_{jX,2}(w_j(X))\tau_{jX,1}(w_j(X))\tau^2_{jX,0}(w_j(X)) + 2\tau^3_{jX,1}(w_j(X))\tau_{jX,0}(w_j(X))}{\tau^4_{jX,0}(w_j(X))}\\
&= \frac{\tau_{jX,3}(w_j(X))}{\tau_{jX,0}(w_j(X))} - 3\left(\frac{\tau_{jX,2}(w_j(X))}{\tau_{jX,0}(w_j(X))}\right)\left(\frac{\tau_{jX,1}(w_j(X))}{\tau_{jX,0}(w_j(X))}\right) + 2\left(\frac{\tau_{jX,1}(w_j(X))}{\tau_{jX,0}(w_j(X))}\right)^3\\
&= \mathrm{E}\big(\beta^3|X, j(i) = j\big) - 3\,\mathrm{E}\big(\beta|X, j(i) = j\big)\,\mathrm{E}\big(\beta^2|X, j(i) = j\big) + 2\,\mathrm{E}\big(\beta|X, j(i) = j\big)^3\\
&= \mathrm{E}\big([\beta - \mathrm{E}(\beta|X, j(i) = j)]^3\big|X, j(i) = j\big).
\end{aligned}$$

Fourth Derivative. The fourth derivative of a firm's labor supply curve $\partial^4 s_j(X)/\partial w_j^4(X)$ is:

$$\begin{aligned}
\frac{\partial^4 s_j(X)}{\partial w_j^4(X)} &= \frac{\partial}{\partial w_j(X)}\left(\frac{\tau_{jX,3}(w_j(X))\tau_{jX,0}^3(w_j(X)) - 3\tau_{jX,2}(w_j(X))\tau_{jX,1}(w_j(X))\tau_{jX,0}^2(w_j(X)) + 2\tau_{jX,1}^3(w_j(X))\tau_{jX,0}(w_j(X))}{\tau_{jX,0}^4(w_j(X))}\right) \\
&= \frac{\tau_{jX,4}(w_j(X))\tau_{jX,0}^7(w_j(X)) - 4\tau_{jX,3}(w_j(X))\tau_{jX,1}(w_j(X))\tau_{jX,0}^6(w_j(X)) - 3\tau_{jX,2}^2(w_j(X))\tau_{jX,0}^6(w_j(X))}{\tau_{jX,0}^8(w_j(X))} \\
&\quad + \frac{12\tau_{jX,2}(w_j(X))\tau_{jX,1}^2(w_j(X))\tau_{jX,0}^5(w_j(X)) - 6\tau_{jX,1}^4(w_j(X))\tau_{jX,0}^4(w_j(X))}{\tau_{jX,0}^8(w_j(X))} \\
&= \frac{\tau_{jX,4}(w_j(X))}{\tau_{jX,0}(w_j(X))} - 4\left(\frac{\tau_{jX,3}(w_j(X))}{\tau_{jX,0}(w_j(X))}\right)\left(\frac{\tau_{jX,1}(w_j(X))}{\tau_{jX,0}(w_j(X))}\right) - 3\left(\frac{\tau_{jX,2}(w_j(X))}{\tau_{jX,0}(w_j(X))}\right)^2 \\
&\quad + 12\left(\frac{\tau_{jX,2}(w_j(X))}{\tau_{jX,0}(w_j(X))}\right)\left(\frac{\tau_{jX,1}(w_j(X))}{\tau_{jX,0}(w_j(X))}\right)^2 - 6\left(\frac{\tau_{jX,1}(w_j(X))}{\tau_{jX,0}(w_j(X))}\right)^4 \\
&= \mathrm{E}\big(\beta^4|X, j(i)=j\big) - 4\,\mathrm{E}\big(\beta|X, j(i)=j\big)\,\mathrm{E}\big(\beta^3|X, j(i)=j\big) - 3\,\mathrm{E}\big(\beta^2|X, j(i)=j\big)^2 \\
&\quad + 12\,\mathrm{E}\big(\beta|X, j(i)=j\big)^2\,\mathrm{E}\big(\beta^2|X, j(i)=j\big) - 6\,\mathrm{E}\big(\beta|X, j(i)=j\big)^4 \\
&= \mathrm{E}\big([\beta - \mathrm{E}(\beta|X, j(i)=j)]^4\big|X, j(i)=j\big) - 3\,\mathrm{Var}\big(\beta|X, j(i)=j\big)^2.
\end{aligned}$$

Fifth Derivative. The fifth derivative of a firm's labor supply curve $\partial^5 s_j(X)/\partial w_j^5(X)$ is:

$$\begin{aligned}
\frac{\partial^5 s_j(X)}{\partial w_j^5(X)} &= \frac{\partial}{\partial w_j(X)}\left(\frac{\tau_{jX,4}(w_j(X))\tau_{jX,0}^7(w_j(X)) - 4\tau_{jX,3}(w_j(X))\tau_{jX,1}(w_j(X))\tau_{jX,0}^6(w_j(X)) - 3\tau_{jX,2}^2(w_j(X))\tau_{jX,0}^6(w_j(X))}{\tau_{jX,0}^8(w_j(X))}\right. \\
&\qquad\left. + \frac{12\tau_{jX,2}(w_j(X))\tau_{jX,1}^2(w_j(X))\tau_{jX,0}^5(w_j(X)) - 6\tau_{jX,1}^4(w_j(X))\tau_{jX,0}^4(w_j(X))}{\tau_{jX,0}^8(w_j(X))}\right) \\
&= \frac{\tau_{jX,5}(w_j(X))\tau_{jX,0}^{15}(w_j(X))}{\tau_{jX,0}^{16}(w_j(X))} - \frac{60\tau_{jX,2}(w_j(X))\tau_{jX,1}^3(w_j(X))\tau_{jX,0}^{12}(w_j(X)) - 24\tau_{jX,1}^5(w_j(X))\tau_{jX,0}^{11}(w_j(X))}{\tau_{jX,0}^{16}(w_j(X))} \\
&\quad - \frac{5\tau_{jX,4}(w_j(X))\tau_{jX,1}(w_j(X))\tau_{jX,0}^{14}(w_j(X)) - 20\tau_{jX,3}(w_j(X))\tau_{jX,1}^2(w_j(X))\tau_{jX,0}^{13}(w_j(X))}{\tau_{jX,0}^{16}(w_j(X))} \\
&\quad - \frac{10\tau_{jX,3}(w_j(X))\tau_{jX,2}(w_j(X))\tau_{jX,0}^{14}(w_j(X)) - 30\tau_{jX,2}^2(w_j(X))\tau_{jX,1}(w_j(X))\tau_{jX,0}^{13}(w_j(X))}{\tau_{jX,0}^{16}(w_j(X))} \\
&= \frac{\tau_{jX,5}(w_j(X))}{\tau_{jX,0}(w_j(X))} - 5\left(\frac{\tau_{jX,4}(w_j(X))}{\tau_{jX,0}(w_j(X))}\right)\left(\frac{\tau_{jX,1}(w_j(X))}{\tau_{jX,0}(w_j(X))}\right) + 20\left(\frac{\tau_{jX,3}(w_j(X))}{\tau_{jX,0}(w_j(X))}\right)\left(\frac{\tau_{jX,1}(w_j(X))}{\tau_{jX,0}(w_j(X))}\right)^2 + 24\left(\frac{\tau_{jX,1}(w_j(X))}{\tau_{jX,0}(w_j(X))}\right)^5 \\
&\quad - 10\left(\frac{\tau_{jX,3}(w_j(X))}{\tau_{jX,0}(w_j(X))}\right)\left(\frac{\tau_{jX,2}(w_j(X))}{\tau_{jX,0}(w_j(X))}\right) + 30\left(\frac{\tau_{jX,2}(w_j(X))}{\tau_{jX,0}(w_j(X))}\right)^2\left(\frac{\tau_{jX,1}(w_j(X))}{\tau_{jX,0}(w_j(X))}\right) - 60\left(\frac{\tau_{jX,2}(w_j(X))}{\tau_{jX,0}(w_j(X))}\right)\left(\frac{\tau_{jX,1}(w_j(X))}{\tau_{jX,0}(w_j(X))}\right)^3 \\
&= \mathrm{E}\big(\beta^5|X, j(i)=j\big) - 5\,\mathrm{E}\big(\beta|X, j(i)=j\big)\,\mathrm{E}\big(\beta^4|X, j(i)=j\big) + 20\,\mathrm{E}\big(\beta|X, j(i)=j\big)^2\,\mathrm{E}\big(\beta^3|X, j(i)=j\big) \\
&\quad - 10\,\mathrm{E}\big(\beta^2|X, j(i)=j\big)\,\mathrm{E}\big(\beta^3|X, j(i)=j\big) + 30\,\mathrm{E}\big(\beta|X, j(i)=j\big)\,\mathrm{E}\big(\beta^2|X, j(i)=j\big)^2 \\
&\quad - 60\,\mathrm{E}\big(\beta|X, j(i)=j\big)^3\,\mathrm{E}\big(\beta^2|X, j(i)=j\big) + 24\,\mathrm{E}\big(\beta|X, j(i)=j\big)^5 \\
&= \mathrm{E}\big([\beta - \mathrm{E}(\beta|X, j(i)=j)]^5\big|X, j(i)=j\big) - 10\,\mathrm{Var}\big(\beta|X, j(i)=j\big)\,\mathrm{E}\big([\beta - \mathrm{E}(\beta|X, j(i)=j)]^3\big|X, j(i)=j\big).
\end{aligned}$$

$\square$

A.2. *Equilibrium Wage Equation*

In equilibrium, each firm chooses a set of wages $\{W_j(\chi, \varphi)\}_{\chi,\varphi}$ to maximize profit Π_j, where:

$$\Pi_j = T_j \left(\sum_{\chi \in \mathcal{X}} \theta_{j\chi} \left(\int \varphi D_j(\chi, \varphi) d\varphi \right)^{\rho_j} \right)^{\frac{1-\alpha_j}{\rho_j}} - \sum_{\chi \in \mathcal{X}} \left(\int W_j(\chi, \varphi) D_j(\chi, \varphi) d\varphi \right).$$

Let $L_j(\chi, \varphi)$ denote the equilibrium employment of skill group $X = (\chi, \varphi)$ at firm j, defined as the amount of labor hired when $D_j(\chi, \varphi) = S_j(\chi, \varphi)$. The firm's first order condition is:

$$\begin{aligned} \frac{\partial \Pi_j}{\partial W_j(\chi, \varphi)} = \; & \varphi T_j(1-\alpha_j)\theta_{j\chi} \left(L_j^{\text{eff}}(\chi) \right)^{\rho_j - 1} \left(\sum_{\chi' \in \mathcal{X}} \theta_{j\chi'} \left(L_j^{\text{eff}}(\chi') \right)^{\rho_j} \right)^{\frac{1-\alpha_j}{\rho_j} - 1} \frac{\partial L_j(\chi, \varphi)}{\partial W_j(\varphi, \chi)} \\ & - L_j(\chi, \varphi) - W_j(\chi, \varphi) \frac{\partial L_j(\chi, \varphi)}{\partial W_j(\varphi, \chi)} = 0, \quad \text{for all } (\chi, \varphi) \in \mathcal{X} \times \mathbb{R}, \end{aligned}$$

where $L_j^{\text{eff}}(\chi) = \int \varphi L_j(\chi, \varphi) d\varphi$ denotes the efficiency units of labor for a given skill type χ. Defining $\boldsymbol{\varepsilon}_j(X)$ as the labor supply elasticity $\partial \log L_j(X) / \partial \log W_j(X)$, this condition becomes:

$$W_j(\chi, \varphi) = \frac{\boldsymbol{\varepsilon}_j(\chi, \varphi)}{1 + \boldsymbol{\varepsilon}_j(\chi, \varphi)} \times \varphi T_j(1-\alpha_j)\theta_{j\chi} \left(L_j^{\text{eff}}(\chi) \right)^{\rho_j - 1} \left(\sum_{\chi' \in \mathcal{X}} \theta_{j\chi'} \left(L_j^{\text{eff}}(\chi') \right)^{\rho_j} \right)^{\frac{1-\alpha_j}{\rho_j} - 1}.$$

The wage markdown at firm j for workers with skills X is: $\frac{W_j(X)}{\partial Y_j / \partial L_j(X)} = \boldsymbol{\varepsilon}_j(X) / (1 + \boldsymbol{\varepsilon}_j(X))$.

Lemmas 1 and 2 establish the existence of an equilibrium corresponding to a unique set of profit-maximizing wages $[W_j(X)]_X$ for each firm j. In this equilibrium, the firm's problem has an interior solution, which satisfies the first-order condition. Taking logs, the condition is:

$$\begin{aligned} w_j(\chi, \varphi) = \; & \log \varphi + \log T_j + \log(1-\alpha_j) + \log \theta_{j\chi} - \log \left(1 + \boldsymbol{\varepsilon}_j^{-1}(\chi, \varphi) \right) \\ & - (1-\rho_j) \log L_j^{\text{eff}}(\chi) + \frac{1-\alpha_j-\rho_j}{\rho_j} \log \sum_{\chi' \in \mathcal{X}} \theta_{j\chi'} \left(L_j^{\text{eff}}(\chi') \right)^{\rho_j}. \end{aligned}$$

A.3. *Firm Production*

After plugging in the labor supply constraint, a firm j's value added from hiring workers is:

$$Y_j = T_j \left(\sum_{\chi \in \mathcal{X}} \theta_{j\chi} \left(\int \varphi L_j(\chi, \varphi) d\varphi \right)^{\rho_j} \right)^{\frac{1-\alpha_j}{\rho_j}}.$$

For workers with skills $X = (\chi, \varphi)$, the marginal product of labor at firm j is $\frac{\partial Y_j}{\partial L_j(\chi, \varphi)}$, where:

$$\frac{\partial Y_j}{\partial L_j(\chi, \varphi)} = \varphi T_j(1-\alpha_j)\theta_{j\chi} \left(L_j^{\text{eff}}(\chi) \right)^{\rho_j - 1} \left(\sum_{\chi' \in \mathcal{X}} \theta_{j\chi'} \left(L_j^{\text{eff}}(\chi') \right)^{\rho_j} \right)^{\frac{1-\alpha_j}{\rho_j} - 1}.$$

To better understand the relationship between skill types χ and skill levels φ in the production function, it will be useful to define the standardized unit of labor at the firm to be N_j, where:

$$N_j = \left(\sum_{\chi \in \mathcal{X}} \theta_{j\chi} \left(\int \varphi L_j(\chi, \varphi) d\varphi \right)^{\rho_j} \right)^{1/\rho_j} .$$

For any skill profile $X = (\chi, \varphi)$, the elasticity of N_j with respect to $L_j(\chi, \varphi)$ is defined by:

$$\frac{\partial \log N_j}{\partial \log L_j(\chi, \varphi)} = \underbrace{\frac{\theta_{j\chi} \left(L_j^{\text{eff}}(\chi) \right)^{\rho_j}}{\sum_{\chi' \in \mathcal{X}} \theta_{j\chi'} \left(L_j^{\text{eff}}(\chi') \right)^{\rho_j}}}_{\partial \log N_j / \partial \log L_j^{\text{eff}}(\chi)} \times \underbrace{\frac{\varphi L_j(\chi, \varphi)}{\int \varphi' L_j(\chi, \varphi') d\varphi'}}_{\partial \log L_j^{\text{eff}}(\chi) / \partial \log L_j(\chi, \varphi)} .$$

The elasticities sum to one across worker skill groups, $\sum_{\chi \in \mathcal{X}} \int \frac{\partial \log N_j}{\partial \log L_j(\chi, \varphi)} d\varphi = 1$. They can be interpreted as the share of firm j's effective labor accounted for by workers with skills (χ, φ).

Property A.2. Suppose that $\alpha_j \in (0, 1)$ and $\rho_j < 1$. Then a firm's value added Y_j is concave in labor inputs. Also, firm production exhibits decreasing marginal returns within skill types:

$$\frac{\partial^2 Y_j}{\partial L_j(\chi, \varphi) \partial L_j(\chi, \varphi')} < 0, \quad \text{for all } \chi \in \mathcal{X} \text{ and } \varphi, \varphi' \in \mathbb{R}_{++},$$

and complementarity (or substitutability) between skill types depending on the value of $1 - \rho_j$:

$$\frac{\partial^2 Y_j}{\partial L_j(\chi, \varphi) \partial L_j(\chi', \varphi')} \begin{cases} > 0 & \text{if } \alpha_j < 1 - \rho_j \\ < 0 & \text{if } \alpha_j > 1 - \rho_j, \end{cases} \quad \text{for all } \chi \neq \chi' \text{ and } \varphi, \varphi' \in \mathbb{R}_{++}.$$

Proof. To show that Y_j is concave, first note that it is continuous, strictly increasing, and homogeneous of degree $1 - \alpha_j \in (0, 1]$. Therefore, by Jehle & Reny (2011), Theorem 3.1, it is sufficient to show that Y_j is strictly quasiconcave. Note that Y_j can be written as a composition of two functions, $Y_j = h \circ g(\{L_j^{\text{eff}}(\chi)\}_\chi)$, where $g : \mathbb{R}_{++}^{|\mathcal{X}|} \to \mathbb{R}$ and $h : \mathbb{R} \to \mathbb{R}$ are defined by:

$$g(\{L_j^{\text{eff}}(\chi)\}_\chi) = \sum_{\chi \in \mathcal{X}} \theta_{j\chi} \big(L_j^{\text{eff}}(\chi) \big)^{\rho_j} \quad \text{and} \quad h(x) = T_j x^{(1-\alpha_j)/\rho_j} .$$

The function g has a diagonal Hessian matrix with entries: $\frac{\partial^2 g}{\partial [L_j^{\text{eff}}(\chi)]^2} = -\rho_j(1 - \rho_j)\theta_{j\chi} \big(L_j^{\text{eff}}(\chi) \big)^{\rho_j - 2}$.

I now analyze three cases. First, if $\rho_j \in (0, 1)$, then h is strictly increasing and g is strictly concave, because its Hessian is negative definite. It follows that Y_j is strictly quasiconcave in $\{L_j^{\text{eff}}(\chi)\}_\chi$, as it is a strictly increasing transformation of a strictly concave function. Second, if $\rho_j < 0$, then h is strictly decreasing and g is strictly convex, because its Hessian is positive definite. Therefore, Y_j is still strictly quasiconcave in $\{L_j^{\text{eff}}(\chi)\}_\chi$, as it is a strictly decreasing transformation of a strictly convex function. Third, if ρ_j approaches zero, then Y_j reduces to a Cobb-Douglas production function that is strictly concave in $\{L_j^{\text{eff}}(\chi)\}_\chi$. In all three cases,

Y_j is shown to be strictly quasiconcave in $\{L_j^{\text{eff}}(\chi)\}_\chi$. By Jehle & Reny (2011), Theorem 3.1, this implies that Y_j is concave in $\{L_j^{\text{eff}}(\chi)\}_\chi$. Furthermore, since $L_j^{\text{eff}}(\chi)$ is a linear function of $\{L_j(\chi,\varphi)\}_\varphi$, it follows that a firm's value added Y_j is a concave function of all labor inputs.

To show that $\frac{\partial^2 Y_j}{\partial L_j(\chi,\varphi)\partial L_j(\chi,\varphi')} < 0$, fix some $\chi \in \mathcal{X}$. For any skill levels φ and φ', I write:

$$\begin{aligned}
\frac{\partial^2 Y_j}{\partial L_j(\chi,\varphi)\partial L_j(\chi,\varphi')} &= \varphi T_j(1-\alpha_j)\theta_{j\chi}\Bigg[\varphi'(\rho_j-1)\left(L_j^{\text{eff}}(\chi)\right)^{\rho_j-2}\left(\sum_{\chi\in\mathcal{X}}\theta_{j\chi}\left(L_j^{\text{eff}}(\chi)\right)^{\rho_j}\right)^{\frac{1-\alpha_j}{\rho_j}-1} \\
&\qquad + \varphi'(1-\alpha_j-\rho_j)\theta_{j\chi}\left(L_j^{\text{eff}}(\chi)\right)^{2\rho_j-2}\left(\sum_{\chi\in\mathcal{X}}\theta_{j\chi}\left(L_j^{\text{eff}}(\chi)\right)^{\rho_j}\right)^{\frac{1-\alpha_j}{\rho_j}-2}\Bigg] \\
&= \varphi'\left(L_j^{\text{eff}}(\chi)\right)^{-1}\frac{\partial Y_j}{\partial L_j(\chi,\varphi)}\left[(\rho_j-1)+(1-\alpha_j-\rho_j)\frac{\theta_{j\chi}\left(L_j^{\text{eff}}(\chi)\right)^{\rho_j}}{\sum_{\chi'\in\mathcal{X}}\theta_{j\chi'}\left(L_j^{\text{eff}}(\chi')\right)^{\rho_j}}\right].
\end{aligned}$$

With positive employment, $\varphi'[L_j^{\text{eff}}(\chi)]^{-1}\frac{\partial Y_j}{\partial L_j(\chi,\varphi)} > 0$. So, $\frac{\partial^2 Y_j}{\partial L_j(\chi,\varphi)\partial L_j(\chi,\varphi')} < 0$ if and only if:

$$\begin{aligned}
0 &> (\rho_j-1)+(1-\alpha_j-\rho_j)\frac{\partial \log N_j}{\partial \log L_j^{\text{eff}}(\chi)} \\
&= (\rho_j-1)\left[1-\frac{\partial \log N_j}{\partial \log L_j^{\text{eff}}(\chi)}\right]-\alpha_j\frac{\partial \log N_j}{\partial \log L_j^{\text{eff}}(\chi)} > (\rho_j-1)-\alpha_j\frac{\partial \log N_j}{\partial \log L_j^{\text{eff}}(\chi)},
\end{aligned}$$

where the final inequality follows because $\partial \log N_j/\partial \log L_j^{\text{eff}}(\chi)$ is bounded between 0 and 1, and $\rho_j < 1$ by assumption. Re-arranging terms, this inequality may be re-written as follows:

$$\rho_j < 1+\alpha_j\frac{\partial \log N_j}{\partial \log L_j^{\text{eff}}(\chi)}.$$

Since this holds trivially when $\alpha_j \in (0,1)$ and $\rho_j < 1$, I conclude that $\frac{\partial^2 Y_j}{\partial L_j(\chi,\varphi)\partial L_j(\chi,\varphi')} < 0$.[36]

Finally, fix $\chi,\chi' \in \mathcal{X}$, where $\chi \neq \chi'$, and $\varphi,\varphi' \in \mathbb{R}$. The derivative $\frac{\partial^2 Y_j}{\partial L_j(\chi,\varphi)\partial L_j(\chi',\varphi')}$ is:

$$\frac{\partial^2 Y_j}{\partial L_j(\chi,\varphi)\partial L_j(\chi',\varphi')} = (1-\alpha_j-\rho_j)\underbrace{\varphi\varphi' T_j(1-\alpha_j)\theta_{j\chi}\theta_{j\chi'}\left(L_j^{\text{eff}}(\chi)L_j^{\text{eff}}(\chi')\right)^{\rho_j-1}\left(\sum_{\chi\in\mathcal{X}}\theta_{j\chi}\left(L_j^{\text{eff}}(\chi)\right)^{\rho_j}\right)^{\frac{1-\alpha_j}{\rho_j}-2}}_{(*)}.$$

Observe that the term $(*)$ is strictly positive. Thus, the sign of the derivative is pinned down by $1-\alpha_j-\rho_j$. It is positive whenever $\rho_j < 1-\alpha_j$, and it is negative whenever $\rho_j > 1-\alpha_j$. □

[36] Note that values of ρ_j above one can also satisfy this inequality, particularly if the returns to scale are small.

A.4. Firm Profit

After plugging in the labor supply constraint, the profit function Π_j for a firm j is defined by:

$$\Pi_j = Y_j - \int W_j(X) L_j(X) dX.$$

In equilibrium, wages are determined by setting the derivatives $\partial \Pi_j / \partial W_j(X)$ to zero, where:

$$\frac{\partial \Pi_j}{\partial W_j(X)} = \left(\frac{\partial Y_j}{\partial L_j(X)} - W_j(X) \right) \frac{\partial L_j(X)}{\partial W_j(X)} - L_j(X).$$

To understand when profit is a concave function of wages, I evaluate the Hessian matrix of Π_j.

Property A.3. The Hessian of the profit function Π_j is given by $\mathrm{H}_{\Pi_j} = \mathbf{A}'(\mathrm{H}_{Y_j})\mathbf{A} + \mathbf{B}$, where $\mathrm{H}_{Y_j} = [\frac{\partial^2 Y_j}{\partial L_j(X) \partial L_j(X')}]_{X,X'}$ is the Hessian of Y_j, and $\mathbf{A}$ and $\mathbf{B}$ are diagonal matrices with entries:

$$\mathbf{A}[X,X] = \frac{\partial L_j(X)}{\partial W_j(X)} \quad \text{and} \quad \mathbf{B}[X,X] = \frac{L_j(X)}{W_j(X)} \left(\frac{\partial^2 \log L_j(X)}{\partial \left[\log W_j(X)\right]^2} \boldsymbol{\varepsilon}_j^{-1}(X) - (\boldsymbol{\varepsilon}_j(X) + 1) \right).$$

Proof. The diagonal entries of the matrix H_{Π_j}, denoted by $\mathrm{H}_{\Pi_j}[X,X] = \frac{\partial^2 \Pi_j}{\partial W_j^2(X)}$, are given by:

$$\begin{aligned}
\frac{\partial^2 \Pi_j}{\partial W_j^2(X)} &= \frac{\partial^2 Y_j}{\partial L_j^2(X)} \times \left(\frac{\partial L_j(X)}{\partial W_j(X)} \right)^2 + \frac{\partial Y_j}{\partial L_j(X)} \times \frac{\partial^2 L_j(X)}{\partial W_j^2(X)} - \frac{\partial L_j(X)}{\partial W_j(X)} - \frac{\partial L_j(X)}{\partial W_j(X)} - W_j(X) \frac{\partial^2 L_j(X)}{\partial W_j^2(X)} \\
&= \frac{\partial^2 Y_j}{\partial L_j^2(X)} \times \left(\frac{\partial L_j(X)}{\partial W_j(X)} \right)^2 + \left(\frac{\partial Y_j}{\partial L_j(X)} - W_j(X) \right) \frac{\partial^2 L_j(X)}{\partial W_j^2(X)} - 2 \frac{\partial L_j(X)}{\partial W_j(X)}.
\end{aligned}$$

If the first-order condition binds, then $\frac{\partial Y_j}{\partial L_j(X)} - W_j(X) = W_j(X) / \boldsymbol{\varepsilon}_j(X)$, which implies that:

$$\begin{aligned}
\frac{\partial^2 \Pi_j}{\partial W_j^2(X)} &= \frac{\partial^2 Y_j}{\partial L_j^2(X)} \times \left(\frac{\partial L_j(X)}{\partial W_j(X)} \right)^2 + \frac{W_j(X)}{\boldsymbol{\varepsilon}_j(X)} \times \frac{\partial^2 L_j(X)}{\partial W_j^2(X)} - 2 \frac{\partial L_j(X)}{\partial W_j(X)} \\
&= \frac{\partial^2 Y_j}{\partial L_j^2(X)} \times \left(\frac{\partial L_j(X)}{\partial W_j(X)} \right)^2 + \frac{L_j(X)}{W_j(X)} \left(\frac{W_j^2(X)}{\boldsymbol{\varepsilon}_j(X) L_j(X)} \times \frac{\partial^2 L_j(X)}{\partial W_j^2(X)} - 2\, \boldsymbol{\varepsilon}_j(X) \right) \\
&= \frac{\partial^2 Y_j}{\partial L_j^2(X)} \times \left(\frac{\partial L_j(X)}{\partial W_j(X)} \right)^2 + \frac{L_j(X)}{W_j(X)} \left(\frac{\partial^2 \log L_j(X)}{\partial \left[\log W_j(X)\right]^2} \boldsymbol{\varepsilon}_j^{-1}(X) - (\boldsymbol{\varepsilon}_j(X) + 1) \right),
\end{aligned}$$

where the final equality relies on the observation that $\partial^2 \log L_j(X) / \partial [\log W_j(X)]^2$ equals:

$$\begin{aligned}
\frac{\partial^2 \log L_j(X)}{\partial \left[\log W_j(X)\right]^2} &= \frac{W_j^2(X)}{L_j(X)} \times \frac{\partial^2 L_j(X)}{\partial W_j^2(X)} - \left(\frac{\partial L_j(X)}{\partial W_j(X)} \right)^2 \times \frac{W_j^2(X)}{L_j^2(X)} + \frac{\partial L_j(X)}{\partial W_j(X)} \times \frac{W_j(X)}{L_j(X)} \\
&= \frac{W_j^2(X)}{L_j(X)} \times \frac{\partial^2 L_j(X)}{\partial W_j^2(X)} - \boldsymbol{\varepsilon}_j^2(X) + \boldsymbol{\varepsilon}_j(X).
\end{aligned}$$

The off-diagonal entries of H_{Π_j}, denoted by $\mathrm{H}_{\Pi_j}[X, X'] = \frac{\partial^2 \Pi_j}{\partial W_j(X) \partial W_j(X')}$ for $X \neq X'$, equal:

$$\frac{\partial^2 \Pi_j}{\partial W_j(X) \partial W_j(X')} = \frac{\partial^2 Y_j}{\partial L_j(X) \partial L_j(X')} \times \frac{\partial L_j(X)}{\partial W_j(X)} \times \frac{\partial L_j(X')}{\partial W_j(X')}.$$

Taken together, I can write $\mathrm{H}_{\Pi_j} = \mathbf{A}'(\mathrm{H}_{Y_j})\mathbf{A} + \mathbf{B}$, where H_{Y_j}, $\mathbf{A}$, and $\mathbf{B}$ are all defined above. □

I now highlight two notable special cases where Π_j is a concave function of wages $\{W_j(X)\}_X$.

Example (Homogeneous β). Suppose β is constant within each skill group: $\mathrm{Var}(\beta_i | X_i = X) = 0$. Then by Property A.1, $\partial^2 \log L_j(X) / \partial [\log W_j(X)]^2 = \mathrm{Var}(\beta_i | X_i = X, j(i) = j) = 0$. In this case, $\mathbf{B}$ is a negative definite matrix with entries $\mathbf{B}[X, X] = -[L_j(X)/W_j(X)] \times [\boldsymbol{\varepsilon}_j(X) + 1]$. Since H_{Y_j} is negative semidefinite by Property A.2, H_{Π_j} is also negative semidefinite, because:

$$\begin{aligned}
v' \mathrm{H}_{\Pi_j} v &= v' \big(\mathbf{A}'(\mathrm{H}_{Y_j})\mathbf{A} + \mathbf{B} \big) v \\
&= v' \mathbf{A}'(\mathrm{H}_{Y_j})\mathbf{A} v + v' \mathbf{B} v \\
&= (\mathbf{A} v)'(\mathrm{H}_{Y_j})(\mathbf{A} v) + v' \mathbf{B} v \leq 0,
\end{aligned}$$

for any vector v. Therefore, the firm's profit function Π_j must be concave in wages $\{W_j(X)\}_X$.

Example (Firm is a Price-taker). Suppose the firm does not exercise wage setting power, so it sets $W_j(X) = \frac{\partial Y_j}{\partial L_j(X)}$ for all X. Then $\mathbf{B}$ is negative definite with entries $\mathbf{B}[X, X] = -2 \frac{\partial L_j(X)}{\partial W_j(X)}$. As in the previous case, H_{Π_j} is negative semidefinite, and the firm's profit function is concave.

A.5. Worker Rents

Worker rents R_i^w are defined so that $u_{ij}\big(W_j(X_i) - R_i^w, a_j(X_i)\big) = \max_{k \neq j(i)} u_{ik}\big(W_k(X_i), a_k(X_i)\big)$. As a share of a worker's earnings, these rents can be expressed as a strictly increasing function of the utility difference between a worker i's chosen firm $j(i)$ and the next-best alternative:

$$\frac{R_i^w}{W_{j(i)}(X_i)} = 1 - \exp\left(- \frac{u_{ij(i)}\big(W_{j(i)}(X_i), a_{j(i)}(X_i)\big) - \max_{k \neq j(i)} u_{ik}\big(W_k(X_i), a_k(X_i)\big)}{\beta} \right).$$

I now derive the distribution of these rent shares among workers with skills $X_i = X$ at firm j. I show that, if β is homogeneous, then these shares are independent of a worker's skills and firm.

Property A.4. For workers with skills X at firm j, the density of $\omega_i = R_i^w / W_{j(i)}(X_i)$ equals:

$$f_{\omega|X,j}(\omega | X_i = X, j(i) = j) = \mathrm{E}\Big(\beta_i (1 - \omega)^{\beta_i - 1} \;\Big|\; X_i = X, j(i) = j \Big).$$

Whenever β is homogeneous, ω_i will be independent of both a worker's skills X_i and firm $j(i)$.

Proof. Let $W_j(X)$ be the wage that a firm j pays workers with skills X. For any $W \leq W_j(X)$, the density of these workers who would be willing to accept their current firm at wage W is:

$$\frac{\partial L_j(X,W)}{\partial W} \times \frac{1}{L_j(X, W_j(X))}.$$

By the change-of-variables formula, the conditional density of ω given $(X_i = X, j(i) = j)$ is:

$$\begin{aligned}
f_{\omega|X,j}(\omega|X_i = X, j(i) = j) &= -\frac{\partial L_j(X,(1-\omega)W_j(X))}{\partial \omega} \times \frac{1}{L_j(X, W_j(X))} \\
&= -\frac{\partial}{\partial \omega}\left(\int \frac{\exp\big(\beta \log\big((1-\omega)W_j(X)\big) + a_j(X)\big) f_{\beta|X}(\beta|X) f_X(X)}{I(\beta, X)} d\beta\right) \times \frac{1}{L_j(X, W_j(X))} \\
&= \left(\int \beta(1-\omega)^{\beta-1} \frac{\exp\big(\beta \log W_j(X) + a_j(X)\big) f_{\beta|X}(\beta|X) f_X(X)}{I(\beta, X)} d\beta\right) \times \frac{1}{L_j(X, W_j(X))} \\
&= \frac{\mathrm{E}\Big(\beta(1-\omega)^{\beta-1} \times \exp\big(\beta \log W_j(X) + a_j(X)\big)/I(\beta, X) \;\Big|\; X\Big)}{\mathrm{E}\Big(\exp\big(\beta \log W_j(X) + a_j(X)\big)/I(\beta, X) \;\Big|\; X\Big)} \\
&= \mathrm{E}\Big(\beta_i(1-\omega)^{\beta_i-1} \;\Big|\; X_i = X, j(i) = j\Big).
\end{aligned}$$

If β is homogeneous, then this density simplifies to $f_{\omega|X,j}(\omega|X_i = X, j(i) = j) = \beta(1-\omega)^{\beta-1}$. Since this function does not depend on a worker's skills $X_i = X$ or firm $j(i) = j$, I can write:

$$f_{\omega|X,j}(\omega|X_i = X, j(i) = j) = f_\omega(\omega), \quad \text{for all } X \text{ and } j.$$

□

When β is heterogeneous, the share of wages that workers earn as rents systematically varies across firms and skill groups. To characterize this variation, I compute the average rent share, $\mathrm{E}(R_i^w/W_{j(i)}(X_i)|X_i = X, j(i) = j)$, for workers with skills X at firm j. I also show that (fixing skills) workers who sort into higher paying firms receive a smaller share of their wage as rents.

Property A.5. Among workers with skills X employed at firm j, the expected rent share is:

$$\mathrm{E}\left(\frac{R_i^w}{W_{j(i)}(X_i)} \;\Big|\; X_i = X, j(i) = j\right) = \mathrm{E}\left(\frac{1}{1+\beta_i} \;\Big|\; X_i = X, j(i) = j\right).$$

If $\mathrm{Var}(\beta_i|X_i = X) > 0$, then $\mathrm{E}(R_i^w/W_{j(i)}(X_i)|X_i = X, j(i) = j)$ is strictly decreasing in $W_j(X)$.

Proof. Average worker rents $\mathrm{E}(R_i^w|X, j(i) = j)$ are derived by integrating $W_j(X) - W$ with respect to the density of workers who would be willing to accept their current job at wage W.

$$\begin{aligned}
\mathrm{E}(R_i^w|X, j(i) = j) &= \int_0^{W_j(X)} \big(W_j(X) - W\big)\left(\frac{\partial L_j(X,W)}{\partial W} \times \frac{1}{L_j(X, W_j(X))}\right) dW \\
&= -\frac{W_j(X)}{L_j(X, W_j(X))} \times \int_0^1 \omega\left(\frac{\partial}{\partial \omega}\int \frac{\exp\big(\beta \log\big((1-\omega)W_j(X)\big) + a_j(X)\big) f_{\beta|X}(\beta|X) f_X(X)}{I(\beta, X)} d\beta\right) d\omega \\
&= \frac{W_j(X)}{L_j(X, W_j(X))} \times \int_0^1 \omega\left(\int \frac{\beta(1-\omega)^{\beta-1} \exp\big(\beta \log W_j(X) + a_j(X)\big) f_{\beta|X}(\beta|X) f_X(X)}{I(\beta, X)} d\beta\right) d\omega.
\end{aligned}$$

The final equality relies on the assumption that firms view themselves to be strategically small in the market. Changing the order of integration, the average worker rents can be re-written as:

$$\begin{aligned}
\mathrm{E}(R_i^w|X, j(i)=j) &= \frac{W_j(X)}{L_j(X, W_j(X))} \times \int \frac{1}{I(\beta, X)} \exp\big(\beta \log W_j(X) + a_j(X)\big) f_{\beta|X}(\beta|X) f_X(X) \left(\int_0^1 \omega\beta(1-\omega)^{\beta-1} d\omega\right) d\beta \\
&= \frac{W_j(X)}{L_j(X, W_j(X))} \times \int \frac{1}{I(\beta, X)} \left(\frac{1}{1+\beta}\right) \exp\big(\beta \log W_j(X) + a_j(X)\big) f_{\beta|X}(\beta|X) f_X(X) d\beta \\
&= W_j(X) \times \frac{\mathrm{E}\Big(\frac{1}{1+\beta} \times \exp\big(\beta \log W_j(X) + a_j(X)\big)/I(\beta, X) \mid X\Big)}{\mathrm{E}\big(\exp\big(\beta \log W_j(X) + a_j(X)\big)/I(\beta, X) \mid X\big)} \\
&= W_j(X) \times \mathrm{E}\left(\frac{1}{1+\beta_i} \;\middle|\; X_i = X, j(i) = j\right).
\end{aligned}$$

To see how $\mathrm{E}(R_i^w/W_{j(i)}(X_i)|X_i = X, j(i) = j)$ depends on $W_j(X)$, I evaluate the elasticity:

$$\frac{\partial\left(\frac{\mathrm{E}(R_i^w|j(i)=j, X_i=X)}{W_j(X)}\right)}{\partial \log W_j(X)} = \mathrm{Cov}\left(\beta_i, \frac{1}{1+\beta_i} \;\middle|\; j(i) = j, X_i = X\right) \leq 0.$$

As long as $\mathrm{Var}(\beta_i|X_i = X) > 0$, the covariance term is strictly negative. Hence, fixing skills, workers at higher paying firms tend to receive a smaller share of their wage in the form of rents. □

A.6. Employer Rents

Employer rents come in the form of excess profits that firms obtain by exploiting their wage-setting power. To calculate these rents, I consider a counterfactual setting where firms act as price-takers in the market, facing perfectly-elastic labor supply curves. I then define the rents at firm j to be the difference between the realized and counterfactual profits $R_j^e = \Pi_j - \Pi_j^{\mathrm{pt}}$.

Property A.6. Let Y_j^{pt} be a firm j's value added when it is a price taker. The firm's rents are:

$$R_j^e = Y_j \times \left[\sum_{\chi \in \mathcal{X}} \int \left(\frac{1 + \alpha_j\, \boldsymbol{\varepsilon}_j(\chi, \varphi)}{1 + \boldsymbol{\varepsilon}_j(\chi, \varphi)}\right) \frac{\partial \log N_j}{\partial \log L_j(\chi, \varphi)} d\varphi - \alpha_j \left(\frac{Y_j^{\mathrm{pt}}}{Y_j}\right)\right].$$

If β is homogeneous, then all firms receive the same rents as a share of their profits, provided that they have the same returns to scale, α_j, and that labor is perfectly substitutable, $\rho_j = 1$.

Proof. For any firm j, the profit Π_j that the firm obtains in the monopsonistic equilibrium is:

$$\begin{aligned}
\Pi_j &= Y_j - \sum_{\chi \in \mathcal{X}} \left(\int \left(\frac{\boldsymbol{\varepsilon}_j(\chi, \varphi)}{1 + \boldsymbol{\varepsilon}_j(\chi, \varphi)}\right) \left(\frac{\partial Y_j}{\partial L_j(\chi, \varphi)}\right) L_j(\chi, \varphi) d\varphi\right) \\
&= Y_j - \sum_{\chi \in \mathcal{X}} \left(\int \left(\frac{\boldsymbol{\varepsilon}_j(\chi, \varphi)}{1 + \boldsymbol{\varepsilon}_j(\chi, \varphi)}\right) \varphi T_j (1-\alpha_j) \theta_{j\chi} \left(L_j^{\mathrm{eff}}(\chi)\right)^{\rho_j - 1} \left(\sum_{\chi' \in \mathcal{X}} \theta_{j\chi'} \left(L_j^{\mathrm{eff}}(\chi')\right)^{\rho_j}\right)^{\frac{1-\alpha_j}{\rho_j} - 1} L_j(\chi, \varphi) d\varphi\right).
\end{aligned}$$

Simplifying terms, I derive the following expression for a monopsonistic firm's realized profit:

$$
\begin{aligned}
\Pi_j &= Y_j \times \left(1 - (1-\alpha_j) \sum_{\chi \in \mathcal{X}} \left(\frac{\theta_{j\chi} \left(L_j^{\text{eff}}(\chi)\right)^{\rho_j}}{\sum_{\chi' \in \mathcal{X}} \theta_{j\chi'} \left(L_j^{\text{eff}}(\chi')\right)^{\rho_j}} \right) \left(\int \left(\frac{\boldsymbol{\varepsilon}_j(\chi,\varphi)}{1+\boldsymbol{\varepsilon}_j(\chi,\varphi)} \right) \left(\frac{\varphi L_j(\chi,\varphi)}{\int \varphi' L_j(\chi,\varphi') d\varphi'} \right) d\varphi \right) \right). \\
&= Y_j \times \left(1 - (1-\alpha_j) \sum_{\chi \in \mathcal{X}} \int \left(\frac{\boldsymbol{\varepsilon}_j(\chi,\varphi)}{1+\boldsymbol{\varepsilon}_j(\chi,\varphi)} \right) \frac{\partial \log N_j}{\partial \log L_j(\chi,\varphi)} d\varphi \right) \\
&= Y_j \times \sum_{\chi \in \mathcal{X}} \int \left(\frac{1+\alpha_j \, \boldsymbol{\varepsilon}_j(\chi,\varphi)}{1+\boldsymbol{\varepsilon}_j(\chi,\varphi)} \right) \frac{\partial \log N_j}{\partial \log L_j(\chi,\varphi)} d\varphi.
\end{aligned}
$$

Next, if the firm is a price-taker in the labor market, then its profit Π_j^{pt} is defined as follows:

$$
\Pi_j^{\text{pt}} = \max_{\{D_j^{\text{pt}}(\chi,\varphi)\}_{\chi,\varphi}} T_j \left(\sum_{\chi \in \mathcal{X}} \theta_{j\chi} \left(\int \varphi D_j^{\text{pt}}(\chi,\varphi) d\varphi \right)^{\rho_j} \right)^{\frac{1-\alpha_j}{\rho_j}} - \sum_{\chi \in \mathcal{X}} \left(\int W_j^{\text{pt}}(\chi,\varphi) D_j^{\text{pt}}(\chi,\varphi) d\varphi \right).
$$

Taking first-order conditions with respect to labor demand yields the following wage equation:

$$
W_j^{\text{pt}}(\chi,\varphi) = \varphi T_j (1-\alpha_j) \theta_{j\chi} \left(\int \varphi D_j^{\text{pt}}(\chi,\varphi) d\varphi \right)^{\rho_j - 1} \left(\sum_{\chi \in \mathcal{X}} \theta_{j\chi} \left(\int \varphi D_j^{\text{pt}}(\chi,\varphi) d\varphi \right)^{\rho_j} \right)^{\frac{1-\alpha_j}{\rho_j} - 1}.
$$

In equilibrium, the firm's labor demand equals its labor supply evaluated at the wage $W_j^{\text{pt}}(X)$. Let $L_j^{\text{pt}}(X)$ be the market-clearing quantity of labor, so that $D_j^{\text{pt}}(X) = S_j\big(X, W_j^{\text{pt}}(X)\big)$. I write:

$$
\begin{aligned}
\Pi_j^{\text{pt}} &= T_j \left(\sum_{\chi \in \mathcal{X}} \theta_{j\chi} \left(\int \varphi L_j^{\text{pt}}(\chi,\varphi) d\varphi \right)^{\rho_j} \right)^{\frac{1-\alpha_j}{\rho_j}} \times \left(1 - (1-\alpha_j) \sum_{\chi \in \mathcal{X}} \int \frac{\partial \log N_j}{\partial \log L_j(\chi,\varphi)} d\varphi \right) \\
&= \alpha_j T_j \left(\sum_{\chi \in \mathcal{X}} \theta_{j\chi} \left(\int \varphi L_j^{\text{pt}}(\chi,\varphi) d\varphi \right)^{\rho_j} \right)^{\frac{1-\alpha_j}{\rho_j}} \\
&= \alpha_j Y_j^{\text{pt}},
\end{aligned}
$$

where the second equality above follows because the elasticities $\frac{\partial \log N_j}{\partial \log L_j(\chi,\varphi)}$ aggregate to one. Given these expressions for the firm's realized and counterfactual profits, Π_j and Π_j^{pt}, I obtain:

$$
R_j^e = Y_j \times \left[\sum_{\chi \in \mathcal{X}} \int \left(\frac{1+\alpha_j \, \boldsymbol{\varepsilon}_j(\chi,\varphi)}{1+\boldsymbol{\varepsilon}_j(\chi,\varphi)} \right) \frac{\partial \log N_j}{\partial \log L_j(\chi,\varphi)} d\varphi - \alpha_j \left(\frac{Y_j^{\text{pt}}}{Y_j} \right) \right].
$$

If β is homogeneous across workers, then the labor supply elasticity $\boldsymbol{\varepsilon}_j(X)$ is constant across all firms j and skill groups X. In this case, a firm j's rents as a share of its profits are given by:

$$\frac{R_j^e}{\Pi_j} = 1 - \frac{\alpha_j(1+\boldsymbol{\varepsilon})}{1+\alpha_j\,\boldsymbol{\varepsilon}} \times \frac{Y_j^{\mathrm{pt}}}{Y_j} = 1 - \frac{\alpha_j(1+\boldsymbol{\varepsilon})}{1+\alpha_j\,\boldsymbol{\varepsilon}} \times \left(\frac{T_j\left(\sum_{\chi\in\mathcal{X}} \theta_{j\chi}\left(\int \varphi L_j^{\mathrm{pt}}(\chi,\varphi)d\varphi\right)^{\rho_j}\right)^{\frac{1-\alpha_j}{\rho_j}}}{T_j\left(\sum_{\chi\in\mathcal{X}} \theta_{j\chi}\left(\int \varphi L_j(\chi,\varphi)d\varphi\right)^{\rho_j}\right)^{\frac{1-\alpha_j}{\rho_j}}} \right)$$

$$= 1 - \frac{\alpha_j(1+\boldsymbol{\varepsilon})}{1+\alpha_j\,\boldsymbol{\varepsilon}} \times \left(\frac{\sum_{\chi\in\mathcal{X}} \theta_{j\chi}\left(\int \varphi\left(\frac{W_j^{\mathrm{pt}}(\chi,\varphi)}{W_j(\chi,\varphi)}\right)^{\beta} L_j(\chi,\varphi)d\varphi\right)^{\rho_j}}{\sum_{\chi\in\mathcal{X}} \theta_{j\chi}\left(\int \varphi L_j(\chi,\varphi)d\varphi\right)^{\rho_j}} \right)^{\frac{1-\alpha_j}{\rho_j}}.$$

If labor is perfectly substitutable, so that $\rho_j = 1$, then the wage ratio $W_j^{\mathrm{pt}}(\chi,\varphi)/W_j(\chi,\varphi)$ is:

$$\frac{W_j^{\mathrm{pt}}(\chi,\varphi)}{W_j(\chi,\varphi)} = \frac{1+\boldsymbol{\varepsilon}}{\boldsymbol{\varepsilon}} \left(\frac{\sum_{\chi\in\mathcal{X}} \theta_{j\chi} \int \varphi L_j^{\mathrm{pt}}(\chi,\varphi)d\varphi}{\sum_{\chi\in\mathcal{X}} \theta_{j\chi} \int \varphi L_j(\chi,\varphi)d\varphi} \right)^{-\alpha_j}$$

$$= \frac{1+\boldsymbol{\varepsilon}}{\boldsymbol{\varepsilon}} \left(\frac{\sum_{\chi\in\mathcal{X}} \theta_{j\chi} \int \varphi\left(\frac{W_j^{\mathrm{pt}}(\chi,\varphi)}{W_j(\chi,\varphi)}\right)^{\beta} L_j(\chi,\varphi)d\varphi}{\sum_{\chi\in\mathcal{X}} \theta_{j\chi} \int \varphi L_j(\chi,\varphi)d\varphi} \right)^{-\alpha_j}$$

$$= \frac{1+\boldsymbol{\varepsilon}}{\boldsymbol{\varepsilon}} \left(\frac{W_j^{\mathrm{pt}}(\chi,\varphi)}{W_j(\chi,\varphi)}\right)^{\beta} \left(\frac{\sum_{\chi\in\mathcal{X}} \theta_{j\chi} \int \varphi L_j(\chi,\varphi)d\varphi}{\sum_{\chi\in\mathcal{X}} \theta_{j\chi} \int \varphi L_j(\chi,\varphi)d\varphi} \right)^{-\alpha_j} = \frac{1+\boldsymbol{\varepsilon}}{\boldsymbol{\varepsilon}} \left(\frac{W_j^{\mathrm{pt}}(\chi,\varphi)}{W_j(\chi,\varphi)}\right)^{\beta} = \left(\frac{1+\boldsymbol{\varepsilon}}{\boldsymbol{\varepsilon}}\right)^{\frac{1}{1-\beta}},$$

where the third equality holds since $W_j^{\mathrm{pt}}(\chi,\varphi)/W_j(\chi,\varphi)$ is constant within each firm j. Substituting this expression, I can write R_j^e/Π_j as a deterministic function of the returns to scale α_j.

$$\frac{R_j^e}{\Pi_j} = 1 - \frac{\alpha_j(1+\boldsymbol{\varepsilon})}{1+\alpha_j\,\boldsymbol{\varepsilon}} \times \left(\frac{\sum_{\chi\in\mathcal{X}} \theta_{j\chi} \int \varphi\left(\frac{1+\boldsymbol{\varepsilon}}{\boldsymbol{\varepsilon}}\right)^{\frac{1}{1-\beta}} L_j(\beta,\chi)d\varphi}{\sum_{\chi\in\mathcal{X}} \theta_{j\chi} \int \varphi L_j(\chi,\varphi)d\varphi} \right)^{1-\alpha_j}$$

$$= 1 - \frac{\alpha_j(1+\boldsymbol{\varepsilon})}{1+\alpha_j\,\boldsymbol{\varepsilon}} \times \left(\frac{1+\boldsymbol{\varepsilon}}{\boldsymbol{\varepsilon}}\right)^{\frac{1-\alpha_j}{1-\beta}}.$$

□

A.7. *Pass-through of TFP Shocks to Wages*

In the next property, I derive an expression for the pass-through of a hypothetical shock to firm productivity (specifically, the TFP parameter T_j) on the wages $\{W_j(X)\}_X$ of different workers at the firm. I also show how the pass-through simplifies in a model where β is homogeneous.

Property A.7. The vector of elasticities of firm j's wages $\{W_j(X)\}_X$ with respect its TFP T_j is $\left[\frac{\partial \log W_j(X)}{\partial \log T_j}\right]_X = (I - \boldsymbol{\Gamma}_j - \mathbf{K}_j)^{-1}\mathbf{1}$, where $\boldsymbol{\Gamma}_j$ and $\mathbf{K}_j$ are both matrices in $\mathbb{R}^{|X|\times|X|}$, with entries:

$$\mathbf{\Gamma}_j[X, X'] = \mathbb{1}\{X = X'\}\frac{\frac{\partial^2 \log L_j(X)}{\partial[\log W_j(X)]^2}}{\boldsymbol{\varepsilon}_j(X)[1 + \boldsymbol{\varepsilon}_j(X)]}$$
$$\mathbf{K}_j[X, X'] = \left((1 - \alpha_j - \rho_j)\frac{\partial \log N_j}{\log L_j(\chi', \varphi')} - \mathbb{1}\{\chi = \chi'\}(1 - \rho_j)\frac{\partial \log L_j^{\text{eff}}(\chi')}{\partial \log L_j(\chi', \varphi')}\right)\boldsymbol{\varepsilon}_j(\chi', \varphi').$$

If β is homogeneous, then $\frac{\partial \log W_j(X)}{\partial \log T_j}$ is constant across all firms j and skill groups X, provided that firms have the same returns to scale, α_j, and that labor is perfectly substitutable, $\rho_j = 1$.

Proof. For any skill group X, the elasticity of $W_j(X)$ with respect to firm TFP is given by:

$$\begin{aligned}
\frac{\partial \log W_j(\chi, \varphi)}{\partial \log T_j} &= 1 + \frac{\partial\left(\frac{\boldsymbol{\varepsilon}_j(\chi,\varphi)}{1+\boldsymbol{\varepsilon}_j(\chi,\varphi)}\right)}{\partial \log T_j} - (1 - \rho_j)\frac{\partial \log L_j^{\text{eff}}(\chi)}{\partial \log T_j} + (1 - \alpha_j - \rho_j)\frac{\partial \log N_j}{\partial \log T_j} \\
&= 1 + \frac{\partial\left(\frac{\boldsymbol{\varepsilon}_j(\chi,\varphi)}{1+\boldsymbol{\varepsilon}_j(\chi,\varphi)}\right)}{\partial \log W_j(\chi, \varphi)}\frac{\partial \log W_j(\chi, \varphi)}{\partial \log T_j} - (1 - \rho_j)\int \frac{\partial \log L_j^{\text{eff}}(\chi)}{\partial \log L_j(\chi, \varphi')}\boldsymbol{\varepsilon}_j(\chi, \varphi')\frac{\partial \log W_j(\chi, \varphi')}{\partial \log T_j}d\varphi' \\
&\quad + (1 - \alpha_j - \rho_j)\sum_{\chi' \in X}\left(\int \frac{\partial \log N_j}{\partial \log L_j(\chi', \varphi')}\boldsymbol{\varepsilon}_j(\chi', \varphi')\frac{\partial \log W_j(\chi', \varphi')}{\partial \log T_j}d\varphi'\right) \\
&= 1 + \left(\frac{1}{\boldsymbol{\varepsilon}_j(\chi, \varphi)[1 + \boldsymbol{\varepsilon}_j(\chi, \varphi)]}\right)\frac{\partial^2 \log L_j(\chi, \varphi)}{\partial[\log W_j(\chi, \varphi)]^2}\frac{\partial \log W_j(\chi, \varphi)}{\partial \log T_j} \\
&\quad + \sum_{\chi' \in X}\left[(1 - \alpha_j - \rho_j)\frac{\partial \log N_j}{\log L_j^{\text{eff}}(\chi')} - (1 - \rho_j)\mathbb{1}\{\chi = \chi'\}\right]\int \frac{\partial \log L_j^{\text{eff}}(\chi')}{\partial \log L_j(\chi', \varphi')}\boldsymbol{\varepsilon}_j(\chi', \varphi')\frac{\partial \log W_j(\chi', \varphi')}{\partial \log T_j}d\varphi'.
\end{aligned}$$

So, in each firm j, the elasticities solve a linear system, which can be written in matrix form as:

$$\left[\frac{\partial \log W_j(X)}{\partial \log T_j}\right]_X = \mathbf{1} + \mathbf{\Gamma}_j\left[\frac{\partial \log W_j(X)}{\partial \log T_j}\right]_X + \mathbf{K}_j\left[\frac{\partial \log W_j(X)}{\partial \log T_j}\right]_X,$$

with $\mathbf{\Gamma}_j$ and $\mathbf{K}_j$ defined above. If $(I - \mathbf{\Gamma}_j - \mathbf{K}_j)^{-1}$ exists, then $\left[\frac{\partial \log W_j(X)}{\partial \log T_j}\right]_X = (I - \mathbf{\Gamma}_j - \mathbf{K}_j)^{-1}\mathbf{1}$.

If β is homogeneous across workers, the labor supply elasticity $\boldsymbol{\varepsilon}_j(X)$ is constant across firms j and skill groups X, and the second derivative $\partial^2 \log L_j(X)/\partial[\log W_j(X)]^2$ equals zero. So, if workers are perfectly substitutable in a firm's production function, so that $\rho_j = 1$, then:

$$\begin{aligned}
\frac{\partial \log W_j(\chi, \varphi)}{\partial \log T_j} &= 1 - \alpha_j\,\boldsymbol{\varepsilon}\sum_{\chi' \in X}\int \frac{\partial \log N_j}{\log L_j(\chi', \varphi')}\frac{\partial \log W_j(\chi', \varphi')}{\partial \log T_j}d\varphi' \\
&= 1 - \alpha_j\,\boldsymbol{\varepsilon}\,\frac{\partial \log W_j(\chi, \varphi)}{\partial \log T_j} = \frac{1}{1 - \alpha_j\,\boldsymbol{\varepsilon}}.
\end{aligned}$$

The pass-through is constant across firms and skills if firms share the same returns to scale.

□

A.8. *Allocative Inefficiency*

Social welfare is defined to be the sum of workers' utilities net of amenities and firms' profits:

$$\mathcal{W} = \mathrm{E}\left(\max_j \left\{u_{ij}\left(W_j(X_i), a_j(X_i)\right)\right\}\right) + \log \sum_{j=1}^{J} \Pi_j.$$

Using the formula for the expectation of a maximum over T1EV random variables, I write:

$$\begin{aligned}
\mathcal{W} &= \mathrm{E}\left(\max_j \left\{\beta_i \log W_j(X_i) + a_j(X_i) + \epsilon_{ij}\right\}\right) + \log \sum_{j=1}^{J} \Pi_j \\
&= \int \mathrm{E}\left[\max_j \left\{\beta_i \log W_j(X_i) + a_j(X_i) + \epsilon_{ij}\right\} \Big| \beta_i = \beta, X_i = X\right] f_{\beta,X}(\beta_i, X) d(\beta_i, X) + \log \sum_{j=1}^{J} \Pi_j \\
&= \int \left(\log \sum_{j=1}^{J} \exp\left[\beta \log W_j(X) + a_j(X)\right] + \gamma\right) f_{\beta,X}(\beta, X) d(\beta, X) + \log \sum_{j=1}^{J} \Pi_j \\
&= \int \log I(\beta, X) f_{\beta,X}(\beta, X) d(\beta, X) + \gamma + \log \sum_{j=1}^{J} \Pi_j,
\end{aligned}$$

where $\gamma \approx 0.5772$ is the Euler-Mascheroni constant.[37] The social planner seeks to maximize welfare by solving $\mathcal{W}^* = \max_{\{j(i)\}_i} \mathcal{W}$. The first order condition of the planner's problem is:

$$\frac{\partial \mathcal{W}}{\partial L_j(X)} = 0,$$

for all X and j. The following property characterizes when the optimal allocation is achieved.

Property A.8. The allocation of labor that solves the social planner's problem, denoted by $\{L_j^*(X)\}_{j,X}$, coincides with the equilibrium allocation in a competitive (Walrasian) economy.

Proof. The planner's solution is characterized by setting $\frac{\partial \mathcal{W}}{\partial L_j(X)}$ to zero for all X and j, where:

$$\frac{\partial \mathcal{W}}{\partial L_j(X)} = \frac{\partial \left[\int \log I(\beta, X) f_{\beta|X}(\beta|X) f_X(X) d\beta\right]}{\partial L_j(X)} + \frac{\partial \log \sum_{j=1}^{J} \Pi_j}{\partial L_j(X)}$$

[37] This property is proven in Small & Rosen (1981). Even without T1EV errors, expected maximal utility is:

$$\begin{aligned}
\mathrm{E}\left(\max_j \left\{u_{ij}\left(W_j(X_i), a_j(X_i)\right)\right\}\right) &= \sum_{j=1}^{J} \left(\int \mathrm{E}\left[\beta_i \log W_j(X_i) + a_j(X_i) + \epsilon_{ij} \Big| j(i) = j, X_i = X\right] L_j(X) dX\right) \\
&= \sum_{j=1}^{J} \left(\int \left(\boldsymbol{\varepsilon}_j(X) \log W_j(X) + a_j(X)\right) L_j(X) dX\right) + \mathrm{E}\left[\epsilon_{ij} \big| j(i) = j\right].
\end{aligned}$$

Taking derivatives and rearranging terms, this optimality condition can be written as follows:

$$\begin{aligned}
\frac{\partial \mathcal{W}}{\partial L_j(X)} &= \int \frac{\partial \log I(\beta, X))}{\partial L_j(X)} f_{\beta|X}(\beta|X) f_X(X) d\beta + \frac{\partial \Pi_j}{\partial L_j(X)} \left(\sum_{j=1}^{J} \Pi_j \right)^{-1} \\
&= \int \frac{\beta \exp\left(\beta \log W_j(X) + a_j(X)\right)}{I(\beta, X)} \left(\frac{\partial \log W_j(X)}{\partial L_j(X)} \right) f_{\beta|X}(\beta|X) f_X(X) d\beta + \frac{\partial \Pi_j}{\partial L_j(X)} \left(\sum_{j=1}^{J} \Pi_j \right)^{-1}. \\
&= \left(\frac{\partial \log W_j(X)}{\partial \log L_j(X)} \right) \mathrm{E}(\beta|X, j(i) = j) + \frac{\partial \Pi_j}{\partial L_j(X)} \left(\sum_{j=1}^{J} \Pi_j \right)^{-1} \\
&= \left(\frac{\partial \log W_j(X)}{\partial \log L_j(X)} \right) \mathrm{E}(\beta|X, j(i) = j) + \frac{\partial \left[Y_j - \int W_j(X') L_j(X') dX' \right]}{\partial L_j(X)} \left(\sum_{j=1}^{J} \Pi_j \right)^{-1} \\
&= \left(\frac{\partial \log W_j(X)}{\partial \log L_j(X)} \right) \mathrm{E}(\beta|X, j(i) = j) + \frac{\frac{\partial Y_j}{\partial L_j(X)} - W_j(X)}{\sum_{j=1}^{J} \Pi_j} - \frac{\int \frac{\partial \log W_j(X')}{\partial \log L_j(X)} W_j(X') dX'}{\sum_{j=1}^{J} \Pi_j}.
\end{aligned}$$

If all firms are price-takers, then they take wages as given, so: $\partial \log W_j(X')/\partial \log L_j(X) = 0$ for all $X, X' \in \mathcal{X} \times \mathbb{R}_+$. In this case, the solution to the social planner's problem reduces to:

$$\frac{\partial \mathcal{W}}{\partial L_j(X)} = 0 \quad \Longleftrightarrow \quad W_j(X) = \frac{\partial Y_j}{\partial L_j(X)},$$

for all X and j, which coincides with the equilibrium wage condition in a Walrasian economy.

□

Property A.8 implies that the first-best allocation is implemented in a competitive (Walrasian) economy. To compute the optimal welfare $\mathcal{W}^*$, I therefore consider a counterfactual setting where all firms are price takers in the labor market. In this setting, social welfare is given by:

$$\mathcal{W}^* = \int \log I^*(\beta, X) f_{\beta,X}(\beta, X) d(\beta, X) + \gamma + \log \sum_{j=1}^{J} \alpha_j Y_j^*.$$

I measure allocative inefficiency by comparing $\mathcal{W}^*$ to welfare in a monopsonistic equilibrium:

$$\mathcal{W} = \int \log I(\beta, X) f_{\beta,X}(\beta, X) d(\beta, X) + \gamma + \log \sum_{j=1}^{J} Y_j \left(\int \left(\frac{1 + \alpha_j \, \boldsymbol{\varepsilon}_j(X)}{1 + \boldsymbol{\varepsilon}_j(X)} \right) \omega_j(X) dX \right).$$

When estimating the welfare cost of monopsony, I express welfare in dollar terms. To do so, I compute utility in log-wage-equivalent units, dividing each worker's utility by her marginal utility of log earnings (β) so welfare changes are interpreted as proportional changes in wages. To implement the first-best optimal allocation, a planner can give wage-specific wage subsidies to workers, where the shape of the subsidy curve depends on the distribution of preferences.

B. Model Extensions

B.1. Micro-foundation for the Worker's Indirect Utility Function

I now present a simple micro-foundation for the indirect utility function of the worker, where each worker i chooses a firm j to maximize utility from consuming goods and leisure. Let C_{ij} denote the worker's expected consumption from working at the firm and let H_{ij} denote the worker's expected time spent working. A worker i's utility from choosing a firm j equals:

$$u_{ij} = f_i(C_{ij}, H_{ij}).$$

Let $\bar{H}_j(X_i)$ denote the scheduled work hours for employees with skills X_i at firm j, accounting for paid time off, overtime, vacation leave, and other benefits. Let $\tilde{H}_{ij}$ denote the idiosyncratic component of hours worked, accounting for commuting time and worker-firm match factors. Assume that $f_i : \mathbb{R}^2_+ \to \mathbb{R}$ is log-additive in consumption and scheduled hours, such that:

$$f_i(C_{ij}, H_{ij}) = a_i + \kappa_i \log C_{ij} - \eta \log \bar{H}_j(X_i) - \tilde{f}(\tilde{H}_{ij}).$$

The marginal rate of substitution between log consumption and log work hours equals $-\kappa_i/\eta$. Define the worker's budget constraint as $C_{ij} = W_j(X_i)$, where $W_j(X_i)$ denotes total earnings. Additionally, define $a_j(X) = -\log \bar{H}_j(X_i)$, $\beta_i = \kappa_i/\eta$, and $\epsilon_{ij} = -\eta^{-1}\tilde{f}(\tilde{H}_{ij})$. It follows that:

$$u_{ij} = \frac{a_i}{\eta} + \beta_i \log W_j(X_i) + a_j(X_i) + \epsilon_{ij},$$

which corresponds to utility specification (1) in the paper.[38] Note that this utility function can be extended to include non-labor income. For example, let V_i denote the worker's other forms of income. The budget constraint becomes $C_{ij} = W_j(X_i) + V_i$. If V_i is observed in data, then I can re-define earnings to be $\tilde{W}_{ij}(X_i) = W_j(X_i) + V_i$, and the same utility specification applies.

B.2. Capital and Monopolistic Competition in the Product Market

I now present an extension of the model that includes capital and monopolistic competition in the product market. Consider a monopolistic firm j with the following production function:

$$Q_j = T_j K_j^{\eta_j} N_j^{1-\alpha_j},$$

where K_j denotes capital. In a monopolistic product market, the revenue curve is $Y_j = Q_j^{1-\kappa_j}$. For each skill vector X, labor is hired according to the labor supply curve $L_j(X)$ and capital is rented at some fixed price r_j. The firm's profit function has the following representation:

$$\begin{aligned}\Pi_j &= Q_j^{1-\kappa_j} - \sum_{\chi \in \mathcal{X}} \left(\int W_j(\chi, \varphi) L_j(\chi, \varphi) d\varphi \right) - r_j K_j \\ &= \tilde{T}_j K_j^{\tilde{\eta}_j} N_j^{1-\tilde{\alpha}_j} - \sum_{\chi \in \mathcal{X}} \left(\int W_j(\chi, \varphi) L_j(\chi, \varphi) d\varphi \right) - r_j K_j,\end{aligned}$$

[38] The term $\frac{a_i}{\eta}$ does not vary across firms. Therefore, it does not impact the worker's employment decisions.

where $\tilde{T}_j = T_j^{1-\kappa_j}$, $\tilde{\eta}_j = \eta_j(1-\kappa_j)$, and $\tilde{\alpha}_j = \alpha_j + \kappa_j(1-\alpha_j)$. I now show that both perfect and monopolistic competition in the product market deliver the same profit function. As a first step, I derive the first order condition of the firm's problem with respect to capital K_j. I write:

$$K_j = \left(\frac{r_j}{\tilde{\eta}_j \tilde{T}_j N_j^{1-\tilde{\alpha}_j}}\right)^{\frac{1}{\tilde{\eta}_j - 1}}.$$

Plugging this condition into the firm's profit function, I obtain the following:

$$\begin{aligned}
\Pi_j &= \tilde{T}_j \left(\frac{r}{\tilde{\eta}_j \tilde{T}_j N_j^{1-\tilde{\alpha}_j}}\right)^{\frac{\tilde{\eta}_j}{\tilde{\eta}_j - 1}} N_j^{1-\tilde{\alpha}_j} - \sum_{\chi \in \mathcal{X}} \left(\int W_j(\chi,\varphi) L_j(\chi,\varphi) d\varphi \right) - r \left(\frac{r}{\tilde{\eta}_j \tilde{T}_j N_j^{1-\tilde{\alpha}_j}}\right)^{\frac{1}{\tilde{\eta}_j - 1}} \\
&= \left[\tilde{T}_j \left(\frac{r}{\tilde{\eta}_j \tilde{T}_j}\right)^{\frac{\tilde{\eta}_j}{\tilde{\eta}_j - 1}} - r \left(\frac{r}{\tilde{\eta}_j \tilde{T}_j}\right)^{\frac{1}{\tilde{\eta}_j - 1}} \right] N_j^{-\frac{1-\tilde{\alpha}_j}{\tilde{\eta}_j - 1}} - \sum_{\chi \in \mathcal{X}} \left(\int W_j(\chi,\varphi) L_j(\chi,\varphi) d\varphi \right).
\end{aligned}$$

Note that this is just a reinterpretation of the original problem, where the firm's profit equals:

$$\Pi_j = \hat{T}_j N_j^{1-\hat{\alpha}_j} - \sum_{\chi \in \mathcal{X}} \left(\int W_j(\chi,\varphi) L_j(\chi,\varphi) d\varphi \right), \quad \text{where:} \quad \begin{cases} \hat{T}_j &= \tilde{T}_j \left(\frac{r}{\tilde{\eta}_j \tilde{T}_j}\right)^{\frac{\tilde{\eta}_j}{\tilde{\eta}_j - 1}} - r \left(\frac{r}{\tilde{\eta}_j \tilde{T}_j}\right)^{\frac{1}{\tilde{\eta}_j - 1}}. \\ \hat{\alpha}_j &= \dfrac{\tilde{\eta}_j - \tilde{\alpha}_j}{\tilde{\eta}_j - 1}. \end{cases}$$

B.3. Properties of the Labor Supply Curve

To see how the results in equations (9) and (10) extend to general error structures, I establish two properties. First, in Property A.9, I show that a firm's labor-supply elasticity can be expressed as the average β-specific elasticity among the firm's incumbent workers. This property shows how workforce composition shapes the labor supply elasticities that firms internalize.

Property A.9. The own-wage elasticity of labor supply, $\frac{\partial \log S_j(X)}{\partial \log W_j(X)}$, takes the following form:

$$\frac{\partial \log S_j(X)}{\partial \log W_j(X)} = \mathrm{E}\left(\frac{\partial \log \mathrm{P}(j(i) = j|\beta, X)}{\partial \log W_j(X)} \,\middle|\, X, j(i) = j \right).$$

Proof. The derivative of a firm's labor supply $\log S_j(X)$ with respect to $\log W_j(X)$ is given by:

$$\begin{aligned}
\frac{\partial \log S_j(X)}{\partial \log W_j(X)} &= \frac{W_j(X)}{S_j(X)} \times \frac{\partial}{\partial W_j(X)} \int \mathrm{P}(j(i) = j|\beta, X) f_{\beta,X}(\beta, X) d\beta \\
&= S_j^{-1}(X) \int \frac{\partial \log \mathrm{P}(j(i) = j|\beta, X)}{\partial \log W_j(X)} \mathrm{P}(j(i) = j|\beta, X) f_{\beta,X}(\beta, X) d\beta \\
&= \frac{\mathrm{E}\left(\frac{\partial \log \mathrm{P}(j(i) = j|\beta, X)}{\partial \log W_j(X)} \mathrm{P}(j(i) = j|\beta, X) \middle| X \right)}{\mathrm{P}(j(i) = j|X)}.
\end{aligned}$$

By the Law of Iterated Expectations, this firm-specific elasticity can be expressed as follows:

$$\begin{aligned}
\frac{\partial \log S_j(X)}{\partial \log W_j(X)} &= \frac{\sum_{k=1}^{J} \mathrm{E}\left(\frac{\partial \log \mathrm{P}(j(i)=j|\beta,X)}{\partial \log W_j(X)} \mathrm{P}(j(i)=j|\beta,X)|X, j(i)=k\right) \times \mathrm{P}(j(i)=k|X)}{\mathrm{P}(j(i)=j|X)} \\
&= \frac{\mathrm{E}\left(\frac{\partial \log \mathrm{P}(j(i)=j|\beta,X)}{\partial \log W_j(X)}\Big|X, j(i)=j\right) \times \mathrm{P}(j(i)=j|X) + \sum_{k\neq j} 0 \times \mathrm{P}(j(i)=k|X)}{\mathrm{P}(j(i)=j|X)} \\
&= \mathrm{E}\left(\frac{\partial \log \mathrm{P}(j(i)=j|\beta,X)}{\partial \log W_j(X)}\Big|X, j(i)=j\right).
\end{aligned}$$

□

Next, in Property A.10, I show that the own-wage derivative of the firm's labor supply elasticity is equal to the sum of two terms: (1) the average second derivative of $\log \mathrm{P}(j(i)=j|\beta,X)$ with respect to $\log W_j(X)$ among the firm's incumbent workers, and (2) the variance of the β-specific elasticities among its incumbent workers. The second term is positive, and it captures sorting on individual heterogeneity: when a firm posts a higher wage, it attracts workers who are more responsive to wages (higher β), which leads it to face a larger labor supply elasticity.

Property A.10. The second derivative of $\log S_j(X)$ with respect to $\log W_j(X)$ takes the form:

$$\frac{\partial^2 \log S_j(X)}{\partial \left[\log W_j(X)\right]^2} = \mathrm{E}\left(\frac{\partial^2 \log \mathrm{P}(j(i)=j|\beta,X)}{\partial \left[\log W_j(X)\right]^2}\Big|X, j(i)=j\right) + \mathrm{Var}\left(\frac{\partial \log \mathrm{P}(j(i)=j|\beta,X)}{\partial \log W_j(X)}\Big|X, j(i)=j\right).$$

Proof. To simplify notation, let $\boldsymbol{\varepsilon}_j(X) = \frac{\partial \log S_j(X)}{\partial \log W_j(X)}$ and $\boldsymbol{\varepsilon}_j(\beta,X) = \frac{\partial \log \mathrm{P}(j(i)=j|\beta,X)}{\partial \log W_j(X)}$. I write:

$$\begin{aligned}
\frac{\partial^2 \log S_j(X)}{\partial \left[\log W_j(X)\right]^2} &= \frac{\partial}{\partial \log W_j(X)}\left[\frac{\int \boldsymbol{\varepsilon}_j(\beta,X)\,\mathrm{P}(j(i)=j|\beta,X) f_{\beta,X}(\beta|X)d\beta}{\mathrm{P}(j(i)=j|X)}\right] \\
&= \frac{\int \frac{\partial\left[\boldsymbol{\varepsilon}_j(\beta,X)\,\mathrm{P}(j(i)=j|\beta,X)\right]}{\partial \log W_j(X)} f_{\beta|x}(\beta|X)d\beta - \boldsymbol{\varepsilon}_j(X)\int \boldsymbol{\varepsilon}_j(\beta,X)\,\mathrm{P}(j(i)=j|\beta,X) f_{\beta|x}(\beta|X)d\beta}{\mathrm{P}(j(i)=j|X)} \\
&= \frac{\int \frac{\partial\left[\boldsymbol{\varepsilon}_j(\beta,X)\,\mathrm{P}(j(i)=j|\beta,X)\right]}{\partial \log W_j(X)} f_{\beta|x}(\beta|X)d\beta}{\mathrm{P}(j(i)=j|X)} - \boldsymbol{\varepsilon}_j^2(X) \\
&= \frac{\int \frac{\partial \boldsymbol{\varepsilon}_j(\beta,X)}{\partial \log W_j(X)}\,\mathrm{P}(j(i)=j|\beta,X) f_{\beta|x}(\beta|X)d\beta}{\mathrm{P}(j(i)=j|X)} + \frac{\int \boldsymbol{\varepsilon}_j^2(\beta,X)\,\mathrm{P}(j(i)=j|\beta,X) f_{\beta|x}(\beta|X)d\beta}{\mathrm{P}(j(i)=j|X)} - \boldsymbol{\varepsilon}_j^2(X) \\
&= \mathrm{E}\left(\frac{\partial \boldsymbol{\varepsilon}_j(\beta,X)}{\partial \log W_j(X)}\Big|X, j(i)=j\right) + \mathrm{E}\left(\boldsymbol{\varepsilon}_j^2(\beta,X)\Big|X, j(i)=j\right) - \mathrm{E}\left(\boldsymbol{\varepsilon}_j(\beta,X)\Big|X, j(i)=j\right)^2 \\
&= \mathrm{E}\left(\frac{\partial \boldsymbol{\varepsilon}_j(\beta,X)}{\partial \log W_j(X)}\Big|X, j(i)=j\right) + \mathrm{Var}\left(\boldsymbol{\varepsilon}_j(\beta,X)\Big|X, j(i)=j\right).
\end{aligned}$$

□

C. Proofs of Main Results

C.1. Equilibrium Properties of the Model

This appendix section establishes equilibrium properties relating to existence and uniqueness, which are necessary for the identification arguments. I start by proving equilibrium existence.

Lemma 1. There exists an equilibrium involving strictly positive wages and employment.

Proof. The taste shocks ϵ_{ij} have strictly positive density, so $\mathrm{P}(j(i) = j|\beta, X)$ is strictly positive for all β, X and j. Therefore, an equilibrium (if it exists) involves strictly positive employment.

I restrict attention to equilibria with non-negative wages, so that $W_j(X) \geq 0$ for all X and j. For a firm to be profitable, its wages cannot exceed the revenue that it earns per unit of labor. Since $\partial^2 Y_j/\partial L_j^2(X) < 0$ for all X and j, this means that there exists a strict upper bound on the wage vectors $\{W_j(\mathbf{X})\}_j$. Thus, the set of feasible wage vectors $\{W_j(\mathbf{X})\}_j$ is contained in a convex, compact subset of Euclidean space. Every equilibrium must lie within the interior of this subspace, since $\lim_{W_j(X)\to 0} \partial\Pi_j/\partial W_j(X) > 0$ and $\lim_{W_j(X)\to\infty} \partial\Pi_j/\partial W_j(X) < 0$ for all X and j, implying that firms can always raise their profit by deviating from a corner solution. Given these properties, I can restrict attention to wages that satisfy the first-order conditions:

$$W_j(X) = \frac{\boldsymbol{\varepsilon}_j(X)}{1 + \boldsymbol{\varepsilon}_j(X)} \times \frac{\partial Y_j}{\partial L_j(X)}, \quad \text{for all } X \text{ and } j.$$

The right-hand-side is a continuously differentiable function of $W_j(X)$ that is bounded within the set of feasible wages, since $0 < [\boldsymbol{\varepsilon}_j(X)/(1 + \boldsymbol{\varepsilon}_j(X))] \times \partial Y_j/\partial L_j(X) < \partial Y_j/\partial L_j(X)$, where $\partial Y_j/\partial L_j(X)$ is finite for all $L_j(X) > 0$. By Brouwer's fixed point theorem, there is a solution to this system of equations, which corresponds to an equilibrium with strictly positive wages. □

Identification of firm-specific labor supply elasticities, as laid out in Section 4.A, requires that an infinitesimal, isolated productivity shock to one firm does not generate discrete jumps in that firm's wages and labor. One way to guarantee this property is to show that, conditional on the wages at rival firms, $k \neq j$, each firm j's profit maximization problem admits a unique, locally stable solution.[39] In the following lemma, I show that this property holds almost surely.

[39]This property is immediate if profit Π_j is concave in wages $\{W_j(X)\}_X$, as this guarantees a unique solution to the firm's first-order condition. As shown in appendix section A.4, concavity holds under two special cases: the competitive (Walrasian) benchmark and a model with constant wage–amenity tradeoffs, such that $\beta_i = \beta$. However, the property does not apply in general, which means that it cannot be leveraged for the proof. From inspection of the Hessian matrix of the profit function, H_{Π_j}, I can conclude that Π_j is concave in wages whenever $\partial^2 \log L_j(X)/\partial[\log W_j(X)]^2 < \boldsymbol{\varepsilon}_j(X)(\boldsymbol{\varepsilon}_j(X) + 1)$ for all X. In a logit model, $\boldsymbol{\varepsilon}_j(X)$ equals $\mathrm{E}(\beta|X, j(i) = j)$ and $\partial^2 \log L_j(X)/\partial[\log W_j(X)]^2$ equals $\mathrm{Var}(\beta|X, j(i) = j)$. Thus, concavity of Π_j is more likely to hold if workers have higher marginal utilities of (log) earnings and/or if the dispersion of these marginal utilities is low.

Lemma 2. Conditional on the wages of rival firms, $\{W_k(\mathbf{X})\}_{k\neq j}$, firm j's profit-maximization problem will have a unique optimal wage schedule $W_j(\mathbf{X})$ for almost every value of (α_j, ρ_j).

Proof. Let $W_j(\mathbf{X})$ be firm j's wage vector. The first-order condition requires $\partial\Pi_j/\partial W_j(X) = 0$ for every X. If $W_j(\mathbf{X})$ satisfies this condition, then the Hessian $[\partial^2\Pi_j/\partial W_j(X)\partial W_j(X')]_{X,X'}$ is equal to $\mathrm{H}_{\Pi_j} = \mathbf{A}'(\mathrm{H}_{Y_j})\mathbf{A} + \mathbf{B}$, where H_{Y_j}, $\mathbf{A}$, and $\mathbf{B}$ are derived in Properties A.2 and A.3. This Hessian matrix satisfies two properties. First, $\mathbf{A}'(\mathrm{H}_{Y_j})\mathbf{A}$ and $\mathbf{B}$ are both nonsingular, as $\mathbf{A}'(\mathrm{H}_{Y_j})\mathbf{A}$ is symmetric, negative definite and $\mathbf{B}$ is diagonal. Second, given rival firms' wages $\{W_k(\mathbf{X})\}_{k\neq j}$, amenities $\{a_k(\mathbf{X})\}_k$, and worker distributions $(F_X, \{F_{\beta|X}\}_X, F_\epsilon)$, the matrices $\mathbf{A}$ and $\mathbf{B}$ are deterministic functions of $W_j(\mathbf{X})$. So, for any wage vector $W_j(\mathbf{X})$, H_{Π_j} depends on the firm's technology parameters only through H_{Y_j}. It follows that, for almost all (α_j, ρ_j) in $(0,1)\times(-\infty,1)$, the matrix H_{Π_j}, evaluated at $W_j(\mathbf{X})$, has a nonzero determinant and is therefore nonsingular. By the inverse function theorem, any solution $W_j(\mathbf{X})$ to the first-order condition is locally unique (and hence isolated) with probability one. Because all fixed points are isolated and the set of fixed points is compact, there can only exist finitely many of them.

Not every critical point $W_j(\mathbf{X})$ of Π_j is necessarily a global maximizer of the firm's profit function. I now show that, for almost all parameter values, there is a unique global maximizer, corresponding to a unique optimal wage schedule for the firm. Suppose that two distinct wage vectors both maximized Π_j. Since these vectors are (almost surely) locally stable fixed points of the first-order condition, the implicit function theorem delivers well-defined marginal effects of $\frac{1-\alpha_j}{\rho_j}$ on Π_j at each one. Because the marginal effects differ, any arbitrarily small change in $\frac{1-\alpha_j}{\rho_j}$ yields different profit levels at these two wage vectors, so that they can no longer both be global maximizers. Thus, for almost all (α_j, ρ_j), the firm's problem has a unique solution. $\square$

Throughout the paper, I consider two key counterfactuals. First, I analyze the competitive (Walrasian) equilibrium, defined as the equilibrium in which firms are price-takers, acting as if they face perfectly elastic labor supply curves. Second, I analyze a version of the model in which workers' wage-amenity tradeoffs are homogeneous, so $\beta_i = \beta$ for all i. In both cases, it is important to show that the equilibrium is unique; otherwise, multiple counterfactual equilibria could exist, complicating the interpretation of the estimates. In the following lemma, I show that the equilibrium is indeed unique under either of these counterfactual specifications. For simplicity, I only provide a proof for the case where there is a single skill type, i.e., $|\chi| = 1$.

Lemma 3. Let $|\chi| = 1$. There exists a unique equilibrium under each of the two counterfactual specifications of the model: the Walrasian benchmark and the benchmark where β is constant.

Proof. Under each counterfactual specification, the wage markdown is constant: $\frac{\boldsymbol{\varepsilon}_j(X)}{1+\boldsymbol{\varepsilon}_j(X)} = \frac{\boldsymbol{\varepsilon}}{1+\boldsymbol{\varepsilon}}$.[40]

[40] For the Walrasian benchmark, this claim holds trivially: firms do not markdown wages, so $\frac{\boldsymbol{\varepsilon}_j(X)}{1+\boldsymbol{\varepsilon}_j(X)} = 1$. For the benchmark with constant β, it follows from Property A.1, which demonstrates $\boldsymbol{\varepsilon}_j(X) = \mathrm{E}(\beta|X, j(i) = j)$.

Thus, letting $|\chi| = 1$, I can define the effective wage at firm j as $W_j^{\text{eff}}(\varphi) = W_j(\varphi)/\varphi$, where:

$$\begin{aligned}
W_j^{\text{eff}}(\varphi) &= \frac{\boldsymbol{\varepsilon}}{1+\boldsymbol{\varepsilon}} T_j(1-\alpha_j)\left(\int \varphi L_j\big(\varphi, \mathbf{W}(\varphi)\big)d\varphi\right)^{-\alpha_j} \\
&= \frac{\boldsymbol{\varepsilon}}{1+\boldsymbol{\varepsilon}} T_j(1-\alpha_j)\left(\int \varphi \int \frac{\exp\big(\beta \log W_j(\varphi) + a_j(\varphi)\big)}{\sum_{k=1}^J \exp\big(\beta \log W_k(\varphi) + a_k(\varphi)\big)} f_{\beta,\varphi}(\beta,\varphi)d\beta d\varphi\right)^{-\alpha_j} \\
&= \frac{\boldsymbol{\varepsilon}}{1+\boldsymbol{\varepsilon}} T_j(1-\alpha_j)\left(\int \varphi \int \frac{\exp\big(\beta \log W_j^{\text{eff}}(\varphi) + a_j(\varphi)\big)}{\sum_{k=1}^J \exp\big(\beta \log W_k^{\text{eff}}(\varphi) + a_k(\varphi)\big)} f_{\beta,\varphi}(\beta,\varphi)d\beta d\varphi\right)^{-\alpha_j} \\
&= \frac{\boldsymbol{\varepsilon}}{1+\boldsymbol{\varepsilon}} T_j(1-\alpha_j)\left(\int \varphi L_j\big(\varphi, \mathbf{W}^{\text{eff}}(\varphi)\big)d\varphi\right)^{-\alpha_j}.
\end{aligned}$$

Because the right-hand-side is constant within each firm, I can write $W_j^{\text{eff}}(\varphi) = W_j^{\text{eff}}$ for all j. Hence, an equilibrium is characterized by a solution to $g(\mathbf{W}^{\text{eff}}) = \mathbf{0}$, where I define g so that:

$$g_j(\mathbf{W}^{\text{eff}}) = \frac{\boldsymbol{\varepsilon}}{1+\boldsymbol{\varepsilon}} T_j(1-\alpha_j)[L_j^{\text{eff}}(\mathbf{W}^{\text{eff}})]^{-\alpha_j} - W_j^{\text{eff}}, \quad \text{for } j \in \{1,\dots,J\}.$$

To prove that this equilibrium is unique, I apply Theorem 3.1 in Kennan (2001), which implies that there is at most one equilibrium if g is strictly radially quasi-concave and quasi-increasing. To show that g is strictly radially quasi-concave, fix any $\mathbf{W}^{\text{eff}} > \mathbf{0}$ where $g(\mathbf{W}^{\text{eff}}) = \mathbf{0}$. I write:

$$\begin{aligned}
g_j(\lambda\mathbf{W}^{\text{eff}}) &= \frac{\boldsymbol{\varepsilon}}{1+\boldsymbol{\varepsilon}} T_j(1-\alpha_j)[L_j^{\text{eff}}(\lambda\mathbf{W}^{\text{eff}})]^{-\alpha_j} - \lambda W_j^{\text{eff}} \\
&= \frac{\boldsymbol{\varepsilon}}{1+\boldsymbol{\varepsilon}} T_j(1-\alpha_j)[L_j^{\text{eff}}(\mathbf{W}^{\text{eff}})]^{-\alpha_j} - \lambda W_j^{\text{eff}} \\
&= (1-\lambda)W_j^{\text{eff}} > 0,
\end{aligned}$$

for any $\lambda \in (0,1)$ and $j \in \{1,\dots,J\}$. The second equality above holds since the labor supply curves $L_j(\varphi,\mathbf{W})$ are homogeneous of degree zero. To show that g is quasi-increasing, notice that, for every firm j, $g_j(\lambda\mathbf{W}^{\text{eff}})$ is a decreasing function of effective labor $L_j^{\text{eff}}(\mathbf{W}^{\text{eff}})$, which is itself decreasing in the effective wages of j's rival firms, $\{W_k^{\text{eff}}\}_{k\neq j}$. Therefore, if $\tilde{W}_j^{\text{eff}} = W_j^{\text{eff}}$ and $\tilde{W}_k^{\text{eff}} \geq W_k^{\text{eff}}$ for all $k \neq j$, it must be that $g_j(\tilde{\mathbf{W}}^{\text{eff}}) \geq g_j(\mathbf{W}^{\text{eff}})$. The fact that g is radially quasi-concave, together with monotonicity, allows me to apply Kennan (2001), Theorem 3.1. So, I conclude that there is a unique equilibrium wage vector in each counterfactual economy. $\square$

C.2. *Identification Results*

I prove Proposition 1 in the more general case used in the estimation, where worker skill levels φ are unobserved and the moment conditions are written in terms of effective wages for each skill type χ. The identification of these effective wages is established separately in the proof of Proposition 2. If skill levels φ were observed, the same argument would apply with realized wages replacing effective wages. I omit the proof for that case, as the arguments are analogous.

Proposition 1. Under Assumptions I and II, the following moment conditions are satisfied:

$$\mathrm{E}\left[\ell_{j\tau_1,0}(\chi) - \ell_{j\tau_0,0}(\chi)\big|Z_j = 1, w^{\text{eff}}_{j\tau_0}(\chi) = w\right] = \mathrm{E}\left[\ell_{j\tau_1,0}(\chi) - \ell_{j\tau_0,0}(\chi)\big|Z_j = 0, w^{\text{eff}}_{j\tau_0}(\chi) = w\right]$$

$$\mathrm{E}\left[w^{\text{eff}}_{j\tau_1,0}(\chi) - w^{\text{eff}}_{j\tau_0,0}(\chi)\big|Z_j = 1, w^{\text{eff}}_{j\tau_0}(\chi) = w\right] = \mathrm{E}\left[w^{\text{eff}}_{j\tau_1,0}(\chi) - w^{\text{eff}}_{j\tau_0,0}(\chi)\big|Z_j = 0, w^{\text{eff}}_{j\tau_0}(\chi) = w\right].$$

Proof. Let $\Delta w^{\text{eff}}_{j0}(\chi) = w^{\text{eff}}_{j\tau_1,0}(\chi) - w^{\text{eff}}_{j\tau_0,0}(\chi)$ and $\Delta\ell_{j0}(\chi) = \ell_{j\tau_1,0}(\chi) - \ell_{j\tau_0,0}(\chi)$ denote the change in untreated potential wage and labor outcomes from τ_0 and τ_1 for skill type χ at firm j. In the model, these untreated potential outcomes evolve due to spillover effects that result from the TFP shocks at treated firms. These spillovers operate through two channels: workers' wage indices $\{I(\beta, X)\}_{\beta,X}$ and the joint distribution of worker types $F_{\beta,X}$. So, for the common trends assumption to hold, I must show that treated and untreated firms would be affected in the same way, on average, by any change to $\{I(\beta, X)\}_{\beta,X}$ and $F_{\beta,X}$. I prove this in two steps.

<u>Step 1.</u> *Derive a sufficient statistic for $\Delta w^{\mathit{eff}}_{j0}(\chi)$ and $\Delta\ell_{j0}(\chi)$.*

First, I introduce some new notation. For $\Gamma_j = \left(\rho_j, \alpha_j, \{\theta_{j\chi}\}_{\chi}, \{a_{j\chi} - a_{j\chi'}\}_{\chi,\chi'}\right)'$, I define:

$$g_{\chi,\tau}(w|\Gamma_j) = \log(1-\alpha_j) + \log\theta_{j\chi} - (1-\rho_j)\log h_{\chi,\tau}(w|\Gamma_j) + \frac{1-\alpha_j-\rho_j}{\rho_j}\log\sum_{\chi'\in\mathcal{X}}\theta_{j\chi'}\left(h_{\chi',\tau}(w|\Gamma_j)\right)^{\rho_j}$$

$$h_{\chi,\tau}(w|\Gamma_j) = \int\int\varphi\left(\frac{\exp\left(\beta w + a_{j\chi} - a_{j\chi^*}\right)}{I^{\text{eff}}_{\tau}(\beta,\chi)}\right)f_{\beta|\chi,\tau}(\beta|\chi)f_{\varphi|\chi,\tau}(\varphi)f_{\chi|\tau}(\chi)d\beta d\varphi,$$

for $\tau \in \{\tau_0, \tau_1\}$, where χ^* denotes some "reference" skill type. Since each firm j views itself as strategically small, it takes the wage index $I^{\text{eff}}_{\tau}(\beta,\chi)$ as given. Note that $g_{\chi,\tau}(w|\Gamma_j)$ can be interpreted as the equilibrium wage equation for a price-taking firm (with zero markdowns) where $\log(T_j) = 0$ and $a_{j\chi}$ is normalized relative to the reference type, i.e., $a_{j\chi} = a_{j\chi} - a_{j\chi^*}$. As shown in Appendix C.1, there is a unique fixed point solution to this system. Thus, I can write $g_{\chi,\tau}(w^*|\Gamma_j) = g^*_{\chi,\tau}(\Gamma_j)$. Given this property, the untreated potential wage $w^{\text{eff}}_{j,\tau_1,0}(\chi)$ is:

$$\begin{aligned}
w^{\text{eff}}_{j\tau_1,0}(\chi) &= \log T_j + \log(1-\alpha_j) + \log\theta_{j\chi} - \log\left(1 + \frac{1}{\boldsymbol{\varepsilon}_{j\tau_1,0}(\chi)}\right)\\
&\qquad - (1-\rho_j)\ell^{\text{eff}}_{j\tau_1,0}(\chi) + \frac{1-\alpha_j-\rho_j}{\rho_j}\log\sum_{\chi'\in\mathcal{X}}\theta_{j\chi'}\left(\exp\ell^{\text{eff}}_{j\tau_1,0}(\chi)\right)^{\rho_j}\\
&= \log T_j - \alpha_j a_{j\chi^*} - \log\left(1 + \frac{1}{\boldsymbol{\varepsilon}_{j\tau_1,0}(\chi)}\right) + g^*_{\chi,\tau_1}(\Gamma_j),
\end{aligned}$$

where the second equality follows from the fact that $\ell^{\text{eff}}_{j\tau_1,0}(\chi) = a_{j\chi^*} + \log h_{\chi,\tau}(w^{\text{eff}}_{j\tau_1,0}(\chi)|\Gamma_j)$. Because the wage changes are infinitesimal, I can write $\boldsymbol{\varepsilon}_{j\tau_1,0}(\chi) = \boldsymbol{\varepsilon}_{\tau_1,0}(\chi, w^{\text{eff}}_{j\tau_0,0}(\chi))$, where $\boldsymbol{\varepsilon}_{j\tau_1,0}(\chi, w^{\text{eff}}_{j\tau_0,0}(\chi))$ represents the labor supply elasticity faced by an untreated firm j at time period τ_1, evaluated at the pre-period effective wage $w^{\text{eff}}_{j\tau_0,0}(\chi)$. By this relationship, I obtain:

$$\Delta w^{\text{eff}}_{j0}(\chi) = \underbrace{\log\left(1 + \frac{1}{\boldsymbol{\varepsilon}_{\tau_1,0}(\chi, w^{\text{eff}}_{j\tau_0,0}(\chi))}\right) - \log\left(1 + \frac{1}{\boldsymbol{\varepsilon}_{\tau_0,0}(\chi, w^{\text{eff}}_{j\tau_0,0}(\chi))}\right) + g^*_{\chi,\tau_1}(\Gamma_j) - g^*_{\chi,\tau_0}(\Gamma_j)}_{q_w(w^{\text{eff}}_{j\tau_0,0}(\chi)|\Gamma_j,\tau_0,\tau_1,\chi)}.$$

In addition, I can write the change in untreated potential labor outcomes, $\Delta\ell_{j0}(\chi)$, as follows:

$$\Delta\ell_{j0}(\chi) = \underbrace{\log\left(\frac{\int \frac{\exp[\beta q_w(w^{\text{eff}}_{j\tau_0,0}(\chi)|\Gamma_j,\tau_0,\tau_1,\chi)+\beta w^{\text{eff}}_{j\tau_0,0}(\chi)+a_{j\chi}-a_{j\chi^*}]}{I_{\tau_1}(\beta,X)} f_{\beta|\chi,\tau_1}(\beta|\chi) f_{\chi,\varphi|\tau_1}(\chi,\varphi)d\beta}{\int \frac{\exp[\beta' w^{\text{eff}}_{j\tau_0,0}(\chi)+a_{j\chi}-a_{j\chi^*}]}{I_{\tau_0}(\beta',X)} f_{\beta|\chi,\tau_0}(\beta'|\chi) f_{\chi,\varphi|\tau_0}(\chi,\varphi)d\beta'}\right)}_{q_\ell(w^{\text{eff}}_{j\tau_0,0}(\chi)|\Gamma_j,\tau_0,\tau_1,\chi)}.$$

<u>Step 2.</u> Show that Assumption I implies common trends.

Plugging in the functions $q_w(w^{\text{eff}}_{j\tau_0,0}(\chi)|\Gamma_j,\tau_0,\tau_1,\chi)$ and $q_\ell(w^{\text{eff}}_{j\tau_0,0}(\chi)|\Gamma_j,\tau_0,\tau_1,\chi)$, I obtain:

$$\begin{aligned}
\mathrm{E}\left[\Delta w^{\text{eff}}_{j0}(\chi)\middle|Z_j = 0, w^{\text{eff}}_{j\tau_0}(\chi) = w\right] &= \mathrm{E}\left[q_w(w^{\text{eff}}_{j\tau_0}(\chi)|\Gamma_j,\tau_0,\tau_1,\chi)\middle|Z_j = 0, w^{\text{eff}}_{j\tau_0}(\chi) = w\right]\\
&= \mathrm{E}\left[q_w(w^{\text{eff}}_{j\tau_0}(\chi)|\Gamma_j,\tau_0,\tau_1,\chi)\middle|Z_j = 1, w^{\text{eff}}_{j\tau_0}(\chi) = w\right]\\
&= \mathrm{E}\left[\Delta w^{\text{eff}}_{j0}(\chi)\middle|Z_j = 1, w^{\text{eff}}_{j\tau_0}(\chi) = w\right]
\end{aligned}$$

$$\begin{aligned}
\mathrm{E}\left[\Delta\ell_{j0}(\chi)\middle|Z_j = 0, w^{\text{eff}}_{j\tau_0}(\chi) = w\right] &= \mathrm{E}\left[q_\ell(w^{\text{eff}}_{j\tau_0}(\chi)|\Gamma_j,\tau_0,\tau_1,\chi)\middle|Z_j = 0, w^{\text{eff}}_{j\tau_0}(\chi) = w\right]\\
&= \mathrm{E}\left[q_\ell(w^{\text{eff}}_{j\tau_0}(\chi)|\Gamma_j,\tau_0,\tau_1,\chi)\middle|Z_j = 1, w^{\text{eff}}_{j\tau_0}(\chi) = w\right]\\
&= \mathrm{E}\left[\Delta\ell_{j0}(\chi)\middle|Z_j = 1, w^{\text{eff}}_{j\tau_0}(\chi) = w\right].
\end{aligned}$$

In both equations, the second equality holds by Assumption I. So, I can write $\mathrm{DiD}_{\tau_0,\tau_1}(w|\chi)$ as:

$$\begin{aligned}
\mathrm{DiD}_{\tau_0,\tau_1}(w|\chi) &= \frac{\mathrm{E}\left[\ell_{j\tau_1}(\chi) - \ell_{j\tau_0}(\chi)\middle|Z_j = 1, w^{\text{eff}}_{j\tau_0}(\chi) = w\right] - \mathrm{E}\left[\ell_{j\tau_1}(\chi) - \ell_{j\tau_0}(\chi)\middle|Z_j = 0, w^{\text{eff}}_{j\tau_0}(\chi) = w\right]}{\mathrm{E}\left[w^{\text{eff}}_{j\tau_1}(\chi) - w^{\text{eff}}_{j\tau_0}(\chi)\middle|Z_j = 1, w^{\text{eff}}_{j\tau_0}(\chi) = w\right] - \mathrm{E}\left[w^{\text{eff}}_{j\tau_1}(\chi) - w^{\text{eff}}_{j\tau_0}(\chi)\middle|Z_j = 0, w^{\text{eff}}_{j\tau_0}(\chi) = w\right]}\\
&= \frac{\mathrm{E}\left[\ell_{j\tau_1,1}(\chi) - \ell_{j\tau_0,1}(\chi)\middle|Z_j = 1, w^{\text{eff}}_{j\tau_0}(\chi) = w\right] - \mathrm{E}\left[\ell_{j\tau_1,0}(\chi) - \ell_{j\tau_0,0}(\chi)\middle|Z_j = 0, w^{\text{eff}}_{j\tau_0}(\chi) = w\right]}{\mathrm{E}\left[w^{\text{eff}}_{j\tau_1,1}(\chi) - w^{\text{eff}}_{j\tau_0,1}(\chi)\middle|Z_j = 1, w^{\text{eff}}_{j\tau_0}(\chi) = w\right] - \mathrm{E}\left[w^{\text{eff}}_{j\tau_1,0}(\chi) - w^{\text{eff}}_{j\tau_0,0}(\chi)\middle|Z_j = 0, w^{\text{eff}}_{j\tau_0}(\chi) = w\right]}\\
&= \frac{\mathrm{E}\left[\ell_{j\tau_1,1}(\chi) - \ell_{j\tau_0,1}(\chi)\middle|Z_j = 1, w^{\text{eff}}_{j\tau_0}(\chi) = w\right] - \mathrm{E}\left[\ell_{j\tau_1,0}(\chi) - \ell_{j\tau_0,0}(\chi)\middle|Z_j = 1, w^{\text{eff}}_{j\tau_0}(\chi) = w\right]}{\mathrm{E}\left[w^{\text{eff}}_{j\tau_1,1}(\chi) - w^{\text{eff}}_{j\tau_0,1}(\chi)\middle|Z_j = 1, w^{\text{eff}}_{j\tau_0}(\chi) = w\right] - \mathrm{E}\left[w^{\text{eff}}_{j\tau_1,0}(\chi) - w^{\text{eff}}_{j\tau_0,0}(\chi)\middle|Z_j = 1, w^{\text{eff}}_{j\tau_0}(\chi) = w\right]}\\
&= \frac{\mathrm{E}\left[\ell_{j\tau_1,1}(\chi) - \ell_{j\tau_1,0}(\chi)\middle|Z_j = 1, w^{\text{eff}}_{j\tau_0}(\chi) = w\right]}{\mathrm{E}\left[w^{\text{eff}}_{j\tau_1,1}(\chi) - w^{\text{eff}}_{j\tau_1,0}(\chi)\middle|Z_j = 1, w^{\text{eff}}_{j\tau_0}(\chi) = w\right]}.
\end{aligned}$$

Since the TFP shocks are infinitesimal, the difference $w^{\text{eff}}_{j\tau_1,1}(\chi) - w^{\text{eff}}_{j\tau_0,1}(\chi)$ is also infinitesimal for any firm j. Thus, I can conclude that $\boldsymbol{\varepsilon}_{\tau_1}(\chi, w) = \text{DiD}_{\tau_0,\tau_1}(w|\chi)$ for any wage w, because:

$$\begin{aligned}
\text{E}\left[\ell_{j\tau_1,1}(\chi) - \ell_{j\tau_1,0}(\chi)\middle|Z_j = 1, w^{\text{eff}}_{j\tau_0}(\chi) = w\right] &= \text{E}\left[\boldsymbol{\varepsilon}_{j\tau_1}(\chi) \times \left(w^{\text{eff}}_{j\tau_1,1}(\chi) - w^{\text{eff}}_{j\tau_1,0}(\chi)\right)\middle|Z_j = 1, w^{\text{eff}}_{j\tau_0}(\chi) = w\right] \\
&= \text{E}\left[\boldsymbol{\varepsilon}_{\tau_1}\left(\chi, w^{\text{eff}}_{j\tau_0,1}(\chi)\right) \times \left(w^{\text{eff}}_{j\tau_1,1}(\chi) - w^{\text{eff}}_{j\tau_1,0}(\chi)\right)\middle|Z_j = 1, w^{\text{eff}}_{j\tau_0}(\chi) = w\right] \\
&= \boldsymbol{\varepsilon}_{\tau_1}(\chi, w) \times \text{E}\left[w^{\text{eff}}_{j\tau_1,1}(\chi) - w^{\text{eff}}_{j\tau_1,0}(\chi)\middle|Z_j = 1, w^{\text{eff}}_{j\tau_0}(\chi) = w\right].
\end{aligned}$$

□

Proof of Proposition 2.

Proof. Let $W^{\text{eff}}_{j\tau}(\chi, \varphi) = W_{j\tau}(\chi, \varphi)/\varphi$ denote the effective wage for workers with skill type χ at firm j in period τ. If Assumption II holds, then the equilibrium wage condition can be written as $W^{\text{eff}}_{j\tau}(\chi, \varphi) = g_{j\chi\tau}(\{W^{\text{eff}}_{k\tau}(\chi, \varphi)\}_k)$ for all (j, χ, τ), where $g_{j\chi\tau}$ does not depend on φ. I write:

$$W^{\text{eff}}_{j\tau}(\chi, \varphi) = \frac{\boldsymbol{\varepsilon}_{j\tau}(\chi, \varphi)}{1 + \boldsymbol{\varepsilon}_{j\tau}(\chi, \varphi)} T_{j\tau}(1 - \alpha_j)\theta_{j\chi} \left(L^{\text{eff}}_{j\tau}(\chi)\right)^{\rho_j - 1} \left(\sum_{\chi' \in \mathcal{X}} \theta_{j\chi'} \left(L^{\text{eff}}_{j\tau}(\chi')\right)^{\rho_j}\right)^{\frac{1-\alpha_j}{\rho_j} - 1},$$

where the labor supply elasticity $\boldsymbol{\varepsilon}_{j\tau}(\chi, \varphi)$ equals a (χ, τ)-specific function of $\left\{W^{\text{eff}}_{k\tau}(\chi, \varphi)\right\}_k$:

$$\begin{aligned}
\boldsymbol{\varepsilon}_{j\tau}(\chi, \varphi) &= \int \beta \times \frac{\frac{\exp\left(\beta \log W_{j\tau}(\chi,\varphi) + a_j(\chi,\varphi)\right)}{\sum_{k=1}^J \exp(\beta \log W_{k\tau}(\chi,\varphi) + a_k(\chi,\varphi))} f_{\beta|\chi,\tau}(\beta|\chi) f_{\chi,\varphi|\tau}(\chi, \varphi)}{\int \frac{\exp\left(\beta' \log W_{j\tau}(\chi,\varphi) + a_j(\chi,\varphi)\right)}{\sum_{k=1}^J \exp(\beta' \log W_{k\tau}(\chi,\varphi) + a_k(\chi,\varphi))} f_{\beta|\chi,\tau}(\beta'|\chi) f_{\chi,\varphi|\tau}(\chi, \varphi) d\beta'} d\beta \\
&= \int \beta \times \frac{\frac{\exp\left(\beta \log W^{\text{eff}}_{j\tau}(\chi,\varphi)\right)}{\sum_{k=1}^J \exp\left(\beta \log W^{\text{eff}}_{k\tau}(\chi,\varphi) + a_{k\chi}\right)} f_{\beta|\chi,\tau}(\beta|\chi)}{\int \frac{\exp\left(\beta' \log W^{\text{eff}}_{j\tau}(\chi,\varphi)\right)}{\sum_{k=1}^J \exp\left(\beta' \log W^{\text{eff}}_{k\tau}(\chi,\varphi) + a_{k\chi}\right)} f_{\beta|\chi,\tau}(\beta'|\chi) d\beta'} d\beta.
\end{aligned}$$

As shown above, the equilibrium is fully characterized in terms of effective wages and labor, where φ does not enter any of the determining functions. Therefore, the effective wages do not depend on φ, so that $W^{\text{eff}}_{j\tau}(\chi, \varphi) = W^{\text{eff}}_{j\tau}(\chi)$, which delivers the desired log-additive separability.

□

Proof of Proposition 3.

Proof. Consider two skill types χ and χ'. For each time period $\tilde{\tau} \in \{\tau, \tau'\}$, it must be that:

$$\log\left(\frac{\text{MPL}^{\text{eff}}_{j\tilde{\tau}}(\chi)}{\text{MPL}^{\text{eff}}_{j\tilde{\tau}}(\chi')}\right) = \log\theta_{j\chi} - \log\theta_{j\chi'} - (1 - \rho_j)\left[\log L^{\text{eff}}_{j\tilde{\tau}}(\chi) - \log L^{\text{eff}}_{j\tilde{\tau}}(\chi')\right].$$

Because ρ_j and $\{\theta_{j\chi}\}_\chi$ are invariant over time, the elasticity of substitution may be recovered by computing inter-temporal shifts in the relative marginal products and effective labor shares:

$$(1-\rho_j)^{-1} = \frac{\log\left(\frac{L^{\text{eff}}_{j\tau}(\chi)}{L^{\text{eff}}_{j\tau}(\chi')}\right) - \log\left(\frac{L^{\text{eff}}_{j\tau'}(\chi)}{L^{\text{eff}}_{j\tau'}(\chi')}\right)}{\log\left(\frac{\text{MPL}^{\text{eff}}_{j\tau}(\chi)}{\text{MPL}^{\text{eff}}_{j\tau}(\chi')}\right) - \log\left(\frac{\text{MPL}^{\text{eff}}_{j\tau'}(\chi)}{\text{MPL}^{\text{eff}}_{j\tau'}(\chi')}\right)}.$$

□

Proof of Proposition 4.

Proof. I normalize the firm-specific efficiencies $\{\theta_{j\chi}\}_\chi$ so that $\theta_{j\chi^*} = 1$ for skill type $\chi^* \in \mathcal{X}$. Under this normalization, and given knowledge of ρ_j, these parameters may be computed as:

$$\theta_{j\chi} = \exp\left[\log\left(\frac{\text{MPL}^{\text{eff}}_{j\tau}(\chi)}{\text{MPL}^{\text{eff}}_{j\tau}(\chi^*)}\right) + (1-\rho_j)\log\left(\frac{L^{\text{eff}}_{j\tau}(\chi)}{L^{\text{eff}}_{j\tau}(\chi^*)}\right)\right].$$

A firm's returns to scale $1-\alpha_j$ and total factor productivity $T_{j\tau}$ can then be recovered from the effective marginal products, the effective labor shares, and value added. Specifically, I write:

$$1-\alpha_j = \exp\left[\log \text{MPL}^{\text{eff}}_{j\tau}(\chi) - y_{j\tau} - \log(\theta_{j\chi}) + (1-\rho_j)\log L^{\text{eff}}_{j\tau}(\chi) + \log\sum_{\chi'\in\mathcal{X}}\theta_{j\chi'}\left(L^{\text{eff}}_{j\tau}(\chi')\right)^{\rho_j}\right]$$

$$T_{j\tau} = \exp\left[y_{j\tau} - \frac{1-\alpha_j}{\rho_j}\log\sum_{\chi\in\mathcal{X}}\theta_{j\chi}\left(L^{\text{eff}}_{j\tau}(\chi)\right)^{\rho_j}\right].$$

□

Proof of Proposition 5.

Proof. For some firm j^*, set $a_{j^*\chi} = 0$. Under this normalization, the amenities $\{a_{j\chi}\}_{j\neq j^*}$ are:

$$a_{j\chi} = \log L_{j\tau}\big(\chi, w^{\text{eff}}_{k\tau}(\chi)\big) - \log L_{j^*\tau}(\chi),$$

where $L_{j^*\tau}(\chi)$ is the labor supplied to firm j^* by skill type χ at time τ, and $L_{j\tau}\big(\chi, w^{\text{eff}}_{k\tau}(\chi)\big)$ is the labor supplied to firm j if it posts the same log effective wage as firm k. If the elasticity $\boldsymbol{\varepsilon}_{j\tau}(\chi, w)$ is known to the researcher for all wages w, then $\log L_j(\chi, w^{\text{eff}}_{k\chi})$ is recovered from:

$$a_{j\chi} = \log L_{j\tau}(\chi) + \int_{w^{\text{eff}}_{j\tau}(\chi)}^{w^{\text{eff}}_{j^*\tau}(\chi)} \boldsymbol{\varepsilon}_{j\tau}(\chi, w)dw - \log L_{j^*\tau}(\chi).$$

□

Proof of Proposition 6.

Proof. Suppose that the elasticity curve $\boldsymbol{\varepsilon}_{j\tau}(\chi, w)$ is known to the researcher for skill type χ. Then the firm-specific labor supply curves $\{L_{j\tau}(\chi, w)\}_j$ can be recovered through integration:

$$\log L_{j\tau}(\chi, w) = \log L_{j\tau}(\chi) + \int_{w_{j\tau}^{\text{eff}}(\chi)}^{w} \boldsymbol{\varepsilon}_{j\tau}(\chi, \tilde{w}) d\tilde{w}.$$

Each labor supply curve $L_{j\tau}(\chi, w)$ may be expressed as a Laplace transform $\mathcal{L}\{g\}(s)$, where:

$$\mathcal{L}\{g\}(s) = \int_0^\infty g(t)\exp(-st)dt, \quad \text{such that} \quad \begin{aligned} s &= -w \\ t &= \beta \\ g(t) &= \frac{\exp(a_{j\chi})}{\sum_{k=1}^{J}\exp\left(tw_{k\tau}^{\text{eff}}(\chi) + a_{k\chi}\right)} f_{\beta|\chi,\tau}(t|\chi) f_{\chi|\tau}(\chi). \end{aligned}$$

The transform $\mathcal{L}\{g\}(s)$ is a one-to-one mapping of $g(t)$. Specifically, any two functions $g(t)$ can only share the same Laplace transform if they differ on a set of Lebesgue measure zero. Therefore, when the elasticity $\boldsymbol{\varepsilon}_{j\tau}(\chi, w)$ is point-identified, so is the function $g(\beta)$, where:

$$g(\beta) = \frac{\exp(a_{j\chi})}{\sum_{k=1}^{J}\exp\left(\beta w_{k\tau}^{\text{eff}}(\chi) + a_{k\chi}\right)} f_{\beta|\chi,\tau}(\beta|\chi) f_{\chi|\tau}(\chi).$$

If the amenities $\{a_{j\chi}\}_j$ are identified up-to-location, relative to some reference amenity $a_{j^*\chi}$, then the density $f_{\beta|\chi}(\beta|\chi)$ is point-identified for any β-value from the following relationship:

$$f_{\beta|\chi}(\beta|\chi) = g(\beta) \times \frac{\sum_{k=1}^{J}\exp\left(\beta w_k^{\text{eff}}(\chi) + \tilde{a}_{k\chi} - \tilde{a}_{j^*\chi}\right)}{\exp\left(a_{j\chi} - a_{j^*\chi}\right) f_{\chi|\tau}(\chi)}.$$

□

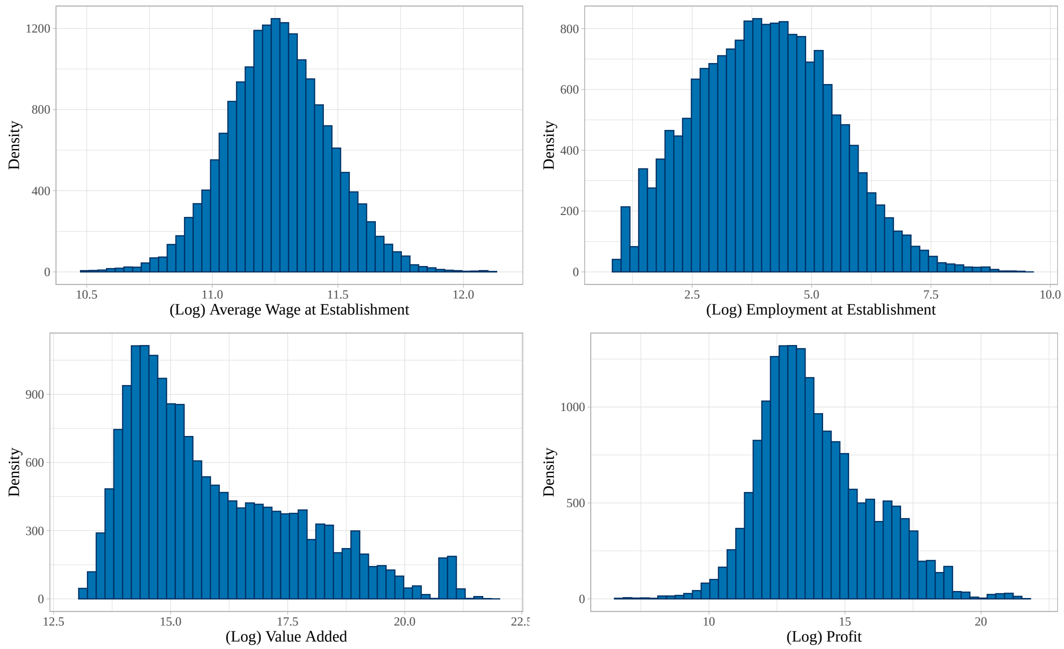
D. Data Preparation, Robustness Analyses, and Additional Results

D.1 Details about the Estimation Sample

Figure A.1 plots the empirical distributions of log wages, employment, value added, and profit for the baseline estimation sample. The sample is restricted to firms with at least five full-time employees, and each firm in the sample is required to remain operational for at least nine consecutive years. Over this period, the firm must continue to report positive revenue and maintain the same set of branches, with each branch staying in the same industry and commuting zone.

The figure reveals that average wages are relatively concentrated across establishments, consistent with Norway's compressed wage structure, while establishment-wide employment is highly dispersed and right-skewed. Value added and profits also exhibit considerably wide dispersion, with long right tails. These distributions closely match those in the full, unrestricted sample, which suggests that the firm panel is representative of the broader firm population.

Figure A.1: Empirical Distributions of Wages, Labor, Value Added, and Profits



Notes. This figure plots histograms of log wages, employment, value added, and profit for the estimation sample. Firms with negative revenue or < 5 workers are omitted; the profit histogram plots firms with non-negative profit.

D.2 Additional Data Sources: Public Procurement Contracts

Public procurement contracts in Norway are enforced by the Public Procurement Act and associated regulations. These laws apply to procurement of all products, services, and construction contracts valued over NOK 100,000, and they cover contracts procured by all public sector entities in Norway. Procurement is generally competitive, and auctions are publicly announced on Doffin.no, Norway's national procurement portal. Notice requirements vary with the contract value, and larger contracts are subject to more extensive announcement and documentation.

Data Collection

Data on public procurement contracts are collected following the methodology of de Frahan et al. (2024), using contract-award announcements from Doffin.no in the years 2003-2018.The resulting data cover more than 60,000 announcements. The data collection process involves three steps. First, researchers obtain HTML files through the Doffin IT department, containing the full history of contract awards over the period. Second, they use standard scraping methods to extract contract-level information, including the winning firm's name, the contract value, date, and product characteristics. Notably, the Doffin files do not report names of losing bidders, and tax identifiers for winning firms are missing in about 90% of award announcements.

In the third step, the researchers implement a fuzzy string matching procedure, following Raffo & Lhuillery (2009), to link firm names in the Doffin.no data with government registers that include both the names and tax identifiers of all Norwegian firms. This matching process yields results for approximately 30,000 contracts. The researchers assess the accuracy of the fuzzy matching procedure by analyzing the 10% of contract awards for which both the name and tax identifier of the winning firm are available, and they find that the algorithm correctly assigns the tax identifier in 93% of cases. The accuracy of this matching procedure does not appear to be correlated with contract characteristics, such as the value or number of bidders.

Sample Construction and Research Design

The external-instrument sample is constructed from the same matched worker-firm data and using the same sample restrictions as in the internal-instrument analysis. I restrict attention to full-time workers and to firms that remain active for nine consecutive years, reporting positive revenue and employing ≥ 5 full-time workers throughout the panel. After linking firms to public procurement awards from Doffin, the sample contains 25,714 firms and 901,811 workers.

The research design leverages the timing of a firm's first procurement award. The firm is classified as treated ($Z_j = 1$) in the year τ in which it first wins a contract recorded in Doffin. Firms first treated in the same calendar year form a "treatment cohort". For each cohort, the control group consists of firms that either first win a procurement contract after $\tau + 3$ or never win one during the sample period. This definition guarantees that control firms are untreated throughout the post-period used to estimate pass-through. Future-winner controls are drawn from the Doffin-linked sample, and never-winner controls are drawn from the full firm panel.

As discussed in Section IV.A, the assumption required for identification is that, conditional on a firm's local labor market (defined as an industry-by-region cell), the timing of its first procurement award is independent of its technology and amenity-difference parameters $\big(\rho_j, \alpha_j, \{\theta_{j\chi}\}_\chi, \{a_j(X) - a_j(X')\}_{X,X'}\big)$. Note that the research design does not require award timing to be fully random, even within a local labor market. In particular, Z_j may depend on the firm's baseline productivity $T_{j\tau_0}$ and on the average value of its amenities $\mathrm{E}(a_j(X) \mid j)$.

D.3 Details about Estimation of Labor Supply Elasticities

To estimate the firm-specific labor supply elasticities, I use a nonparametric Kernel estimator:

$$\widehat{\mathrm{DiD}}_{\tau_0,\tau_1}(w|\chi) = \frac{\sum_j K_{1,j}(w|\chi)\big(\ell_{j\tau_1}(\chi) - \ell_{j\tau_0}(\chi)\big) - \sum_j K_{0,j}(w|\chi)\big(\ell_{j\tau_1}(\chi) - \ell_{j\tau_0}(\chi)\big)}{\sum_j K_{1,j}(w|\chi)\big(w^{\text{eff}}_{j\tau_1}(\chi) - w^{\text{eff}}_{j\tau_0}(\chi)\big) - \sum_j K_{0,j}(w|\chi)\big(w^{\text{eff}}_{j\tau_1}(\chi) - w^{\text{eff}}_{j\tau_0}(\chi)\big)},$$

where I define $K_{z,j}(w|\chi)$ as the Kernel weight for a firm j that has treatment status $z \in \{0, 1\}$. The weights are defined as $K_{1,j}(w|\chi) = \frac{\mathbf{1}\{Z_j = z\}\kappa_h(w^{\text{eff}}_{j\tau_0}(\chi) - w)}{\sum_k \mathbf{1}\{Z_k = z\}\kappa_h(w^{\text{eff}}_{k\tau_0}(\chi) - w)}$, where κ_h is a kernel function with bandwidth h. Common examples of κ_h are the Gaussian, Uniform, and Epanechnikov kernels:

$$\textit{Gaussian:} \quad \kappa_h(u) = \frac{1}{\sqrt{2\pi}} \exp\left[-\frac{1}{2}\left(\frac{u}{h}\right)^2 \right].$$

$$\textit{Uniform:} \quad \kappa_h(u) = \mathbf{1}\{|u| \leq h\}.$$

$$\textit{Epanechnikov:} \quad \kappa_h(u) = \frac{3}{4}\big(1 - (u/h)^2\big)\mathbf{1}\{|u| \leq h\}$$

Here, the tuning parameter h determines the bandwidth of the estimator. As h approaches zero:

$$\lim_{h\to 0} \widehat{\text{DiD}}_{\tau_0,\tau_1}(w|\chi) = \frac{\sum_j \omega_{1,j}(w|\chi)\big(\ell_{j\tau_1}(\chi) - \ell_{j\tau_0}(\chi)\big) - \sum_j \omega_{0,j}(w|\chi)\big(\ell_{j\tau_1}(\chi) - \ell_{j\tau_0}(\chi)\big)}{\sum_j \omega_{1,j}(w|\chi)\big(w^{\text{eff}}_{j\tau_1}(\chi) - w^{\text{eff}}_{j\tau_0}(\chi)\big) - \sum_j \omega_{0,j}(w|\chi)\big(w^{\text{eff}}_{j\tau_1}(\chi) - w^{\text{eff}}_{j\tau_0}(\chi)\big)},$$

where $\omega_{z,j}(w|\chi)$ denotes $\frac{\mathbf{1}\{Z_j=z, w^{\text{eff}}_{j\tau_0}(\chi)=w\}}{\sum_j \mathbf{1}\{Z_j=z, w^{\text{eff}}_{j\tau_0}(\chi)=w\}}$ for $z \in \{0,1\}$. By the weak law of large numbers:

$$\sum_j \omega_{z,j}(w|\chi)\big(\ell_{j\tau_1}(\chi) - \ell_{j\tau_0}(\chi)\big) \xrightarrow{p} \mathrm{E}\left[\ell_{j\tau_1}(\chi) - \ell_{j\tau_0}(\chi)\middle|Z_j = z, w^{\text{eff}}_{j\tau_0}(\chi) = w\right], \quad \text{and:}$$

$$\sum_j \omega_{z,j}(w|\chi)\big(w^{\text{eff}}_{j\tau_1}(\chi) - w^{\text{eff}}_{j\tau_0}(\chi)\big) \xrightarrow{p} \mathrm{E}\left[w^{\text{eff}}_{j\tau_1}(\chi) - w^{\text{eff}}_{j\tau_0}(\chi)\middle|Z_j = z, w^{\text{eff}}_{j\tau_0}(\chi) = w\right],$$

for $z \in \{0,1\}$. From the continuous mapping theorem, I establish consistency of the estimator:

$$\lim_{h\to 0} \widehat{\text{DiD}}_{\tau_0,\tau_1}(w|\chi) \xrightarrow{p} \text{DiD}_{\tau_0,\tau_1}(w|\chi) \quad \text{as } J \to \infty.$$

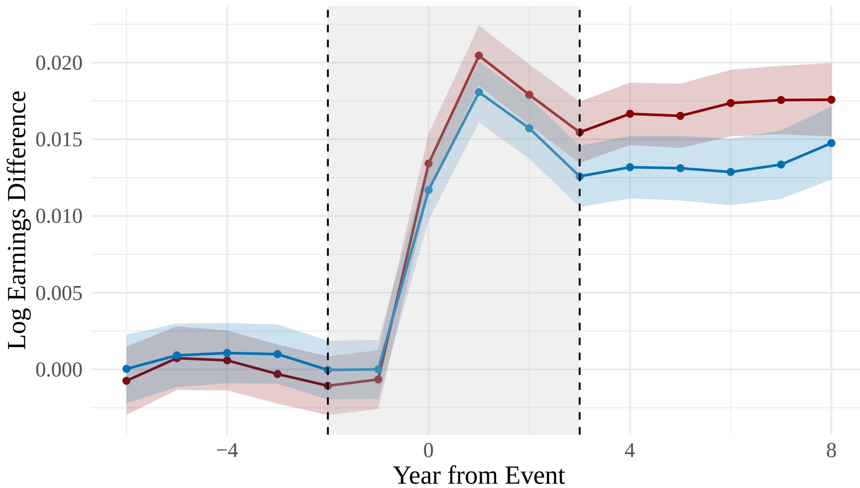
D.4. Robustness Analyses

I provide three sets of robustness analyses. The first assesses the credibility of the IV design. In Figure A.2, I present event-study plots of the pass-through from the instrument Z onto earnings in the years before and after the shock. I find no evidence of pre-trends, and the effect remains stable in the post-period, suggesting that the pass-through is not short-lived. In Figure A.3, I plot reduced form estimates separately by workers' college attainment and experience. Across all skill types, the patterns in Figure 4 persist: the employment response is relatively flat across firms' effective wages, while the wage response falls with effective wages.

The second set of robustness analyses consists of misspecification tests. First, I test whether the instrument affects labor supply on the intensive margin through hours worked, and find no statistically significant effect. Second, I test whether the IV estimates differ between firms with local market shares below versus above 0.05. I find no significant difference, supporting the assumption that firms do not internalize their market shares when setting workers' wages.

The third set of robustness analyses replicates the main IV estimates presented in Figure 5 under alternative specifications. In Figure A.4, I show a version of the estimates where I drop market fixed effects, effectively treating Norway as one single labor market. In Figure A.5, I report estimates based on a restricted sample where firms only operate a single establishment. In Figures A.6, I present estimates for a restricted sample of firms with below-median union membership, addressing concerns that the wage-posting model may be less applicable in highly unionized settings. In Figure A.7, I show estimates for a restricted sample of firms with local market shares below 0.05, addressing concerns about strategic wage setting by firms with larger market shares. In Figures A.8 and A.9, I present estimates constructed using alternative Kernel estimators, in which I implement Uniform and Epanechnikov kernels, respectively. In Figures A.10 and A.11, I report estimates that use alternative bandwidths h. In Figures A.12 and A.13, I present estimates based on the external instruments, described in Appendix D.2.

Figure A.2: Event Study—Pass-through of Productivity Shock to Wages



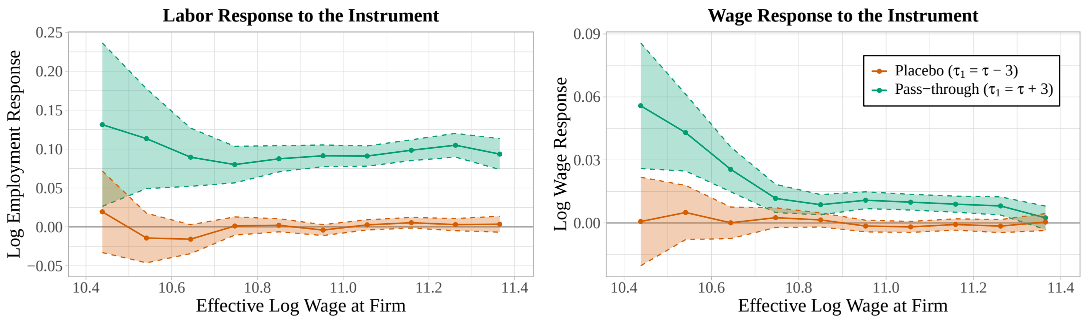
Notes. This figure plots the average differences in log earnings (solid lines) between firms that had above-median versus below-median changes in log value added at event time zero. Estimates are shown with market fixed effects (red) and without market fixed effects (blue). The shaded area indicates time periods where the orthogonality condition need not hold in the identification of the permanent pass-through rate.

Table A.1: Misspecification Tests of Instrument Validity

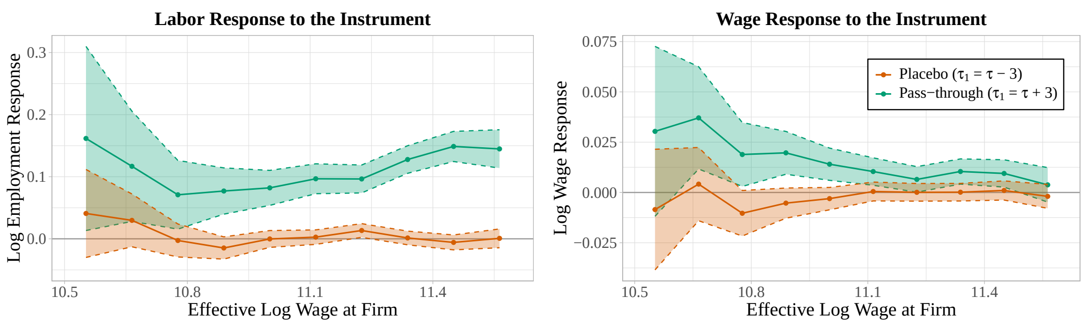
No Market FEs		Market FEs	
Estimated Effect	*p*-value	Estimated Effect	*p*-value
Panel A. Pass-through of Z to Work Hours			
-0.003	0.624	-0.011	0.105
(0.007)		(0.007)	
Panel B. Local Market Share Effect			
-0.144	0.535	1.503	0.183
(1.633)		(1.666)	

Notes. The table presents two misspecification tests. Panel A estimates the pass-through of the internal instrument Z to workers' scheduled hours. Panel B compares estimated firm-specific labor supply elasticities, averaged over firms and skill groups, among firms with local market shares above versus below 0.05; positive estimates imply higher elasticities among high-share firms. Columns report specifications with and without market fixed effects.

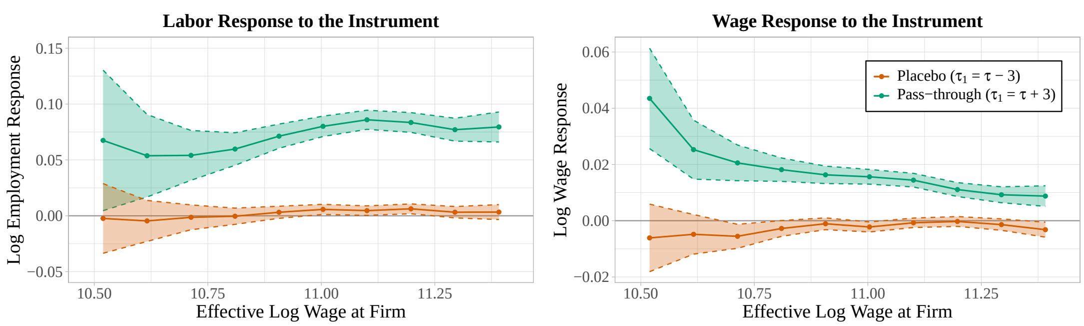
Figure A.3: Average Pass-Through by Education and Experience



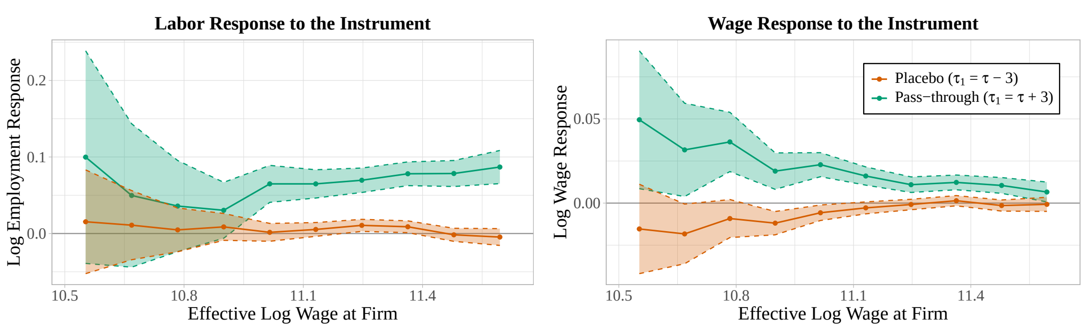
(a) No College Degree, < 10 Years Experience



(b) College Graduate, < 10 Years Experience



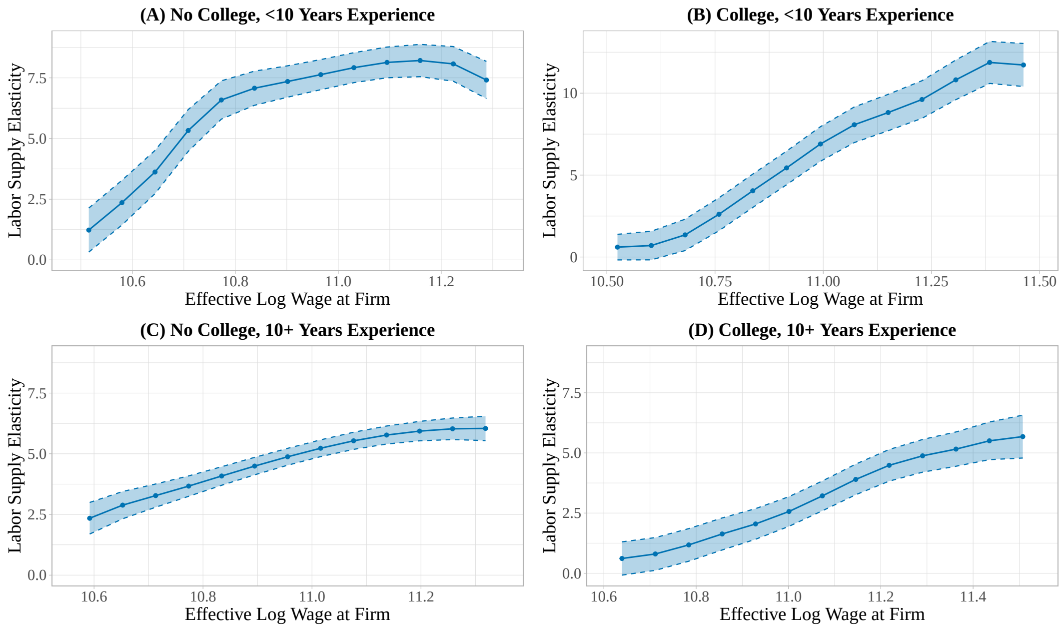
(c) No College Degree, ≥ 10 Years Experience



(d) College Graduate, ≥ 10 Years Experience

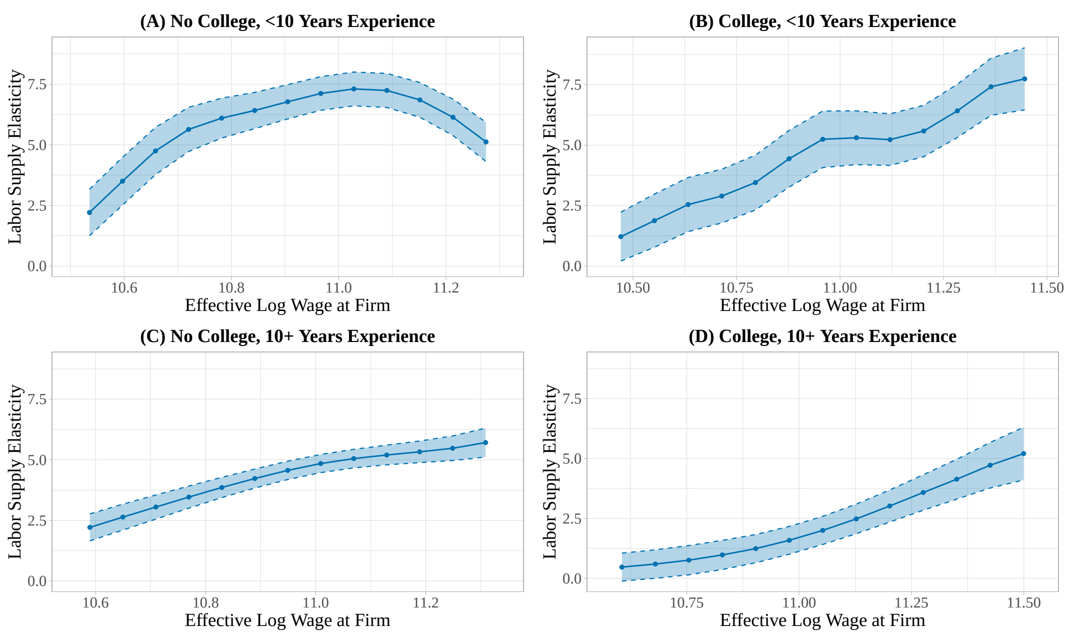
Notes. This figure plots the reduced form estimates by worker education and experience, averaged across years using sample size weights. Market fixed effects are included. Standard errors are based on 500 bootstrap samples.

Figure A.4: IV Estimates of Labor Supply Elasticities—No Market Fixed Effects



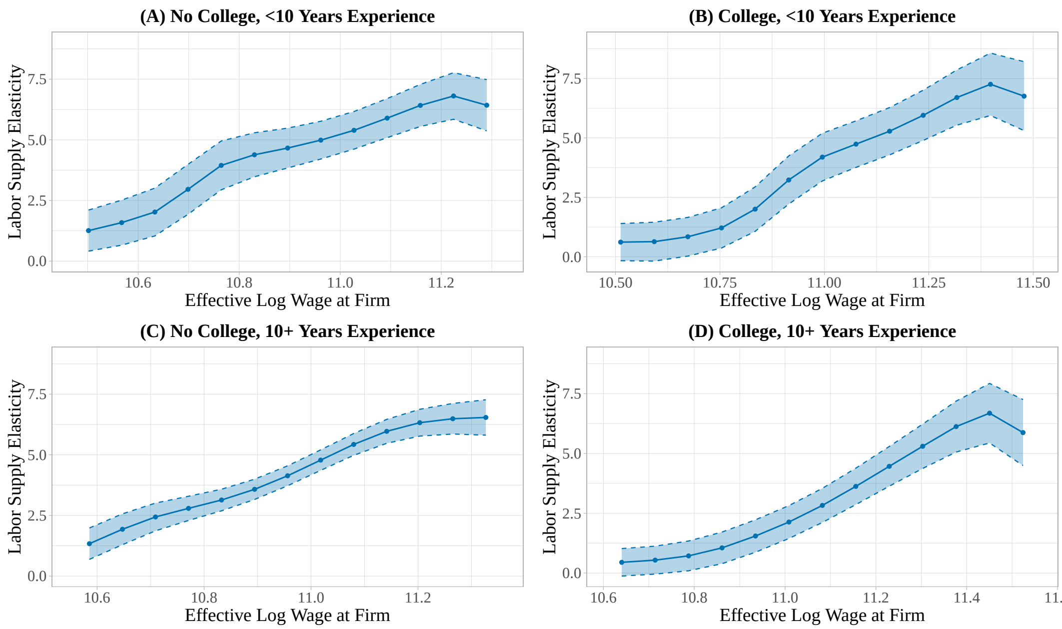
Notes. This figure reports IV estimates of firm-specific labor supply elasticities from the baseline specification re-estimated without any local labor-market fixed effects. Standard errors are based on 500 bootstrap samples.

Figure A.5: IV Estimates of Labor Supply Elasticities—Single-Establishment Firms



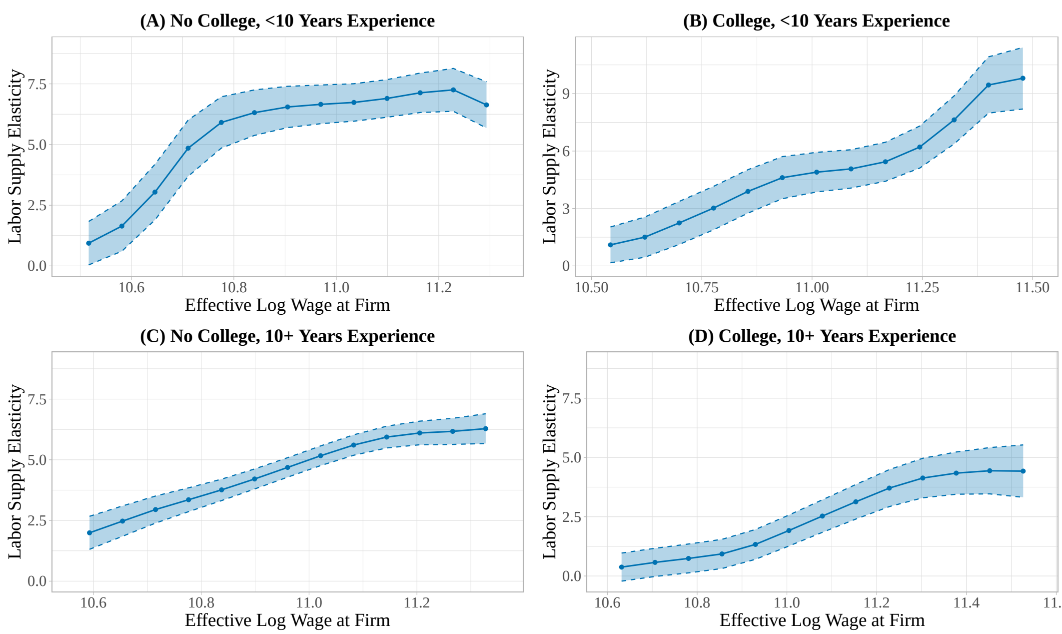
Notes. This figure reports IV estimates of firm-specific labor supply elasticities from the baseline specification re-estimated on the sample of single-establishment firms. Standard errors are based on 500 bootstrap samples.

Figure A.6: IV Estimates of Labor Supply Elasticities—Low Union Membership Rate



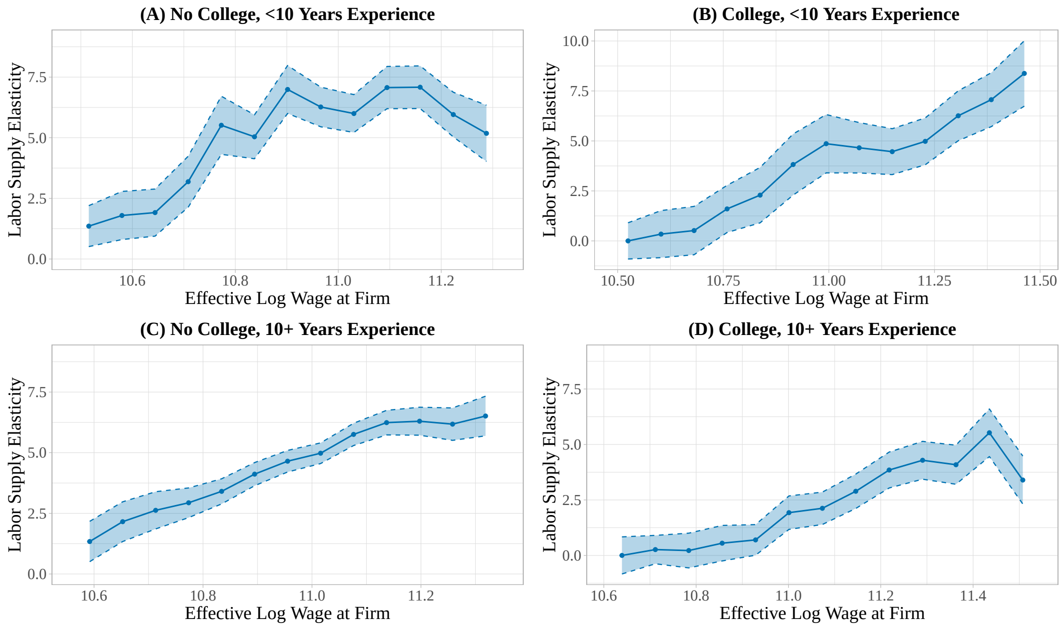
Notes. This figure reports IV estimates of firm-specific labor supply elasticities from the baseline specification re-estimated for sectors with below-median union membership rates. It excludes all public sector firms and all firms in education, healthcare, and manufacturing. Standard errors are based on 500 bootstrap samples.

Figure A.7: IV Estimates of Labor Supply Elasticities—Small Local Market Shares



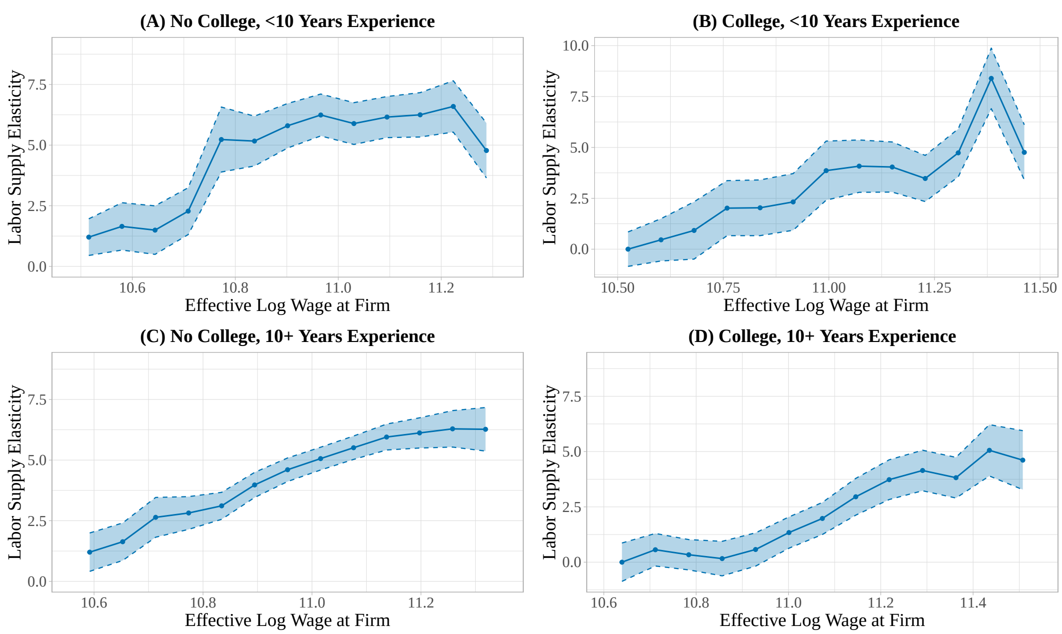
Notes. This figure reports IV estimates of firm-specific labor supply elasticities from the baseline specification estimated on firms with local labor market share <0.05. Standard errors are based on 500 bootstrap samples.

Figure A.8: IV Estimates of Labor Supply Elasticities—Uniform Kernel Estimator



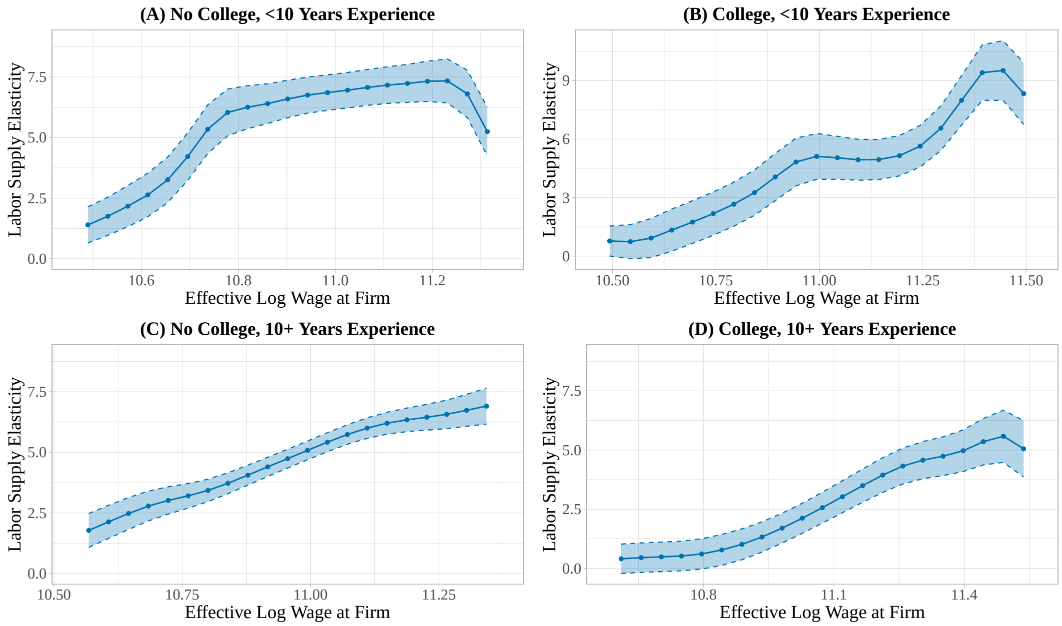
Notes. This figure reports IV estimates of firm-specific labor supply elasticities re-estimated using a uniform kernel to construct the wage-specific weights $K_{z,j}(w)$. Standard errors are based on 500 bootstrap samples.

Figure A.9: IV Estimates of Labor Supply Elasticities—Epanechnikov Kernel Estimator



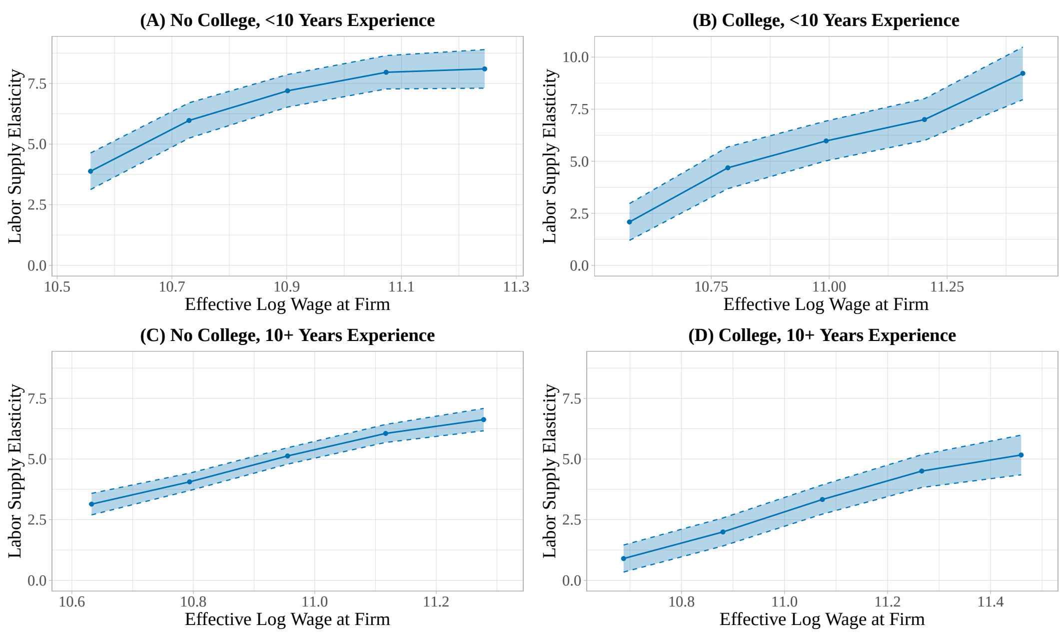
Notes. This figure reports IV estimates of firm-specific labor supply elasticities estimated using an Epanechnikov kernel to construct the wage-specific weights $K_{z,j}(w)$. Standard errors are based on 500 bootstrap samples.

Figure A.10: IV Estimates of Labor Supply Elasticities—Small Kernel Bandwidth



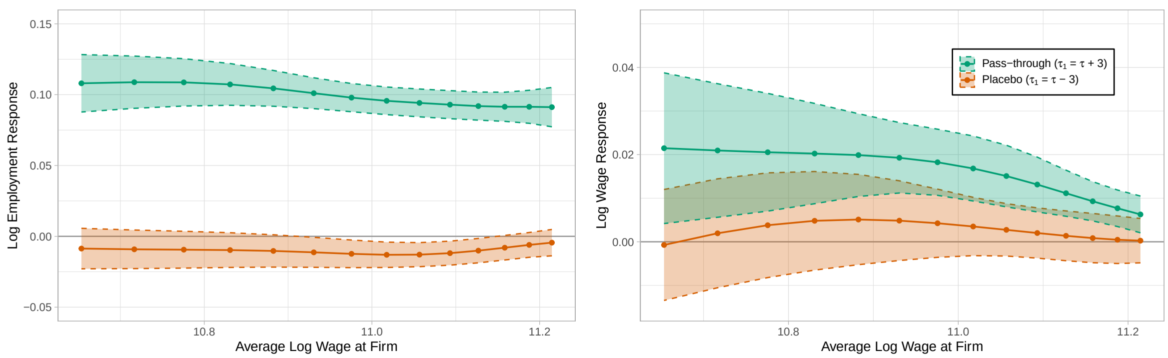
Notes. This figure reports IV estimates of firm-specific labor supply elasticities from the baseline specification re-estimated using a smaller Gaussian-kernel bandwidth, approximately two-thirds of the baseline bandwidth. Estimates are plotted on a finer grid of 21 wage points. Standard errors are based on 500 bootstrap samples.

Figure A.11: IV Estimates of Labor Supply Elasticities—Large Kernel Bandwidth



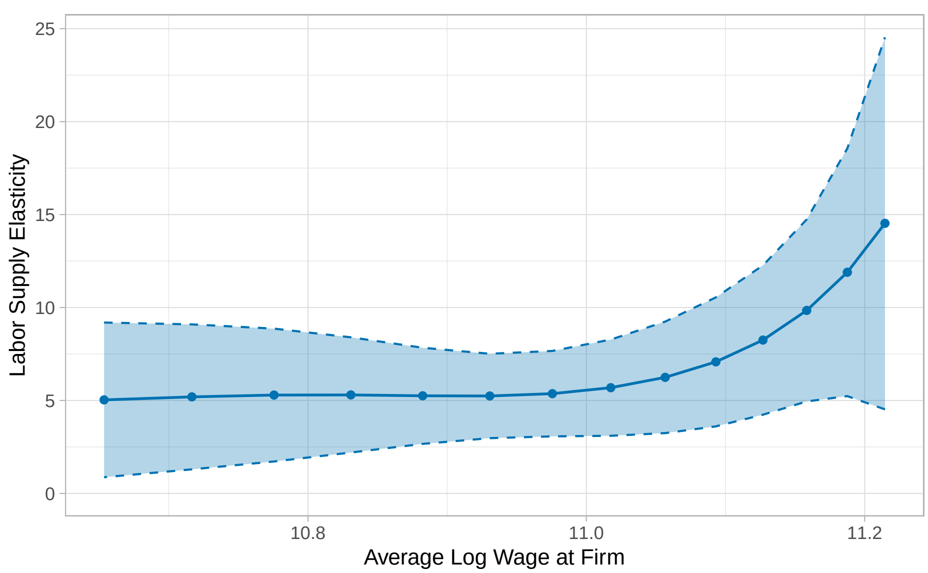
Notes. This figure reports IV estimates of firm-specific labor supply elasticities from the baseline specification re-estimated using a larger Gaussian-kernel bandwidth, approximately four-thirds of the baseline bandwidth. Estimates are plotted on a coarser grid of 5 wage points. Standard errors are based on 500 bootstrap samples.

Figure A.12: Average Pass-through Estimates—Procurement Auction Instrument



Notes. This figure reports first stage and reduced form estimates based on the instrument constructed from public procurement contract wins. The estimates are averaged across skill types and years using sample-size weights. Market fixed effects are included. The instrument is defined following the protocols in de Frahan et al. (2024).

Figure A.13: IV Estimates of Labor Supply Elasticities—Procurement Auction Instrument



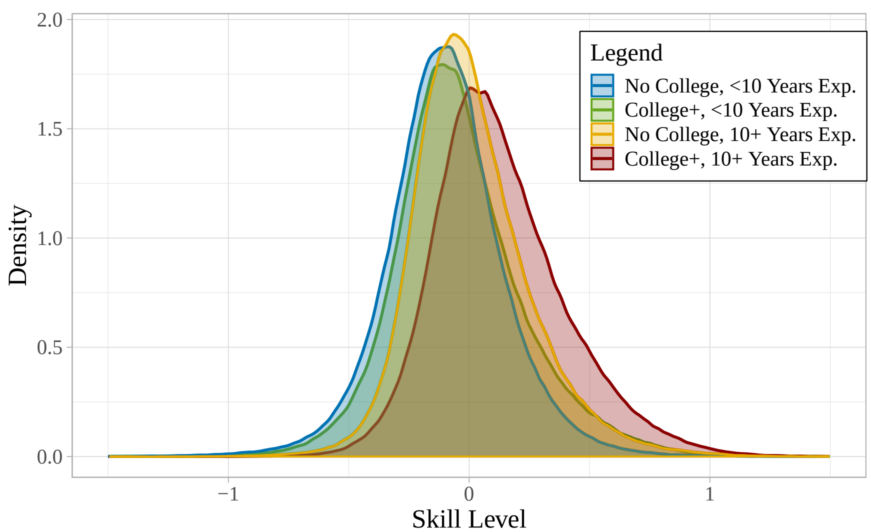
Notes. This figure reports IV estimates of firm-specific labor supply elasticities using the instrument constructed from public procurement contract wins. The estimates are averaged across skill types and years using sample-size weights. Market fixed effects are included. The instrument follows the protocols in de Frahan et al. (2024).

This Appendix subsection provides additional tables and figures not included in the main text.

Structural Parameter Estimates

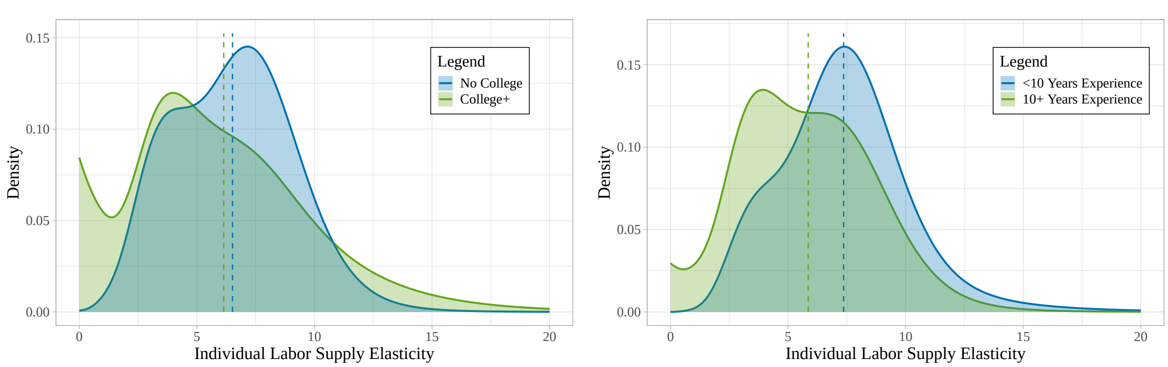
I provide four supplementary figures to further characterize the structural parameter estimates. Figure A.14 plots the estimated density of workers' skill levels φ, separately by college attainment and experience. Figure A.15 plots the corresponding densities of workers' wage-amenity trade-offs, β. Figure A.16 plots the distribution of firms' average non-wage amenities by worker type. Finally, Figure A.17 presents rank-rank plots of firm amenity values for each pair of skill types, with Spearman rank correlations summarizing the similarity of these rankings.

Figure A.14: Density of Worker Skill Levels by Education and Experience



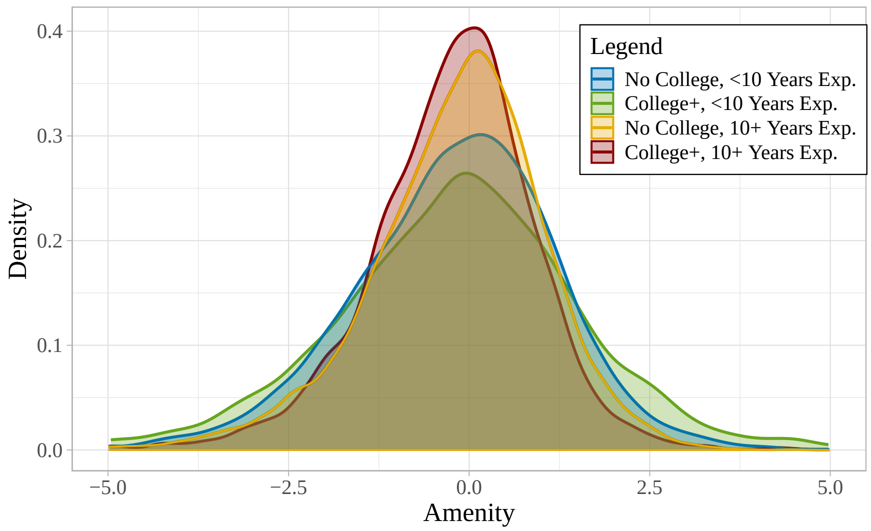
Notes. The figure plots the estimated density of workers' log skill levels $\log\varphi$, separately by college attainment and experience. The skill levels are identified up to the following normalization: $\mathrm{E}(\log\varphi|\chi) = 0$ for each skill type χ.

Figure A.15: Density of Individual Labor Supply Elasticities by Education and Experience



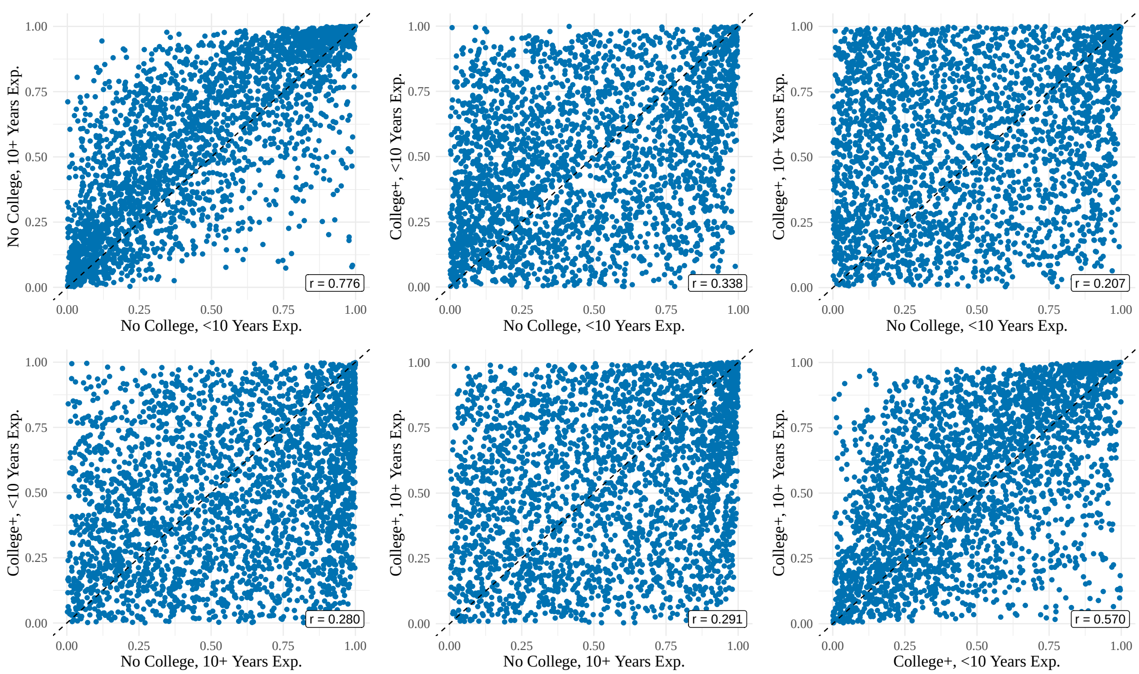
Notes. This figure plots the estimated densities f_β of workers' marginal rates of substitution between wages and amenities by education and experience. The dashed lines in each figure indicate the means of these distributions.

Figure A.16: Density of Non-Wage Amenities by Education and Experience



Notes. This figure plots the estimated density of firms' average non-wage amenities, separately by workers' college attainment and experience. Amenities are identified only up to the normalization $a_{j^*\chi} = 0$, where j^* denotes a reference firm, set in implementation to the firm whose average wage equals the sample median.

Figure A.17: Rank-Rank Plots of Amenities by Education and Experience



Notes. This figure presents rank-rank plots of firms' average non-wage amenities by workers' college attainment and experience, along with the Spearman rank correlations (r) that summarize the similarity of these rankings.

Wage Markdowns and Rent Sharing

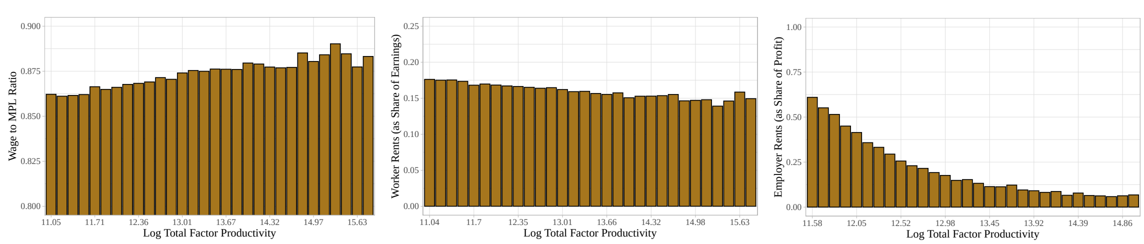
I include four supplementary results to further characterize the incidence of wage markdowns and rents. Table A.2 shows means and variances of estimated wage markdowns by industry and region. Figure A.18 plots estimated firm-level averages of wage markdowns, worker rents, and firm rents against estimated TFP, illustrating how wage setting power varies over the firm productivity distribution. Figure A.19 plots the densities of worker and firm rents and compares them to those under a restricted model with constant β. Figure A.20 plots the estimated distribution of worker rents as a share of earnings by workers' college attainment and experience.

Table A.2: Estimates of Wage Markdowns by Industry and Region

	Firm-Specific Labor Supply Elasticity		Implied Wage-to-MPL Ratio	
	Mean	Std. Dev.	Mean	Std. Dev.
Panel A. Broad Industry Group				
Business/Finance	7.745	2.402	0.877	0.037
Construction	7.053	1.088	0.873	0.022
Education	7.088	1.755	0.870	0.037
Healthcare	6.411	1.745	0.857	0.039
Manufacturing	6.840	1.401	0.868	0.027
Services	6.530	1.499	0.861	0.038
Trade	7.003	1.454	0.871	0.028
Utilities/Transport	7.100	1.345	0.873	0.023
Panel B. Broad Region				
Rural Region	5.725	1.153	0.846	0.031
Urban Region	7.726	3.070	0.874	0.035

Notes. This table reports the means and standard deviations of estimated firm-specific labor supply elasticities and implied wage to marginal product ratios, disaggregated by broad industry and region (urban versus rural).

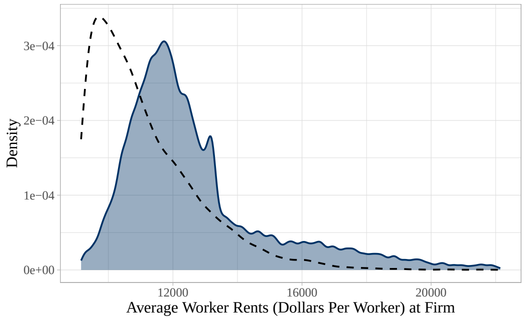
Figure A.18: Incidence of Wage Markdowns and Rents by Firm TFP and Amenities



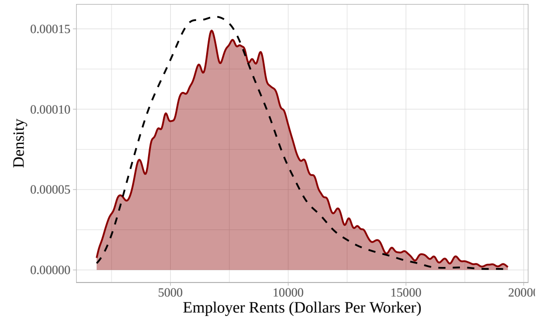
(a) Wage Markdowns (Wage to MPL Ratio) (b) Worker Rents (as Share of Earnings) (c) Firm Rents (as Share of Profit)

Notes. This figure plots firm-level averages of wage markdowns, worker rents, and firm rents against estimated log TFP. Panel (a) reports wage markdowns measured as the wage-to-MPL ratio, with lower values indicating larger markdowns. Panels (b) and (c) report worker rents (as a share of earnings) and firm rents (as a share of profits).

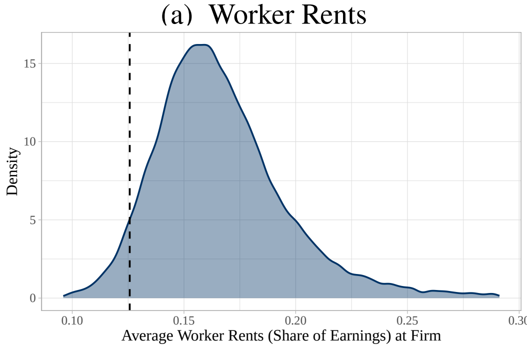
Figure A.19: Estimated Distributions of Worker and Firm Rents



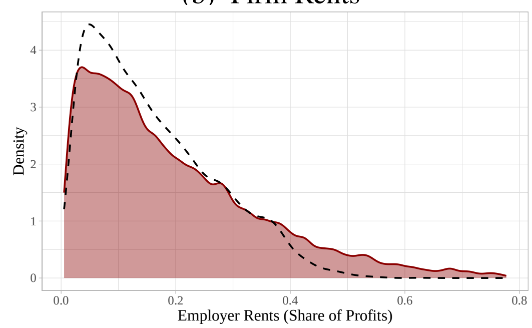
(a) Worker Rents



(b) Firm Rents



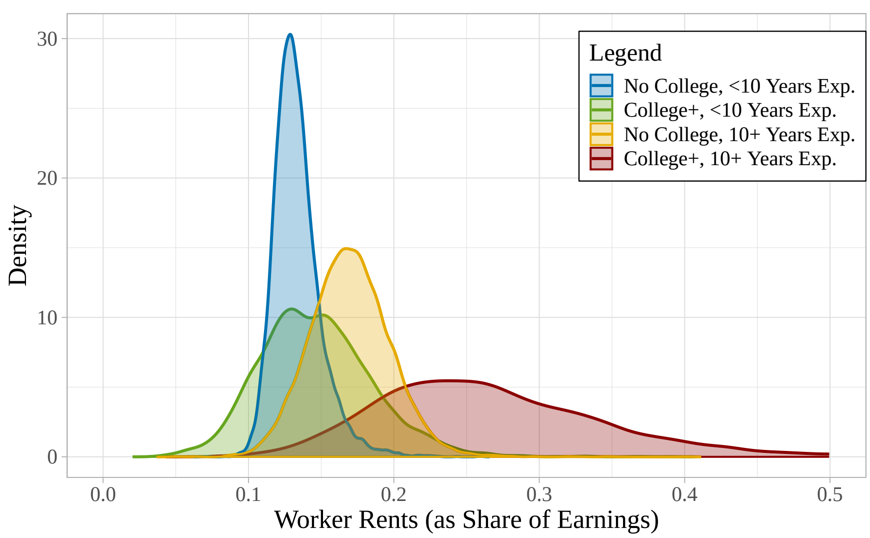
(c) Worker Rents (as Share of Earnings)



(d) Firm Rents (as Share of Profits)

Notes. This figure plots estimated distributions of firm-level average worker rents and firm rents. Panels (a) and (b) show rent levels in 2014 US dollars; panels (c) and (d) scale rents by worker earnings and firm profits. Dotted black lines correspond to estimates from the restricted model with homogeneous β, where $\text{Var}(\beta) = 0$.

Figure A.20: Density of Worker Rents (as Share of Earnings) by Education and Experience

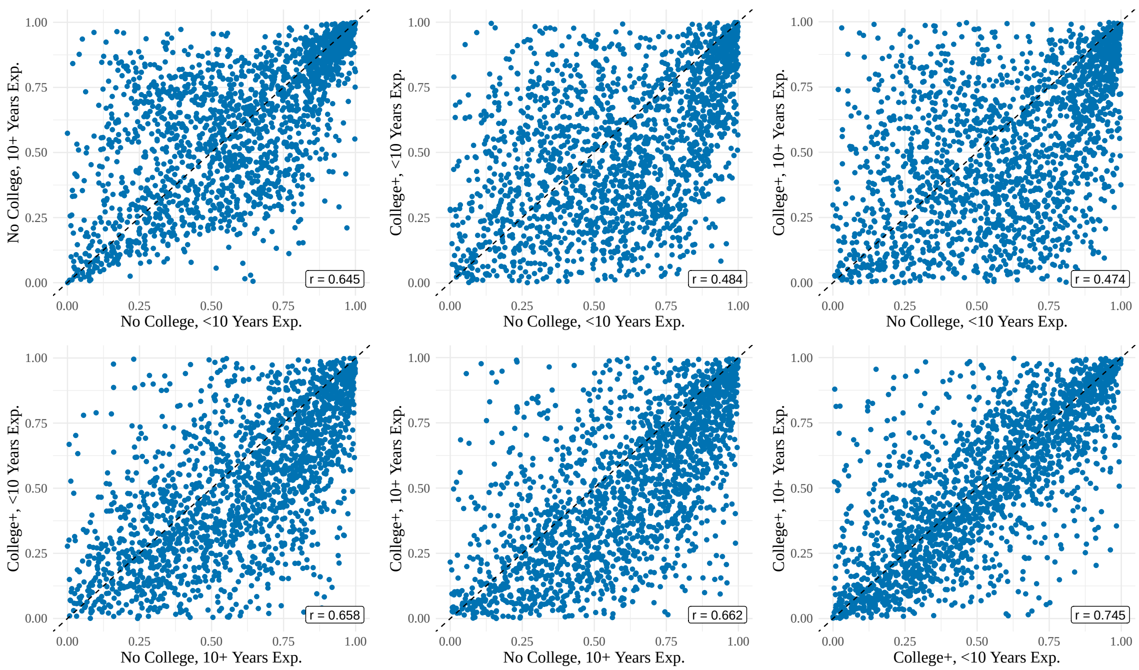


Notes. This figure plots the estimated density of worker rent shares, separately by college attainment and experience.

Testing for a Common Cross-Firm Ranking of Wage Markdowns

I now show results analyzing how the ranking of wage markdowns across firms varies by skill type. Figure A.21 reports rank-rank plots of firm-level markdowns for each pair of skill types, with Spearman rank correlations summarizing the similarity of these rankings. Table A.3 reports formal tests of equality in the rankings, using both paired t-tests and bootstrap rank tests.

Figure A.21: Rank-Rank Plots of Wage Markdowns by Education and Experience



Notes. This figure presents rank-rank plots of wage-to-marginal-product ratios by workers' college attainment and experience, along with the Spearman rank correlations (r) that summarize the similarity of these rankings.

Table A.3: Tests of Equality in Rankings of Wage Markdowns

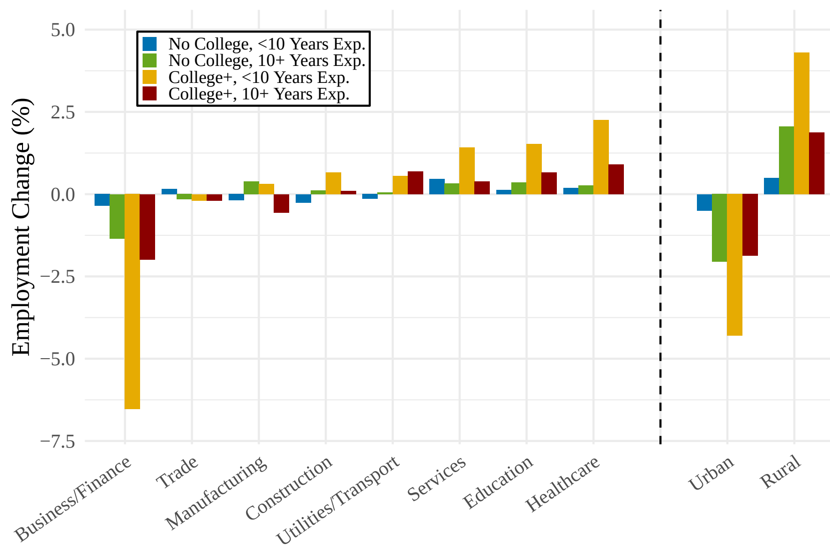
	Correlation Coefficient		Test: p-value	
	Spearman (Rank)	Pearson (Linear)	Paired t-test	Bootstrap Rank Test
No College, Low Exp. / No College, High Exp.	0.645	0.605	0.047	0.608
No College, Low Exp. / College, Low Exp.	0.484	0.220	0.000	0.039
No College, Low Exp. / College, High Exp.	0.474	0.225	0.000	0.020
No College, High Exp. / College, Low Exp.	0.658	0.390	0.000	0.118
No College, High Exp. / College, High Exp.	0.662	0.408	0.000	0.059
College, Low Exp. / College, High Exp.	0.745	0.559	0.007	0.471

Notes. This table compares firm-level rankings of wage markdowns across worker skill types. For each skill type pair, the paired t-test reports a p-value for equality of mean percentile ranks across firms, and the bootstrap rank test reports a p-value based on the mean-squared-difference in percentile ranks across 500 bootstrap samples.

Supplementary Results on Allocative Efficiency

I now present two supplementary figures. Figure A.22 reports the estimated reallocation of labor across industries and regions when moving to a competitive allocation in which labor wedges are eliminated. Figure A.23 plots the estimated mean and the 1st and 99th percentiles of workers' wage markdowns at different points in the realized wage distribution, providing suggestive evidence on the potential efficiency gains from a higher minimum wage in Norway.

Figure A.22: Estimated Reallocation of Labor by Eliminating Labor Wedges



Notes. This figure shows the estimated change in labor shares by sector and region after eliminating labor wedges. It plots the estimated percent change in employment for different types of firms. Positive values indicate sectors or regions that gain labor in the competitive allocation; negative values indicate sectors or regions that lose labor.

Figure A.23: Wage Markdown Distribution by Workers' Earnings

Notes. This figure shows how markdowns vary across the wage distribution. Workers are grouped into bins of realized log wages. For each bin, the figure plots the mean wage markdown, measured as the wage-to-marginal-product ratio, with bars showing the 1st and 99th percentiles of markdowns within the bin. Lower values indicate larger markdowns.

Counterfactual Estimates under Alternative Models

I now present versions of Tables 3 and 4 under two intermediate restrictions on preference heterogeneity. The first sets workers' wage-amenity trade-offs β to be constant within every skill type χ, equal to the corresponding skill-type mean in the full model. This restriction implies that conditional on skills, wage markdowns are constant across firms, leaving no role for wage-setting power in shaping between-firm wage inequality or the allocation of labor across firms.

The second imposes a common labor supply elasticity for all firms within each local labor market, equal to the corresponding market mean in the full model. This restriction allows markdowns to vary across markets, but not across firms within markets, and thus captures only part of the distortionary impact of labor market power. Comparing Panels B and C of Table A.5 with Table 4 shows that these restrictions understate both the distributional effects of market power and the extent of misallocation. These results highlight the importance of allowing for rich heterogeneity in workers' labor supply responses through unrestricted heterogeneity in β.

Table A.4: Estimates of Wage Markdowns and Rents—Intermediate Counterfactuals

	Skill-Type Constant Elasticity			Local-Market Constant Elasticity		
	Mean	Std. Dev.	Variance Decomp. (Within Firm / Between Firm)	Mean	Std. Dev.	Variance Decomp. (Within Firm / Between Firm)
Wage Markdowns	0.873	0.015	100.00% / 0.00%	0.874	0.005	0.00% / 100.00%
	(0.007)	(0.013)		(0.007)	(0.004)	
Worker Rents						
Dollars per Worker	$10,509	$2,424		$10,288	$1,645	
	(651)	(1,287)		(584)	(334)	
Share of Earnings	12.8%	2.8%	100.00% / 0.00%	12.6%	1.7%	70.59% / 29.41%
	(0.8)	(1.5)		(0.7)	(0.5)	
Firm Rents						
Dollars per Worker	$8,357	$3,092		$8,009	$2,787	
	(746)	(968)		(630)	(385)	
Share of Profits	22.1%	19.2%	0.00% / 100.00%	21.6%	19.2%	0.00% / 100.00%
	(1.2)	(0.5)		(1.1)	(0.5)	

Notes. This table reports wage markdowns and rents under two intermediate restrictions on preference heterogeneity. The first sets workers' wage-amenity trade-offs β constant within skill types χ, equal to the corresponding skill-type mean under the full model. The second imposes a constant firm-specific labor supply elasticity within each local labor market, equal to the corresponding market mean under the full model. Variance decompositions report the share of variation attributable to within-firm and between-firm components, calculated via the Law of Total Variance: $\text{Var}(X) = \text{E}(\text{Var}(X|j)) + \text{Var}(\text{E}(X|j))$. Standard errors are bootstrapped using 500 replications.

Table A.5: Impacts on Wage Inequality and Welfare—Intermediate Counterfactuals

	Skill-Type Constant Elasticity			Local-Market Constant Elasticity		
	Monopsonistic Labor Market	Competitive Labor Market	% Difference in Dollars	Monopsonistic Labor Market	Competitive Labor Market	% Difference in Dollars
Panel A. Market Aggregates						
Mean Earnings (Log Dollars)	11.291	11.402	11.74%	11.291	11.402	11.73%
		(0.009)			(0.045)	
Mean Profit (Log Dollars)	14.687	14.431	-22.59%	14.687	14.442	-21.73%
		(0.017)			(0.018)	
Panel B. Earnings Inequality						
Variance in Log Earnings						
Total	0.102	0.103	0.98%	0.102	0.100	-1.97%
		(0.002)			(0.002)	
Between Firms	0.025	0.025	0.00%	0.025	0.023	-8.05%
					(0.001)	
Within Firms	0.076	0.077	1.32%	0.076	0.076	0.00%
		(0.001)				
Gender Wage Gap	14.3%	13.6%	-4.8%	14.3%	13.9%	-3.0%
		(1.0)			(1.2)	
Panel C. Allocative Inefficiency						
Total Welfare (Log Dollars)	12.883	12.883	0.00%	12.376	12.412	3.62%
	(0.066)	(0.080)		(0.148)	(0.181)	
Labor-TFP Correlation	0.178	0.178	0.00%	0.179	0.156	-12.70%
	(0.006)	(0.006)		(0.004)	(0.007)	

Notes. This table reports how wage inequality and welfare differ between the monopsonistic labor market and the counterfactual where all firms are price-takers. Results are reported for two intermediate restrictions on preference heterogeneity: one where workers' wage-amenity trade-offs β are constant within skill types χ, and one where firm-specific labor supply elasticities are constant within local labor markets. In each case, the imposed value is the corresponding group mean under the full model. Standard errors are bootstrapped using 500 replications.